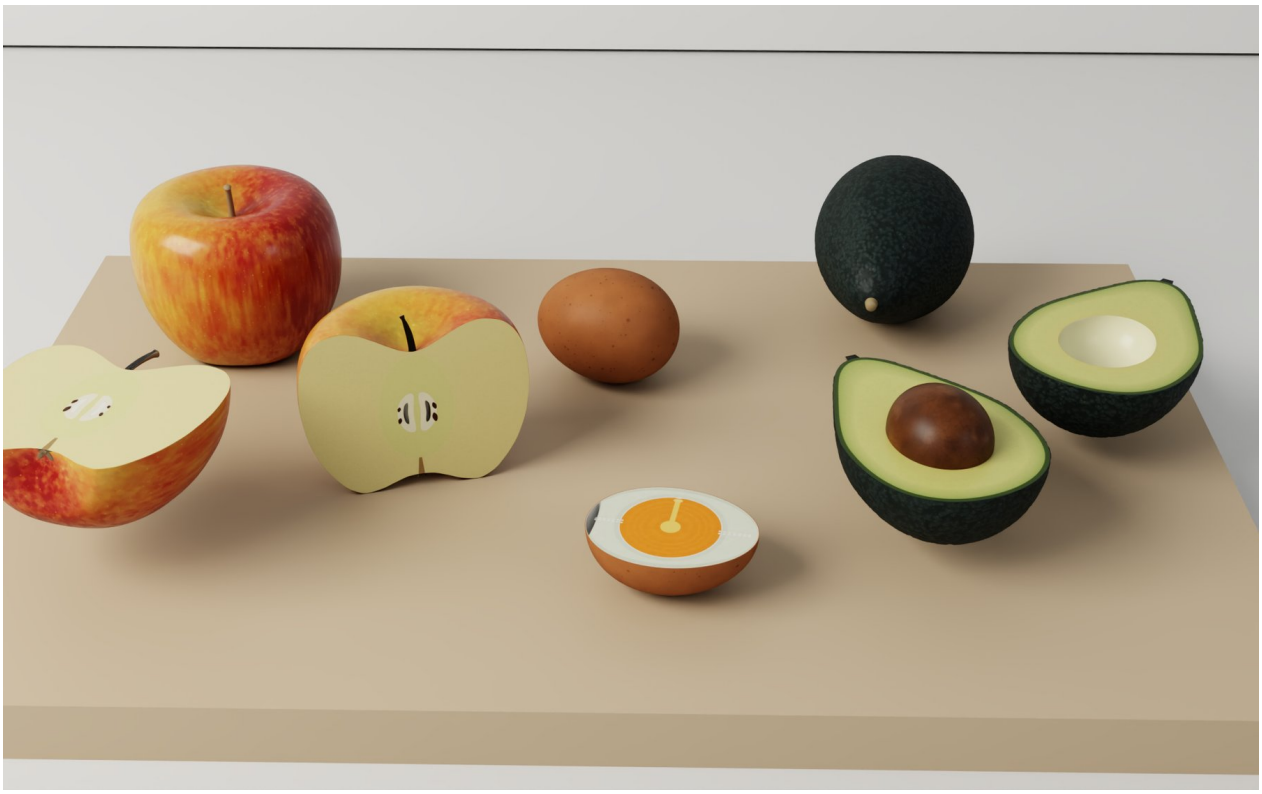


Food that robots can reason about

Semantic digital twins of fruit, vegetables, eggs and bread for robot manipulation in household kitchens: what we have so far

16 kitchen items for the EASE and GARMI apartments, modelled part by part

[Authors and affiliations]

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This is a research twin built from published biology, food-science and cooking sources. It is not a measured model of any particular piece of food. Every quantitative value carries a status label from the project's parameter register (SOURCED, SOURCED-VERIFY, STANDARD, DERIVED, TYPICAL, CALIBRATED, ILLUSTRATIVE; Section 10 and Appendix A). ILLUSTRATIVE values are plausible assumptions of the twin and must not be quoted as facts. Bench results are simulations, not experiments.

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Summary

Purpose. We want models of everyday food that a robot can plan with, act on and learn about: geometric and physical objects in Isaac Sim whose every part is a node of a semantic digital twin (SDT), that knows what it is (*shell*, *yolk*, *seed*, *crust*), how the robot can come to know it (visible, only after cutting, tactile, inferred, hidden), and how it behaves when it is gripped, dropped, squeezed, cut, peeled or cooked. The target scenes are the EASE (IAI Bremen) and GARMI apartments, which already exist as robot-simulation scenes in our CRAM tooling. Each item has three layers: a continuous model of a single, randomly drawn *instance* (its hidden truth), the robot's *beliefs* about it (priors from a parameter register, updated by observations), and a small *discrete* model (attributes, states and actions) that plans and action designators refer to.

What was built. 16 items have a code module and a parameter register: the egg set (egg, apple, avocado), wave 1 (roll, banana, carrot), wave 2 (loaf, onion, orange, potato, tomato) and wave 3 (lemon, cucumber, croissant, baguette, pepper). All 16 are built as USD assets: 27 instances in `assets/food` with 712 part prims in total, each with cut states, a belief layer (`sdt.usda`) and a hidden-truth layer (`truth.usda`). The register holds 564 entries and 205 sources, 166 of which the entries cite by key. There are 30 task designators (egg 7, roll 4, banana 3, carrot 3, loaf 2, tomato 2, potato 2, onion 2, orange 2, avocado 3), seven Isaac Sim benches with results in `test_reports/` (egg grasp and drop, egg opening, avocado, roll and tomato squeeze, banana peel, carrot plank) and two CPU studies (egg cracking and boiling), and the EASE and GARMI apartments and the GARMI robot are imported into USD.

One avocado, end to end. The avocado now goes through the whole loop a robot needs (Section 7): a photo becomes an SDT node; the node is a 4000-particle belief updated by Bayes' rule from photos, squeezes, cuts, twists and the spoon; a vision-based monitor judges peeling against the twin's expectation; an emulator halves and pits the avocado as USD states; the robot answers competence questions from its belief, its task models and mental simulation (including whether a squeeze is worth it); and a PR2 in the CRAM kitchen detects, squeezes, cuts round the seed, twists the halves apart and spoons out the seed, checking every step against what it imagined – scripted, or driven by an operator from the CRAMERA viewer.

From reasoning to acting and to theories. The robot then acts as an agent that enforces the effects its mission requires and prevents the ones it forbids, deciding over the fruit's lifetime (ripening, spoilage, browning), and its knowledge is written down and tested as an avocado manipulation microtheory – the egg microtheory transferred – which the robot also learns from its own failures and which programs its reinforcement learning (Section 8; the full microtheory is `report/avocado_microtheory.pdf`).

Headline results. (1) The egg boiling model (heat conduction through the twin's own tissue map, doneness thresholds CALIBRATED) reproduces 6/6 doneness classes of the hot-start timing table it was calibrated on and 3/3 of the fridge-egg timings; this is a fit, not an independent test. (2) The egg crack study turns kitchen advice into numbers: over 400 register-sampled eggs the best strike on another egg is 0.50 m/s and on a flat counter 0.45 m/s (strike-model constants ILLUSTRATIVE). (3) For the red tomato the PhysX FEM squeeze at 4000 Hz gives 4.1 N/mm against 4.53 N/mm from Hertz theory and 4.5 N/mm measured (ince16, SOURCED-VERIFY); stiffer lanes stay under-converged (green: 6.0 vs Hertz 10.2 N/mm). The same solver limit appears in the avocado and roll benches. (4) Opening a cracked egg over a bowl in Isaac Sim keeps 99.9 % of 7,703 albumen particles and the yolk in the bowl. (5) The carrot bench shows in simulation why cooks cut a plank before slicing: with a slicing stroke the round root rolls away at 1 N already, the planked half only at about 10 N. (6) The avocado's peeling monitor finds 22/24 correct verdicts (skin IoU 0.46) on synthetic peels from RGB-D, and in the PR2 scenario the robot's ripeness belief, misled by the photo, is corrected by an unexpected twist outcome towards the hidden truth (Section 7.6).

What it is not, yet. 224 of 564 register entries (40 %) are ILLUSTRATIVE and another 108 need their source checked. Wave 3 is now built (USD assets with cut states, renders, anatomy plates) but has no task designators or benches, and its research notes are referenced but missing. Only egg, apple and avocado have been placed in a kitchen, and the end-to-end GARMI egg task stopped with a numerical failure of the arm controller. The anatomy plates are waiting for the user's review, which is a gate of the build plan. The physics benches are single runs of simplified bodies (homogeneous FEM, rigid peel segments, kinematic shells).

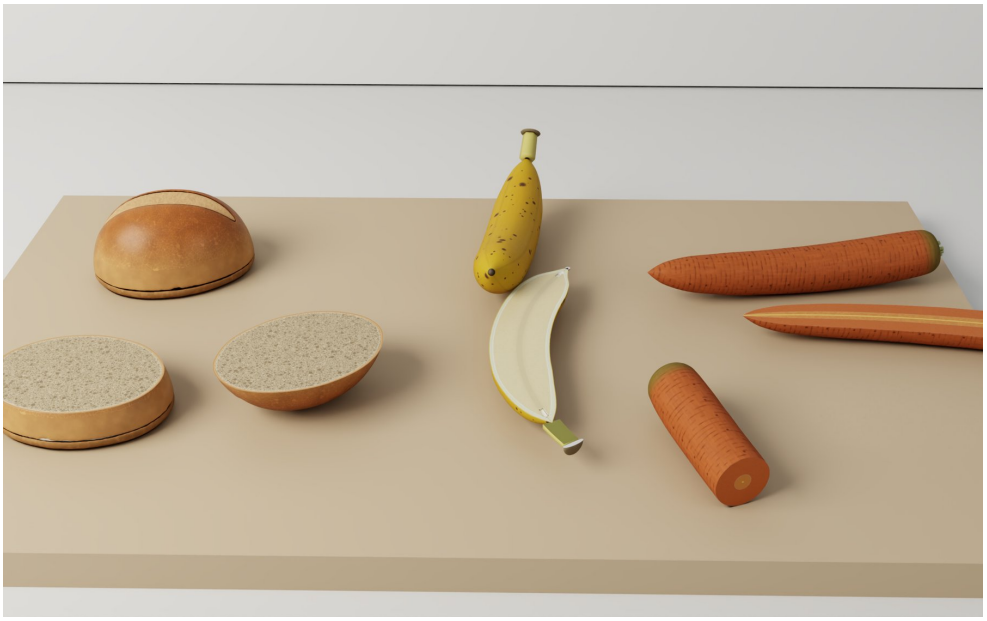


Figure 1: Wave 1 items, path-traced in Isaac Sim 6.2 (existing render): the roll whole and as top and bottom halves, the banana whole and as a lengthwise half, and the carrot whole, as a lengthwise half and as a crown piece.

Table 1: What exists per item. Reg.: register entries; % ILL.: share of ILLUSTRATIVE entries; USD: built instances in assets/food; Des.: task designators. Counted from the repository by this script.

Item	Wave	Reg.	% ILL.	USD	Cut / derived states	Images	Des.	Bench
Hen egg	egg set	36	33	8	cutaway	plate + renders	7	grasp/drop, open, crack, cook
Apple (Gala-like)	egg set	33	24	1	half_a, half_b	plate + renders	0	–
Hass avocado	egg set	48	46	2	half_a, half_b	plate + renders	3	FEM squeeze, peel monitor, emulator, PR2 (CRAM)
Wheat roll (Weizenbrötchen)	wave 1	46	41	1	half_a, half_b, top_half, bottom_half	plate + renders	4	FEM squeeze
Cavendish banana	wave 1	41	37	4	half_a, half_b, tip_half, stalk_half	plate + renders	3	peel
Nantes carrot	wave 1	42	38	1	half_a, half_b, tip_piece, crown_piece	plate + renders	3	plank
Sandwich loaf (Toastbrot)	wave 2	26	42	1	half_a, half_b, slices	plate + renders	2	–
Yellow onion	wave 2	32	47	1	half_a, half_b, top_piece, root_piece	plate + renders	2	–
Sweet orange	wave 2	38	32	1	stem_half, navel_half, half_a, half_b	plate + renders	2	–
Potato	wave 2	36	39	1	half_a, half_b, rose_piece, heel_piece	plate + renders	2	–
Round / truss tomato	wave 2	37	46	1	stem_half, blossom_half, half_a, half_b	plate + renders	2	FEM squeeze
Lemon	wave 3	31	29	1	stem_half, blossom_half, half_a, half_b	plate + renders	0	–
Greenhouse cucumber	wave 3	28	43	1	half_a, half_b, rounds	plate + renders	0	–
Butter croissant	wave 3	21	62	1	top_half, bottom_half, half_a, half_b	plate + renders	0	–
Baguette	wave 3	42	45	1	half_a, half_b, top_half, bottom_half, crostini	plate + renders	0	–
Bell pepper	wave 3	27	37	1	half_a, half_b, top_piece, bottom_piece	plate + renders	0	–

1 Method: a food item as a semantic digital twin

From biology to a part tree. Each item is a Python class (`food/<item>.py`) that draws one instance from the register (`params.sample(kind, seed)`) and builds an *analytic anatomy* for it: meridian curves of revolution, offsets and lofts (`food/geometry.py`, `food/shapes.py`) and a label field that says, for any point in space, which tissue it belongs to (shell, thick albumen, yolk; flavedo, albedo, juice vesicles; crust, crumb). The class's `parts()` returns the anatomical partonomy as a tree of `Part` records: a USD path, an ontology class, an observability class, a documentation string with sources, and the geometry source (a label of the field, an analytic mesh, or a point set for things such as pores, lenticels, gas cells and seeds). The anatomy was researched in biology and food-science sources; the user rejected two early versions (a wrong apple core, an egg-like avocado seed), and the anatomy plates are still awaiting review (Section 11).

Closed meshes and cuts. `food/partgraph.py` turns every labelled part into a closed (watertight) mesh by marching cubes on the label field, and computes volume and mass from the part's density. Cuts are plane splits of every part mesh (manifold3d `split_by_plane`, nudged off the symmetry axis), so a half is again a complete part graph with its own mass, centre of mass and inertia. Table 4 checks mass conservation.

USD part graph. `food/usd_partgraph.py` writes one USD asset per instance. Every prim under `/Food/Parts` is an SDT node with `sdt:represents` (the entity IRI of the physical part), `sdt:class`, `sdt:partOf` and `sdt:observability`, UsdSemantics labels for perception, and `food:mass_g`, `food:volume_cm3`, `food:density_kgm3` from its closed mesh. The whole object is one rigid body whose mass properties are summed over the parts; interior parts are invisible until a cut state exposes them. Two layers sit over the geometry: `sdt.usda` holds the robot's beliefs (priors from the register, observations), and `truth.usda` holds the instance's hidden parameters, which the dynamics models and audits read but the robot never does.

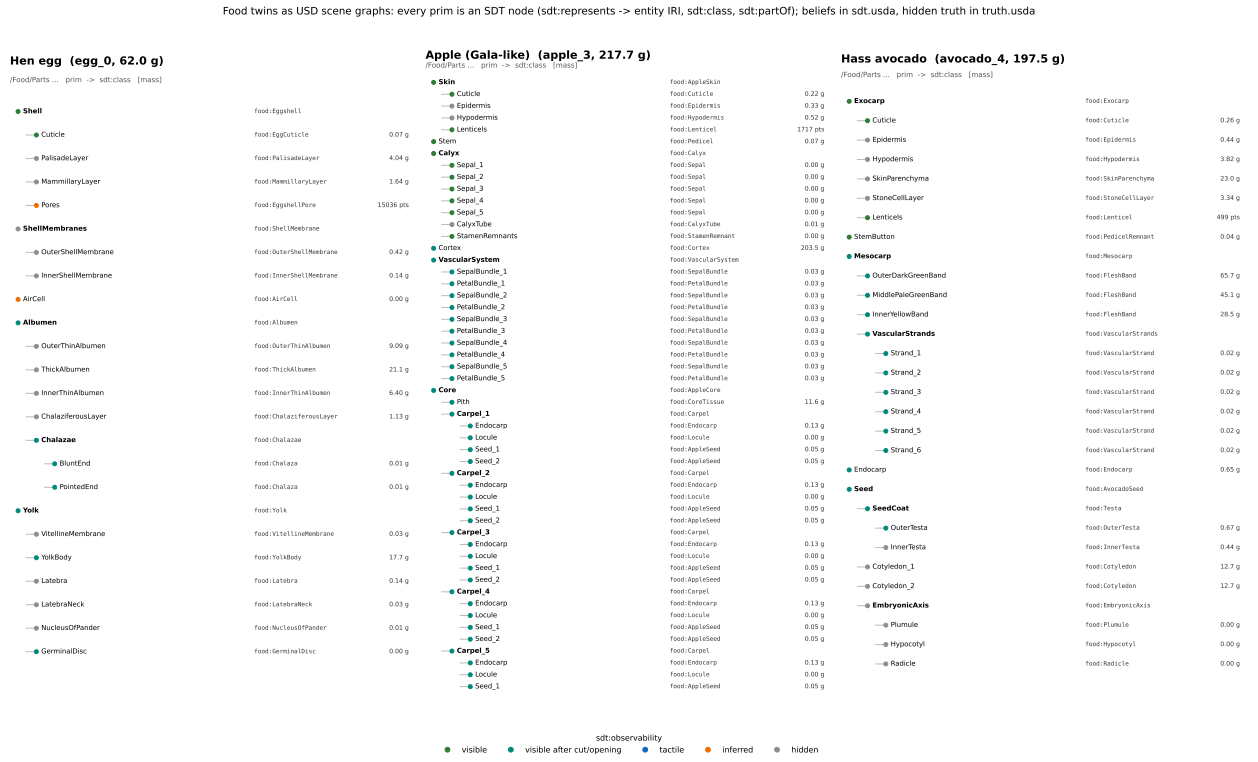


Figure 2: The USD scene graphs of the first three items as SDT part graphs (existing figure from tools/anatomy_doc.py): prim, sdt:class and part mass; the colour of the dot is the observability class.

Excerpt of assets/food/egg_1/whole.usdc (read with pxr by this script)

```
def Xform "Food"   # /Food
    string food:kind = 'egg'
    double food:mass_g = 65.59
    string food:shell_condition = 'intact'
    string sdt:class = 'food:Egg'
```

```
    string sdt:represents = 'https://aicor.knowledge/household/food/egg_1'
    token[] semantics:labels:class = [egg]
    token[] semantics:labels:part = [egg]
def Xform "Shell"   # /Food/Parts/Shell
    string sdt:class = 'food:Eggshell'
    string sdt:observability = 'visible'
    string sdt:partOf = 'https://aicor.knowledge/household/food/egg_1'
    string sdt:represents = 'https://aicor.knowledge/household/food/egg_1/shell'
    string sdt:usdPrimPath = '/Food/Parts/Shell'
    token[] semantics:labels:class = [egg_shell]
    token[] semantics:labels:part = [shell]
def Xform "Cuticle"   # /Food/Parts/Shell/Cuticle
    double food:density_kgm3 = 1300
    double food:mass_g = 0.09072
    double food:volume_cm3 = 0.06978
    string sdt:class = 'food:EggCuticle'
    string sdt:observability = 'visible'
    string sdt:partOf = 'https://aicor.knowledge/household/food/egg_1/shell'
    string sdt:represents = 'https://aicor.knowledge/household/food/egg_1/cuticle'
    string sdt:usdPrimPath = '/Food/Parts/Shell/Cuticle'
    token[] semantics:labels:class = [egg_cuticle]
    token[] semantics:labels:part = [cuticle]
def Xform "GerminalDisc"   # /Food/Parts/Yolk/GerminalDisc
    double food:density_kgm3 = 1030
    double food:mass_g = 0.00096
    double food:volume_cm3 = 0.000932
    string sdt:class = 'food:GerminalDisc'
    string sdt:observability = 'visible after opening'
    string sdt:partOf = 'https://aicor.knowledge/household/food/egg_1/yolk'
    string sdt:represents = 'https://aicor.knowledge/household/food/egg_1/germinal_disc'
    string sdt:usdPrimPath = '/Food/Parts/Yolk/GerminalDisc'
    token[] semantics:labels:class = [egg_germinal_disc]
    token[] semantics:labels:part = [germinal_disc]
```

assets/food/egg_1/sdt.usda (beliefs, lines 9–28)

```
over "Food"
{
    custom string sdt:belief:contents = "in_shell 1.0"
    custom double sdt:belief:mass_g = 65.58658139602177
    custom string sdt:belief:mass_g_source = "weighed"
    custom string sdt:belief:shell_condition = "intact 0.97 | microcracked 0.03 (prior)"
    custom string sdt:epistemicStatus = "prior + observation"

    over "Parts"
    {
        over "Shell"
        {
            custom double sdt:belief:F_break_equatorial_N_mean = 34.92249125484492
            custom double sdt:belief:F_break_equatorial_N_p05 = 21.952609816956432
            custom double sdt:belief:F_break_equatorial_N_p95 = 48.02211933416059
            custom string sdt:epistemicStatus = "prior (register: SOURCED)"
        }

        over "AirCell"
        {
```

assets/food/egg_1/truth.usda (hidden truth, first lines)

```
#usda 1.0
(
    "Simulator ground truth (hidden instance parameters): overs only, never part of the SDT; used by the dynamics models and audits."
)

over "Food"
{
    custom double truth:age_days = 11.762866264385565
    custom double truth:air_cell_height = 3.198358086590051
    custom double truth:albumen_density = 1035
    custom double truth:asymmetry_w = 1.859487302895341
    custom string truth:colour = "brown"
    custom double truth:cuticle_thickness = 9.208033800009677
    custom double truth:F_break_axial = 35.533810468062526
    custom double truth:F_break_equatorial = 29.108367303986665
    custom double truth:friction_shell_rubber = 0.5409584487874936
```

The parameter register. `food/params.py` holds every number the models use, per item, as *(unit, value or distribution, status, source)*. The status rule is carried over from the lab twin: no invented physical parameter is

presented as a fact. Distributions (normal with truncation, uniform, integer, choice, lognormal) are what the hidden-world sampler draws per instance; the robot's belief starts from the same distribution as its prior and learns the instance's values only through observations. Of the 564 entries, 441 carry a number or a distribution; the rest are model notes (formulas, tables, ranges as text) that the code implements.

Table 2: Status labels of the parameter register.

Status	Meaning (food/params.py)
SOURCED	value or range taken from the cited source
SOURCED-VERIFY	value from a source whose details (species, cultivar, protocol) still need checking in the full text
STANDARD	a regulation or grading standard (definitional, not a measurement)
DERIVED	computed from other register entries by a stated physical model
TYPICAL	common textbook value, no citation held in this repository yet
CALIBRATED	free model constant fitted to a cited observation (the fit and its residual are stated)
ILLUSTRATIVE	twin assumption chosen to be plausible; must be calibrated before any quantitative claim

Discrete schema, dynamics and beliefs. `food/schema.py` is the symbolic layer: per item, a class, attributes with domains and observability, and actions with preconditions, effects and *guards* – the continuous quantities, computed by the twin, that decide which effect occurs (for the egg: the grip corridor $F_{\text{slip}} < F < F_{\text{break}}$ and the AICOR H8 admissibility $P(\text{slip}) < \varepsilon_s$, $P(\text{rupture}) < \varepsilon_r$, $P(\text{stable}) > \gamma$). The schema covers the first six items; the wave 2 and wave 3 items expose their discrete state through `discrete()` methods of their truth classes instead. The hidden dynamics are in `food/dynamics.py` (egg, apple, avocado) and `food/<item>_models.py` (the later items); the beliefs in `food/sdt.py` and the `*_beliefs` functions. Task designators (`food/egg_tasks.py`, `wave1_tasks.py`, `wave2_tasks.py`) list goal, parameters, steps, preconditions, effects and failure modes with cues and recoveries, each bound to the hidden model that executes it.

Table 3: The discrete schema (food/schema.py).

Class	Attributes	Actions
Egg	shell_condition, contents, yolk, weight_grade, quality_class, shell_colour, orientation, support, F_break, F_slip	Grasp, Lift, Transport, Place, Drop, CrackOnEdge, OpenUp, Candle
Apple	integrity, bruise, peel, core, cut_face_browning, firmness, orientation, seed_positions	Grasp, Place, Drop, Halve, Core, Peel
Avocado	ripeness_stage, integrity, seed_coat, seed, skin, flesh_condition, cut_face_browning	SqueezeTest, Grasp, Halve, CutRoundSeed, TwistApart, RemoveSeed, Scoop
BreadRoll	freshness, crust_texture, cut_state, toast_level, temperature, topping, support	Grasp, SliceOpen, Reheat, ToastHalves, Store
Banana	ripeness_stage, edible, peel_state, neck, attached_to_hand, bruised, chilling_injury, pulp_browning, integrity	SeparateFromHand, SnapNeck, PeelStrip, CutRounds, Mash
Carrot	firmness_class, peel_state, trim_state, cut_shape, cooked_state, surface_blush, green_shoulder	Scrub, Peel, TopAndTail, Plank, Cut, Snap, Boil

Mass conservation. The table checks, for every built instance, that the part masses add up to the whole and that the pieces of each cut add up to the whole (ratio 1.0000 = conserved).

Table 4: Mass checks of the built instances (assets/food/*/instance.json).

Instance	whole (g)	parts (g)	diff. %	cut pieces / whole	cut failures
apple_3	217.7	217.7	+0.0	half_a+half_b: 1.0000	Core/Carpel_1/Locule, Core/Carpel_3/Locule
avocado_4	197.5	197.5	+0.0	half_a+half_b: 1.0000	none
avocado_photo_burlap	213.0	213.0	-0.0	half_a+half_b: 1.0000	none
baguette_0	213.6	216.0	+1.1	half_a+half_b: 1.0000; top_half+bottom_half: 1.0000; crostini: 1.0000	none
banana_0	117.1	117.1	+0.0	half_a+half_b: 1.0000; tip_half+stalk_half: 1.0000	none
banana_1	130.5	130.5	-0.0	half_a+half_b: 1.0000; tip_half+stalk_half: 1.0000	none
banana_2	119.1	119.1	+0.0	half_a+half_b: 1.0000; tip_half+stalk_half: 1.0000	none
banana_3	113.9	113.9	+0.0	half_a+half_b: 1.0000; tip_half+stalk_half: 1.0000	none
carrot_0	115.0	115.0	-0.0	half_a+half_b: 1.0000; tip_piece+crown_piece: 1.0000	none
croissant_0	51.3	51.3	+0.0	top_half+bottom_half: 1.0000; half_a+half_b: 1.0000	none
cucumber_0	371.0	371.0	+0.0	half_a+half_b: 1.0000; rounds: 1.0000	none
egg_0	62.0	62.0	+0.0	–	none
egg_1	65.6	65.6	+0.0	–	none
egg_2	61.3	61.3	+0.0	–	none
egg_3	64.3	64.3	+0.0	–	none
egg_4	57.1	57.1	+0.0	–	none

Table 4: Mass checks of the built instances (assets/food/*/instance.json). (continued)

Instance	whole (g)	parts (g)	diff. %	cut pieces / whole	cut failures
egg_5	53.4	53.4	+0.0	–	none
egg_6	72.9	72.9	+0.0	–	none
egg_7	62.2	62.2	+0.0	–	none
lemon_0	118.0	118.0	+0.0	stem_half+blossom_half: 1.0000; half_a+half_b: 1.0000	none
loaf_0	494.6	501.5	+1.4	half_a+half_b: 1.0000; slices: 1.0000	none
onion_0	146.5	146.5	-0.0	half_a+half_b: 1.0001; top_piece+root_piece: 1.0000	none
orange_0	193.6	193.6	+0.0	stem_half+navel_half: 1.0000; half_a+half_b: 1.0000	none
pepper_0	163.6	163.6	+0.0	half_a+half_b: 1.0000; top_piece+bottom_piece: 1.0000	none
potato_0	126.4	126.4	+0.0	half_a+half_b: 1.0000; rose_piece+heel_piece: 1.0000	none
roll_0	47.4	48.1	+1.4	half_a+half_b: 1.0000; top_half+bottom_half: 1.0000	none
tomato_15	71.7	71.7	+0.0	stem_half+blossom_half: 1.0000; half_a+half_b: 1.0000	none

Part sums deviate from the whole for baguette_0 (216.0 vs 213.6 g, +1.1 %), loaf_0 (501.5 vs 494.6 g, +1.4 %), roll_0 (48.1 vs 47.4 g, +1.4 %): these are the two breads, whose crust and crumb meshes appear to overlap or are meshed at a different resolution from the whole; the wave 1 checklist in docs/wave1/README.md reports this gate as passed, which this check does not confirm. The apple's lengthwise cut lost two empty locules (cut failures above); their mass is negligible.

2 The hen egg

The egg is the most developed item: anatomy, crack and cook models, a belief that computes the AICOR H8 probabilities, seven task designators, a grasp-and-drop bench and an opening simulation in Isaac Sim. Eight instances are built (`egg_0`..`egg_7`, 53.4–72.9 g); the examples use `egg_1` (65.6 g).

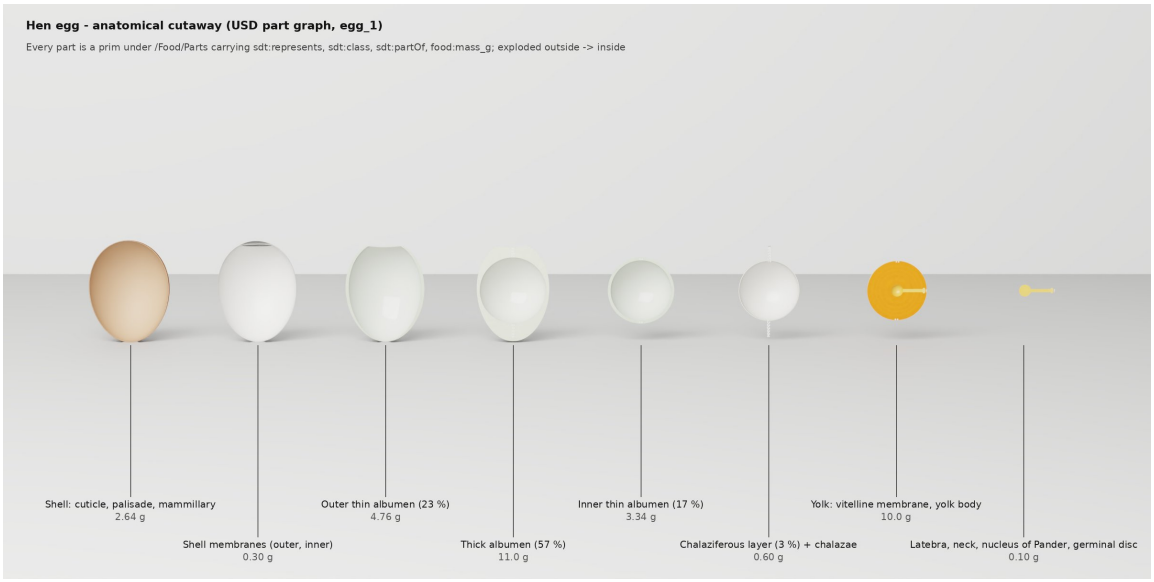


Figure 3: Exploded anatomy plate of egg_1 (existing render): outside to inside, every part is a prim of the USD part graph with its mass.



cutaway state | opened, fried | half shell

Figure 4: Egg renders (existing, path-traced). The cutaway shows the yolk layers, latebra and chalazae.

States and hidden dynamics. The schema's attributes are shell_condition, contents, yolk, weight_grade, quality_class, shell_colour, orientation, support, F_break, F_slip. Shell condition follows the USDA vocabulary (intact, microcracked, cracked, shattered, opened); a micro-crack is invisible and lowers the break force by the register's `microcrack_strength_loss`. `EggTruth` (`food/dynamics.py`) cracks the shell when the effective contact force exceeds the direction-dependent break force; impacts use $F = v\sqrt{k_{\text{eff}}\, m_{\text{eff}}}$ with the Reissner shell stiffness in series with the surface. `food/egg_crack.py` adds strike outcomes on a counter, a bowl rim or another egg and the opening over a vessel; its constants are ILLUSTRATIVE. `food/egg_cook.py` solves axisymmetric heat conduction through the egg's own tissue map. The belief `EggBelief` (`food/sdt.py`) holds particles over crack force, shell thickness and friction and returns the grip corridor and H8 admissibility.

2.1 Bench: grasp and drop (Isaac Sim)

`scripts/sim_egg_bench.py` grips egg_1 with PhysX contacts at commanded pad forces and drops eggs onto four kitchen surfaces; outcomes come from the hidden `EggTruth`, predictions from `EggBelief`. In the grip sweep, the measured pad force equals the commanded one within 0.40 N; the egg slips at 0.5 N, the H8-admissible commands run from 1 to 15 N, and the shell cracks from 25 N (hidden crack force of this egg 23 N). In a validation lane eight instances were gripped at 30 N: 6 of 8 cracked or micro-cracked, while the belief predicted a rupture probability of about 0.34. The eight hidden crack forces (median 29.5 N) sit below the register prior's mean (35 N, SOURCED); with

Table 5: Part tree of egg_1 (instance.json and whole.usdc).

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
Shell	food:Eggshell	visible		
Cuticle	food:EggCuticle	visible		0.09
PalisadeLayer	food:PalisadeLayer	hidden		3.68
MammillaryLayer	food:MammillaryLayer	hidden		1.52
Pores	food:EggshellPore	inferred		12,731 points
ShellMembranes	food:ShellMembrane	hidden		
OuterShellMembrane	food:OuterShellMembrane	hidden		0.45
InnerShellMembrane	food:InnerShellMembrane	hidden		0.15
AirCell	food:AirCell	inferred		0.00
Albumen	food:Albumen	visible after opening		
OuterThinAlbumen	food:OuterThinAlbumen	hidden		9.52
ThickAlbumen	food:ThickAlbumen	hidden		22.01
InnerThinAlbumen	food:InnerThinAlbumen	hidden		6.68
ChalaziferousLayer	food:ChalaziferousLayer	hidden		1.18
Chalazae	food:Chalazae	visible after opening		
BluntEnd	food:Chalaza	visible after opening		0.01
PointedEnd	food:Chalaza	visible after opening		0.01
Yolk	food:Yolk	visible after opening		
VitellineMembrane	food:VitellineMembrane	hidden		0.03
YolkBody	food:YolkBody	visible after opening		20.07
Latebra	food:Latebra	hidden		0.16
LatebraNeck	food:LatebraNeck	hidden		0.03
NucleusOfPander	food:NucleusOfPander	hidden		0.01
GerminalDisc	food:GerminalDisc	visible after opening		0.00

Table 6: Key register entries of the egg.

Register key	Value / distribution	Status	Source and note
length_L	N(57, 1.8) in [52, 63] mm	SOURCED	kop: 58.05 mm; turjaf: 56.12 mm
shape_index_B_over_L	N(0.765, 0.02) in [0.7, 0.81]	SOURCED	kop: 78.25 %; alt: normal 72-76
shell_thickness	N(0.35, 0.03) in [0.28, 0.42] mm	SOURCED	kop: 0.28-0.41 mm; sb: 0.35 mm
shell_E	U(15 .. 43) GPa	SOURCED	sev: 41-43 GPa nanoindentation; sb model 15 GPa
F_break_equatorial	N(35, 8) in [12, 59] N	SOURCED	kop: Hisex 36.43 N short axis, 10-59 N; icf11 34±6 N
F_break_axial	N(37, 9) in [12, 62] N	SOURCED	kop 34.92 N long axis; icf11 41±8 N; alt: highest along the polar axis at the sharp end
shell_contact_stiffness	Reissner k = 4 E t^2 / (R_local sqrt(3(1-nu^2))) N/mm	DERIVED	reissner1946 + shell_E + shell_thickness
yolk_mass_fraction	N(0.3, 0.015) in [0.27, 0.32]	SOURCED	br: 27-32 %; usda 31 %
air_cell_height	1.5 + k(T) * age; k(6 C) = 0.08, k(22 C) = 0.19 mm/d mm	SOURCED-VERIFY	gr: < 4 mm after 28 d at 6 C; > 6 mm after 25 d at 22 C; graded vs eu589/usda
microcrack_band	0.85	ILLUSTRATIVE	F/F_break in [0.85, 1) may leave an invisible micro-crack
impact_effective_mass_fraction	0.53	CALIBRATED	c = sqrt(m_eff)/rate fitted to cohen2025: side-on drops onto a hard surface crack ~45 % at 10 mm (model 45 %), vertical 55 % at 8 mm (model ~57 %); the model population is more heterogeneous than one MIT batch (8 mm side-on: model ~28 % vs 5 %)
impact_rate_factor	1.3	ILLUSTRATIVE	dynamic/static crack force; strength rises with loading rate (only sqrt(m_eff)/rate is identified by the calibration)
crack_strike_model	food/egg_crack.py P: coupled mass 0.05 kg; membrane tears at F/Fc 1.8 (flat), 1.05 (rim), 3.0 (egg-on-egg); shatter at 2.6 / 2.2 / 2.4; shards 0.15 / 0.6 / 0 per F/Fc; yolk puncture 0.06 per shard	ILLUSTRATIVE	structure from atk_egg, egginfo_crack, atk_egg_on_egg; numbers to calibrate
cook_yolk_thresholds	f65 < 0.15 runny, < 0.45 jammy, < 0.85 creamy, else firm	CALIBRATED	alphafoodie timing table (hot start, medium eggs) + Lopez-Alt fridge-egg timings

Table 7: Egg task designators (food/egg_tasks.py; rendered in docs/EGG_TASKS.md).

Designator	Goal	Twin model
PickUpEgg	egg held, shell_condition unchanged	PhysX contacts + EggTruth.static_load; EggBelief.admissible (AICOR H8)
TransportEgg	egg at target support, intact	EggTruth.impact (calibrated to cohen2025)
CrackEgg	shell cracked open, contents in the cup/bowl, yolk whole, no shell inside	food/egg_crack.py strike + open_up + fish_out
SeparateEgg	yolk in yolk bowl (whole), white in white bowl with no yolk	EC.open_up yolk-break model (age, drop height, shards); separation success ILLUSTRATIVE
BoilEgg	yolk doneness = target (runny | jammy | creamy | firm), shell intact, peels cleanly	food/egg_cook.py EggHeat (conduction through the tissue map; thresholds CALIBRATED)
ScrambleEggs	curds set but glossy, no liquid egg, not browned	not yet simulated (coagulation thresholds from opentextbc)
FrySunnySideUp	white set and opaque incl. around the yolk, yolk whole and runny, no shell	not yet simulated

eight eggs this is not conclusive, but it is worth checking whether the truth sampler and the belief draw crack forces from the same effective distribution (the truth applies thickness scaling, which the belief may not). Drops from 1 cm already crack this egg on granite, oak board, silicone mat; on a towel it survives up to 8 cm and cracks at 15 cm (critical heights in the figure). The impact model's effective-mass fraction is CALIBRATED to the drop statistics of Cohen et al. 2025 (cohen2025).

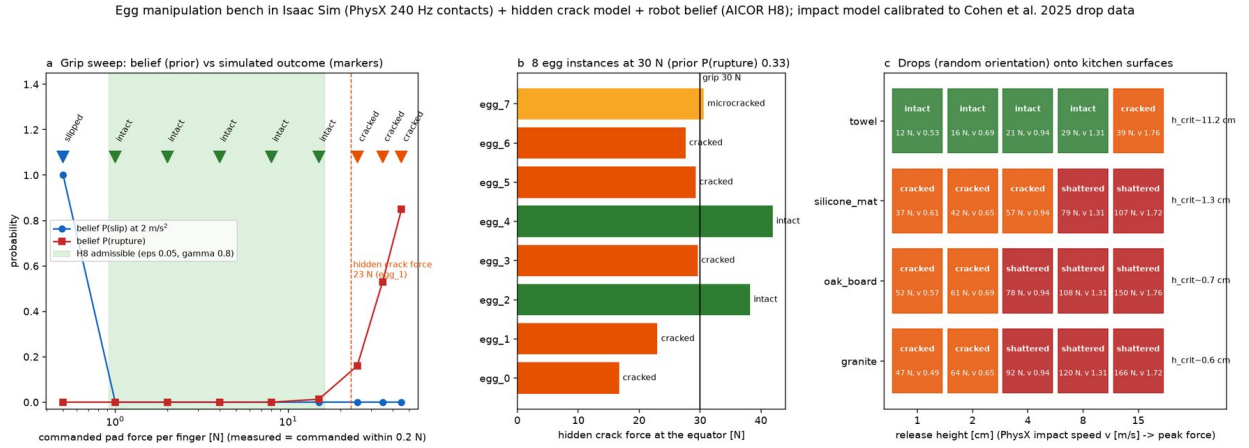


Figure 5: Egg bench (existing figure, tools/plot_egg_bench.py): (a) grip sweep, belief vs simulated outcome; (b) eight instances at 30 N; (c) drops onto kitchen surfaces with peak force and impact speed.

2.2 Study: how hard to crack an egg (CPU)

`tools/egg_crack_study.py` strikes 400 register-sampled eggs at 30 speeds against four tools and counts how often the egg ends up open in the cup with a whole yolk and no shell – directly, and with the recovery people use (crack into a cup first, fish shards out with a half shell). The strike-outcome bands, shard rates and puncture probabilities are ILLUSTRATIVE (register `crack_strike_model`), so the optimum speeds are model answers, not measurements. This report re-ran the computation in memory with the current register (impact effective-mass fraction 0.53): the best speeds and probabilities are identical to the stored study.

Table 8: Egg crack study (test_reports/egg_crack_study.json) and this report's re-run.

Strike surface	best v (m/s)	P with recovery	P direct (best)	90 % window (m/s)	re-run v / P
flat counter (granite)	0.45	0.88	0.75	0.40–0.50	0.45 / 0.88
thin bowl rim (r 1.5 mm)	0.25	0.76	0.50	0.20–0.25	0.25 / 0.76
thick bowl rim (r 4 mm)	0.35	0.77	0.49	0.35–0.40	0.35 / 0.77
another egg	0.50	0.85	0.80	0.45–0.65	0.50 / 0.85

2.3 Study: boiling (CPU)

`tools/egg_cook_study.py` runs `EggHeat` on egg_4 (thermal properties TYPICAL, doneness thresholds on the yolk fraction that reached 65 °C CALIBRATED to the alphafoodie timing table and the Lopez-Alt fridge-egg timings). The table compares the model's doneness class with the source at each source time. Because the thresholds were fitted to these points, agreement shows a consistent fit, not a validation; no independent timing data are held. The cold-start case (water heated to the boil in 9 min, then off the heat, ILLUSTRATIVE profile) is not compared with a source.

Table 9: Egg boiling model vs the cooking sources it was calibrated on (test_reports/egg_cook_study.json).

min	Case	Source class	Model class	Match
4.5	hot start, room-temperature egg	runny	runny	yes
6	hot start, room-temperature egg	jammy	jammy	yes
7	hot start, room-temperature egg	jammy	jammy	yes
8.5	hot start, room-temperature egg	creamy	creamy	yes
10	hot start, room-temperature egg	firm	firm	yes
11	hot start, room-temperature egg	firm	firm	yes
6	hot start, fridge egg	runny	runny	yes
8.5	hot start, fridge egg	creamy	creamy	yes
11	hot start, fridge egg	firm	firm	yes

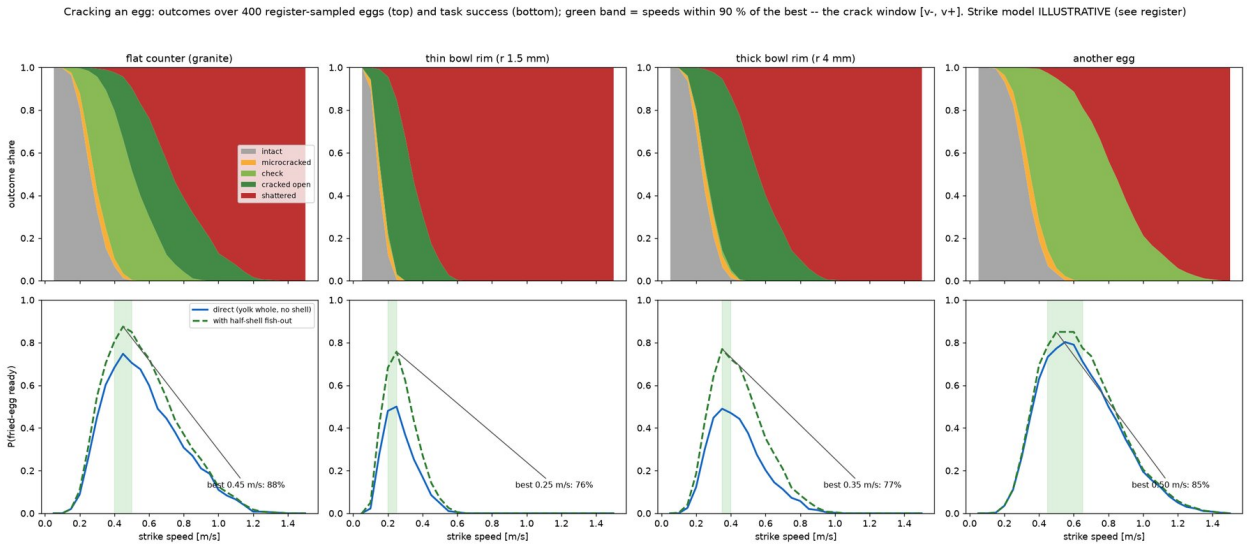


Figure 6: Crack study (existing figure): outcome shares (top) and task success (bottom) over strike speed per tool.

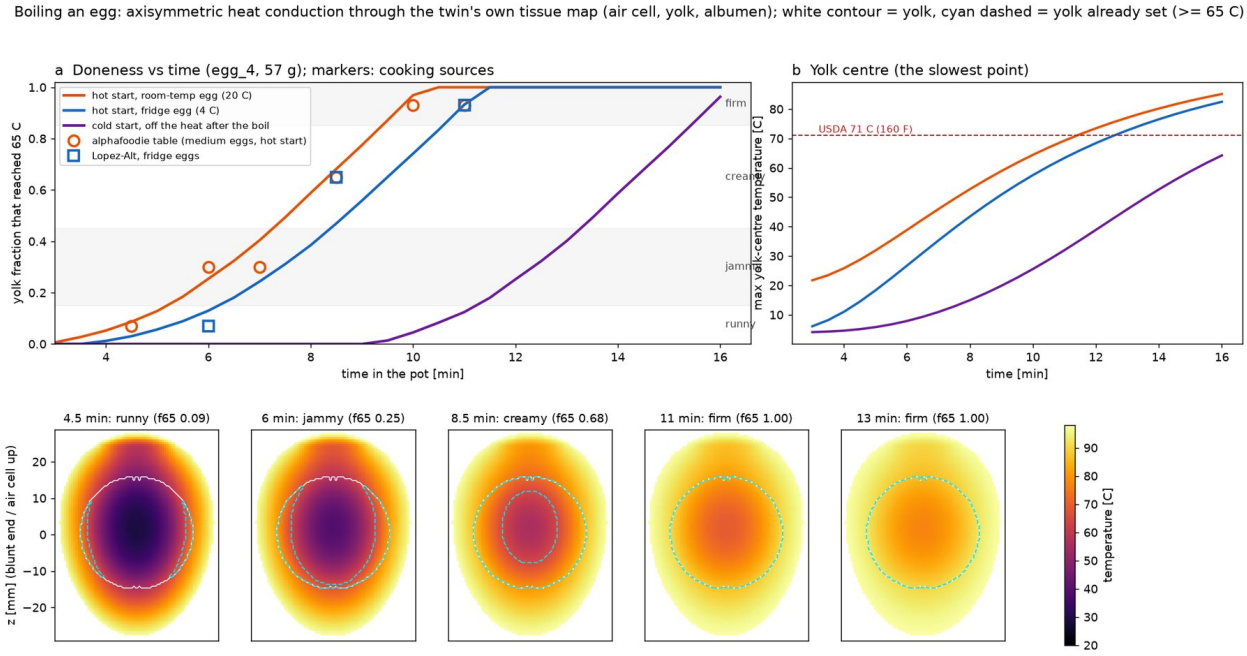


Figure 7: Boiling study (existing figure): doneness over time with the source markers, yolk-centre temperature, and temperature sections at five times.

2.4 Simulation: opening a cracked egg (Isaac Sim)

`scripts/sim_egg_open.py` replaces the cracked egg by its two shell fragments (kinematic, from the crack model), the albumen by 7,703 PBD particles and the yolk by a rigid body, and swings the halves apart over a glass bowl. After 3.95 s, 99.91 % of the particles are in the bowl, none below the table, and the yolk is in the bowl. The yolk cannot break in this simulation (rigid body); yolk breakage comes from the hidden `open_up` model instead. The albumen surface is reconstructed from the particles by marching cubes, because the PhysX isosurface does not work with Fabric.



Figure 8: Opening simulation (existing filmstrip): halves swing apart, the albumen streams, the yolk drops whole and relaxes to its dome.

3 Apple and avocado

3.1 Apple (Gala-like)

One instance is built (`apple_3`, 217.7 g) with two lengthwise halves. The anatomy follows the pome: hypanthial flesh (cortex) round a core of five fused carpels, ten primary vascular bundles, a calyx with five sepals, lenticels as a point set and five named surface regions of the cuticle.

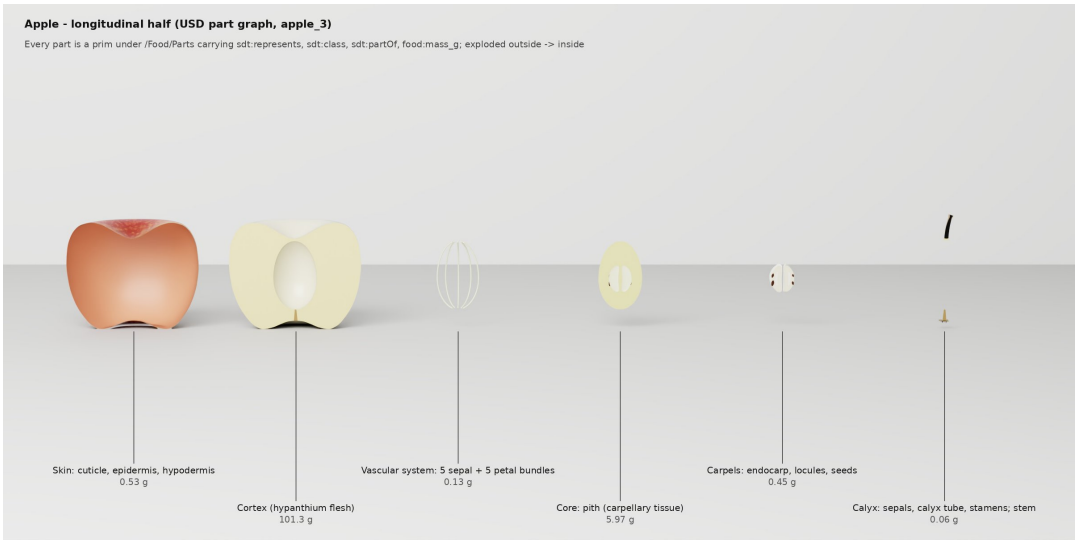
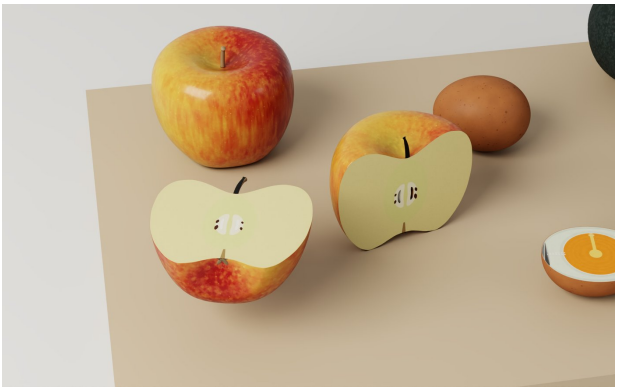
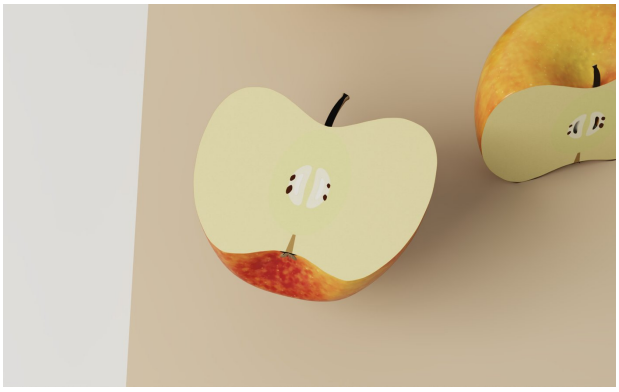


Figure 9: Exploded anatomy plate of apple_3 (existing render).



whole and halves



cut face: core and seeds

Figure 10: Apple renders (existing, path-traced).

States, dynamics, tasks. Schema attributes: integrity, bruise, peel, core, cut_face_browning, firmness, orientation, seed_positions; actions: Grasp, Place, Drop, Halve, Core, Peel. `AppleTruth` bruises under a static load when the Hertz maximum shear 0.31 p_0 exceeds the flesh shear strength, and under impact with a bruise volume linear in absorbed energy above a threshold (fsct2008 anchor, SOURCED-VERIFY); `AppleBelief` tracks the expected bruise after known drops and grips. There are **no apple task designators and no apple bench** yet; the apple's cutting force is ILLUSTRATIVE (no source found).

3.2 Hass avocado

One instance is built (`avocado_4`, 197.5 g); the knife goes round the seed, so `half_a` keeps the whole seed and `half_b` has the seed cavity. The exocarp has five layers including the stone-cell layer that lets ripe Hass peel cleanly, the flesh three colour bands and six vascular strands, and the seed a two-layer testa, two cotyledons and the embryonic axis.

States, dynamics, tasks. Schema attributes: ripeness_stage, integrity, seed_coat, seed, skin, flesh_condition, cut_face_browning; actions: SqueezeTest, Grasp, Halve, CutRoundSeed, TwistApart, RemoveSeed, Scoop. Ripeness

Table 10: Part tree of apple_3 (indexed siblings merged).

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
Skin	`food:AppleSkin`	visible		
Cuticle	`food:Cuticle`	visible		0.22
Epidermis	`food:Epidermis`	hidden		0.33
Hypodermis	`food:Hypodermis`	hidden		0.52
Lenticels	`food:Lenticel`	visible		1,717 points
Stem	`food:Pedicel`	visible		0.07
Calyx	`food:Calyx`	visible		
Sepal_1..5	`food:Sepal`	visible	5	0.01
CalyxTube	`food:CalyxTube`	hidden		0.01
StamenRemnants	`food:StamenRemnant`	visible		0.00
Cortex	`food:Cortex`	visible after cutting		203.51
VascularSystem	`food:VascularSystem`	visible after cutting		
SepalBundle_1..5	`food:SepalBundle`	visible after cutting	5	0.13
PetalBundle_1..5	`food:PetalBundle`	visible after cutting	5	0.13
Core	`food:AppleCore`	visible after cutting		
Pith	`food:CoreTissue`	visible after cutting		11.64
Carpel_1..5	`food:Carpel`	visible after cutting	5	
Endocarp	`food:Endocarp`	visible after cutting	5	0.66
Locule	`food:Locule`	visible after cutting	5	0.00
Seed_1..9	`food:AppleSeed`	visible after cutting	9	0.47

Table 11: Key register entries of the apple.

Register key	Value / distribution	Status	Source and note
`diameter_D`	N(75, 3.5) in [66, 86] mm	SOURCED	wiki_gala: 78 % of Gala fruit 70-80 mm
`flesh_density`	N(860, 40) in [760, 970] kg/m3	SOURCED	wiki_gala 0.86 g/cm3; fuji 955-968 kg/m3 (degruyter)
`flesh_E`	lognormal, median 3, factor 1.7 MPa	SOURCED	fuji 5.0 MPa (consensus f4110d16); quasi-static 4.2-7.8; stored 0.5-1.7 (tjaf2005)
`mt_firmness`	N(75, 8) in [50, 95] N	SOURCED	msu2025: Gala ~18 lb (80 N) at harvest
`flesh_shear_strength`	N(0.4, 0.05) in [0.28, 0.5] MPa	SOURCED	fuji shear 0.45 MPa; failure stress 0.32-0.37 MPa (consensus)
`bruise_energy_threshold`	U(0.004 .. 0.012) J	SOURCED-VERIFY	no bruise below 0.25 m/s impact; bruises from 0.011 J pendulum (consensus)
`bruise_volume_per_J`	0.01 * V_fruit / 0.41 J cm3/J	SOURCED-VERIFY	fsct2008 (about 1 % of fruit volume at 410 mJ); linear in energy
`cutting_force_peak`	N(25, 5) in [12, 40] N	ILLUSTRATIVE	no apple cutting force found
`friction_skin_rubber`	U(0.7 .. 0.95)	SOURCED-VERIFY	fuji vs TPU rubber 0.88 (consensus)
`core_line_radius`	N(14, 1.5) in [11, 18] mm	SOURCED	uspto: core diameter 24.7-36.7 mm
`browning_dE`	2.38 (0 min), 2.68 (30 min), 4.05 (60 min) dE	SOURCED	pmc7022590 (Golden Delicious)

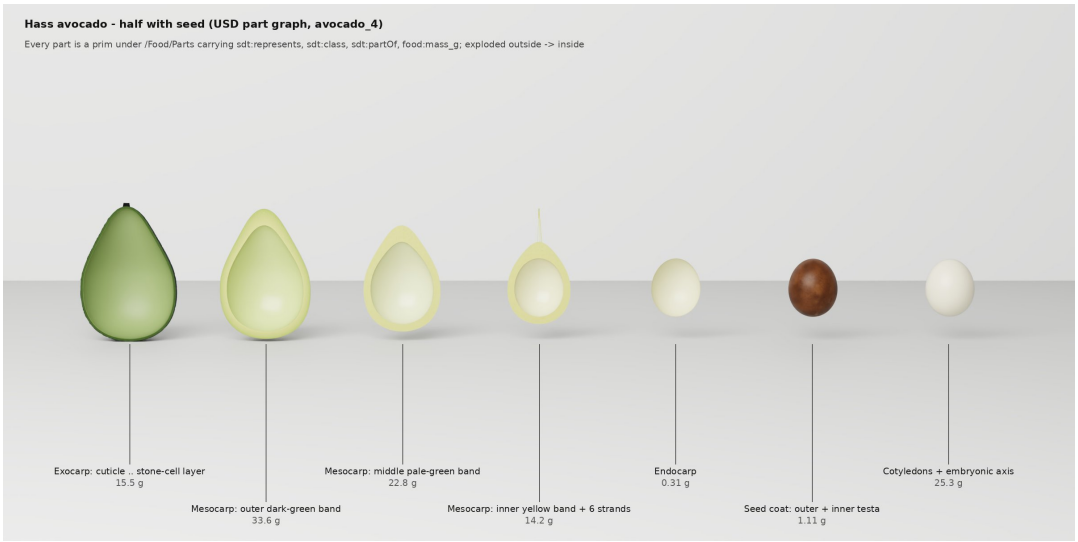
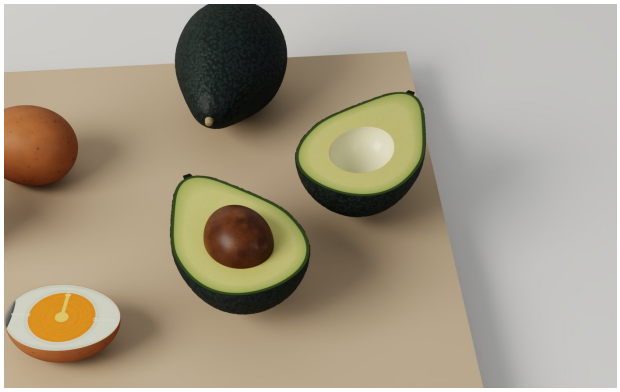


Figure 11: Exploded anatomy plate of avocado_4 (existing render).



whole and halves



half with seed

Figure 12: Avocado renders (existing, path-traced).

Table 12: Part tree of avocado_4 (indexed siblings merged).

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
Exocarp	`food:Exocarp`	visible		
Cuticle	`food:Cuticle`	visible		0.26
Epidermis	`food:Epidermis`	hidden		0.44
Hypodermis	`food:Hypodermis`	hidden		3.82
SkinParenchyma	`food:SkinParenchyma`	hidden		23.02
StoneCellLayer	`food:StoneCellLayer`	hidden		3.34
Lenticels	`food:Lenticel`	visible		499 points
StemButton	`food:PedicelRemnant`	visible		0.04
Mesocarp	`food:Mesocarp`	visible after cutting		
OuterDarkGreenBand	`food:FleshBand`	visible after cutting		65.66
MiddlePaleGreenBand	`food:FleshBand`	visible after cutting		45.14
InnerYellowBand	`food:FleshBand`	visible after cutting		28.52
VascularStrands	`food:VascularStrands`	visible after cutting		
Strand_1..6	`food:VascularStrand`	visible after cutting	6	0.15
Endocarp	`food:Endocarp`	visible after cutting		0.65
Seed	`food:AvocadoSeed`	visible after halving		
SeedCoat	`food:Testa`	visible after halving		
OuterTesta	`food:OuterTesta`	visible after halving		0.67
InnerTesta	`food:InnerTesta`	hidden		0.44
Cotyledon_1..2	`food:Cotyledon`	hidden	2	25.32
EmbryonicAxis	`food:EmbryonicAxis`	hidden		
Plumule	`food:Plumule`	hidden		0.00
Hypocotyl	`food:Hypocotyl`	hidden		0.00
Radicle	`food:Radicle`	hidden		0.00

Table 13: Key register entries of the avocado.

Register key	Value / distribution	Status	Source and note
`length_L`	N(98, 6) in [80, 120] mm	SOURCED-VERIFY	specialty: 7-15 cm; no Hass mm data held
`diameter_D`	N(66, 3.5) in [56, 76] mm	TYPICAL	chosen so mass ~ 200 g (uspto mean 204 g)
`skin_thickness`	N(1.6, 0.15) in [1.2, 2] mm	SOURCED	uspto_pp14239: Hass skin 1.6 mm
`seed_length`	N(40, 2.5) in [34, 47] mm	DERIVED	seed mass 28.4-30.6 g (uspto) at 1.1 g/cm3
`ripeness_stage`	U(1 .. 4.6) 1..5	STANDARD	avo_au_stages; continuous hidden value
`ripening_rate`	N(0.55, 0.1) in [0.3, 0.8] stage/d	SOURCED-VERIFY	ashton2006: firm-ripe ~day 8, soft-ripe ~9.5 at 20 C; landahl: 304 → 2 N in 6 d at 18 C
`penetrometer_force`	stage 1..5 = 118, 46, 15, 8.3, 4.9 N (geometric band centres), log-linear N	SOURCED	avo_au_stages: hard >10, rubbery 2.2-10, breaking 1.1-2.1, firm-ripe 0.7-1.0, ripe 0.4-0.6 kgf (11.1 mm tip, 8 mm)
`flesh_E`	E = 2.29 (F_pen / 118 N)^0.837 → 2.29 .. 0.16 MPa MPa	SOURCED	farinas2021: 2.29±0.75 → 0.16±0.08 MPa over ripening (endpoints); mapping ILLUSTRATIVE
`flesh_poisson`	0.45	ILLUSTRATIVE	near-incompressible soft tissue
`bruise_force_ratio`	0.6	SOURCED-VERIFY	hia2018: a 10 N thumb press bruises firm-ripe fruit (visible within 48 h); ratio of thumb pressure to penetrometer pressure at stage 4

advances with time on the Avocados Australia scale (STANDARD); flesh bruises when contact pressure exceeds a fraction of the penetrometer pressure of the stage; halving around the seed separates cleanly only from stage 3. `AvocadoBelief` updates the ripeness stage from squeeze stiffness and skin darkness. The halving and pitting designators, the probabilistic node and everything the robot does with them follow in Section 7.

3.3 Bench: ripeness by squeezing (Isaac Sim FEM)

`scripts/sim_avocado_squeeze.py` squeezes a PhysX volume-deformable avocado (avocado_4's outer surface, uniform flesh, seed and skin not separate) between a fixed and a force-controlled plate at six ripeness stages; the flesh modulus per stage maps the penetrometer force to the Farinas 2021 modulus range (endpoints SOURCED, mapping ILLUSTRATIVE). The secant stiffness table feeds the belief's stiffness model. As a cross-check this report computes the project's own Hertz model (`food.sdt.hertz_squeeze_stiffness`, two contacts on a sphere of the fruit's diameter 65.4 mm, ν 0.45) at 4 N. The FEM is stiffer than this sphere model by a nearly constant factor of 1.67–1.69 for stages 3–5. The avocado is not a sphere, so the factor is probably a shape effect (not established; in the tomato bench the FEM lies below Hertz instead), and the check confirms the trend over ripeness, not the absolute value. The factor falls to 1.46 at stage 2 and 1.13 at stage 1, and a separate probe run gave 11.9 N/mm for the stage-1 modulus against 7.0 N/mm in the final run (the probe's solver settings are not recorded in its log): the hard lanes are under-converged, as in the tomato and roll benches. An earlier 1000 Hz run (`avocado_squeeze_1000hz.json`) is invalid: its rigid control lane moved 9 mm, so all its stiffnesses were set to a placeholder. With the stiffness table, one simulated squeeze of a hidden

Table 14: Avocado squeeze stiffness at 4 N (test_reports/avocado_squeeze.json; Hertz column DERIVED in this report).

stage	E (MPa)	k FEM (N/mm)	k Hertz (N/mm)	FEM / Hertz
1	2.290	7.03	6.21	1.13
2	1.041	5.36	3.67	1.46
3	0.407	3.31	1.96	1.69
4	0.248	2.37	1.41	1.67
4.5	0.199	2.06	1.22	1.69
5	0.160	1.77	1.05	1.68

stage-4.2 fruit at 4 N (8 % noise) moves the flat prior to a stage estimate of 4.37 ± 0.31, and a second squeeze at 2 N to 4.29 ± 0.23 with $P(\text{ready, stage} \geq 4) = 0.89$ (recomputed here with the plot script's seeds). The bruise-safe plate force falls from 8 N (the largest tested) for hard fruit to 1 N for ripe fruit (panel d), consistent with the Hort Innovation warning that a 10 N thumb press bruises firm-ripe fruit (hia2018, SOURCED).

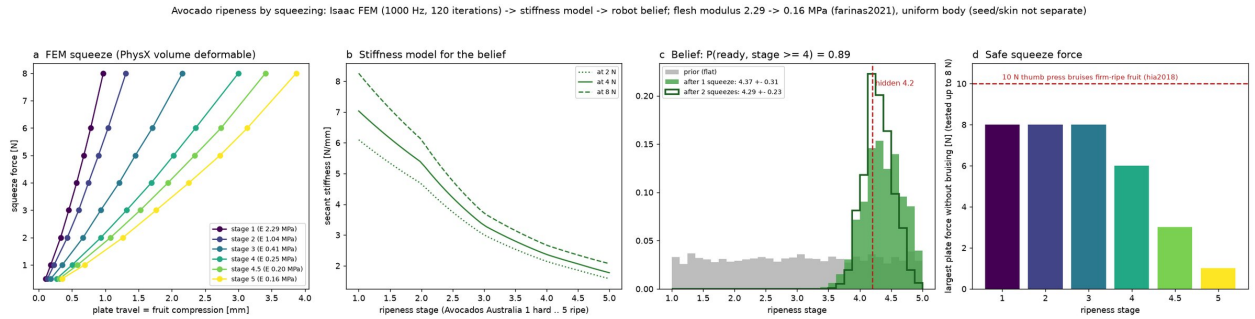


Figure 13: Avocado squeeze study (existing figure, tools/plot_avocado_squeeze.py): FEM force-compression curves, stiffness model, belief update, bruise-safe force.

4 Wave 1: bread roll, banana, carrot

Wave 1 was built with the twelve-step pipeline of the briefing document (scope, research notes, part tree, register, geometry, USD, hidden models, beliefs, designators, bench, renders, review). The research notes are in `docs/wave1/`; the summary page `docs/wave1/README.md` is generated by `tools/wave1_doc.py`.

Table 15: Calibration anchors of the wave-1 hidden-state models (from docs/wave1/README.md).

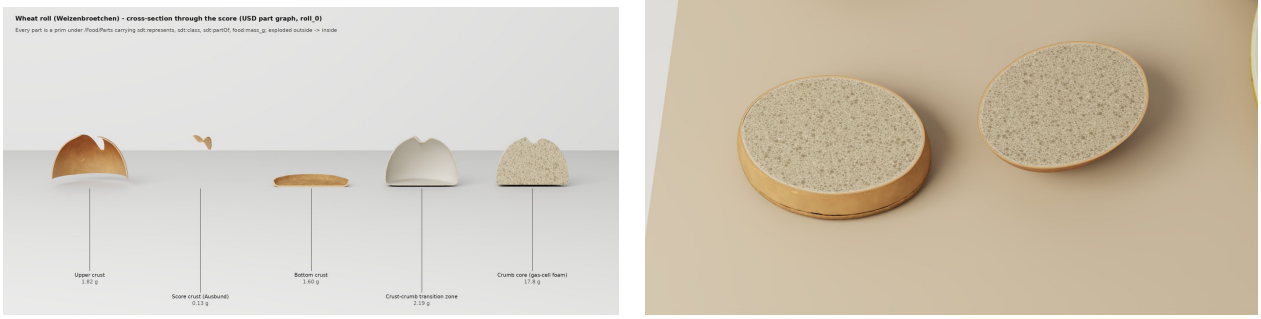
Model	Anchor	Twin	Status
Roll crust water / crispness	hirte_2012: 9.0 % @ 0.5 h, 13.9 % @ 24 h, critical 10.1 % @ 2.1 h	w = 9.73 t^0.112 (10.6 % @ 2.1 h)	CALIBRATED, residual +0.5 points
Roll crumb staling	lagrain_2012 E 20 → 157 kPa (5 h .. 7 d)	log-time interpolation	SOURCED-VERIFY
Carrot boiling	carrot_boil: 6 mm rounds 4-5 min crisp-tender, +1-2 min soft	crisp-tender 4.8 min, tender 6.9 min; whole 31 mm 16 min (source 10-15)	CALIBRATED thresholds on peng14 kinetics
Banana ripening	hou15 1 → 7 in 7 d (20 C, ethylene); ringer21 hue per stage	0.6 stage/d, hue table	SOURCED-VERIFY

Table 16: Wave-1 beliefs scored on simulated observations (from docs/wave1/README.md; 200 simulated instances each).

Item	Observations	Result
Roll	gripper squeeze stiffness (sd 15 %), crackle count (Poisson)	freshness class 66 % (squeeze k + crackles, 200 rolls)
Banana	camera peel hue (sd 3 deg)	stage RMS error 0.74 (one hue reading, 200 fingers)
Carrot	bend test (5 % misread)	limp class 96 % (one bend test, 200 roots)

4.1 Wheat roll (Weizenbrötchen)

A roll is a baked foam, not a plant organ: a dense browned crust round a crumb of gas cells. The twin is the oval single-score Weizenbrötchen; the score (Ausbund) is a separate crust part, the gas cells a point set that paints the cut faces. Built instance: `roll_0`, 47.4 g; states: whole, half_a (23.5 g), half_b (23.9 g), top_half (19.1 g), bottom_half (28.3 g). Research notes: `docs/wave1/ROLL.md`.



anatomy plate sliced open

Figure 14: Roll: exploded anatomy plate and cut halves (existing renders).

States and dynamics. Schema attributes: freshness, crust_texture, cut_state, toast_level, temperature, topping, support. Crust softening is water migration from the crumb (power law CALIBRATED through two hirte_2012 points), crumb staling a log-time interpolation of the Lagrain 2012 modulus (SOURCED-VERIFY, pan bread).

Bench: staling squeeze (Isaac Sim FEM). `scripts/sim_roll_squeeze.py` squeezes a crumb-only FEM roll (crust not separate: GAP) between 30 mm pads at four crumb moduli. Only the fresh lane converges (+4.5 % between solver settings); the stiffer lanes still rise with the solver rate and are lower bounds. The register therefore uses the converged lane scaled linearly in E (`squeeze_k_model`, DERIVED).

Table 17: Part tree of roll_0.

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
Crust	food:BreadCrust	visible		
UpperCrust	food:UpperCrust	visible		3.77
ScoreCrust	food:ScoreCrust	visible		0.29
BottomCrust	food:BottomCrust	visible		3.41
Crumb	food:BreadCrumb	visible after cutting		
TransitionZone	food:CrustCrumbTransition	visible after cutting		4.67
CrumbCore	food:CrumbCore	visible after cutting		35.92
GasCells	food:GasCell	visible after cutting		275,295 points
Dusting	food:FlourDusting	visible		

Table 18: Key register entries of the roll.

Register key	Value / distribution	Status	Source and note
height	N(52, 3) in [44, 60] mm	SOURCED	hirte_2012 ~50 mm; musiienko_2025 50-55 mm
crust_thickness_top	N(1.3, 0.2) in [0.9, 2] mm	SOURCED	hirte_2012: cut-off crust 1.0-1.5 mm
crumb_density	N(180, 15) in [150, 220] kg/m3	DERIVED	roll specific volume 5.9-6.4 ml/g (kurek_2016) → whole ~0.16-0.17 g/cm3, minus the crust share; crumb 0.088-0.252 (zghal_1999, SOURCED-VERIFY)
crumb_porosity	U(0.73 .. 0.81)	SOURCED	hirte_2012 uCT
crumb_E_kPa_by_age	20 @ 5, 70 @ 48, 125 @ 96, 157 @ 168 kPa @ h	SOURCED-VERIFY	lagrain_2012 pan bread, read off figures; log-time interpolation
crumb_poisson	0.103	SOURCED	lagrain_2012
crust_water_model	w(t) = 9.73 t_h^0.112 (40 % RH, 22 C) %	CALIBRATED	power law through hirte_2012 9.0 % @ 0.5 h and 13.9 % @ 24 h; predicts 10.6 % @ 2.1 h (measured: 10.1 % critical reached at 2.1 h) → residual +0.5 points
crust_water_critical	10.1 %	SOURCED	hirte_2012: half the sensory crispness lost
crust_break_force	3 N	SOURCED	hirte_2012: ≥ 3 N with a 1 x 10 mm blade
squeeze_k_model	k = 0.29 E / 20 kPa (30 mm pads across the width, 1 → 3 N) N/mm	DERIVED	Isaac FEM (test_reports/roll_squeeze_summary.json): 20 kPa lane converged (0.276 → 0.29 N/mm, 1000 → 2000 Hz); stiffer lanes under-converged → linear-elastic scaling; crust excluded (GAP)
saw_setpoint	-30 N, 0.02 m/s, ±0.02 m strokes (baguette) N, m/s, m	SOURCED	luo_2026 (robot)

Table 19: Task designators for the wheat roll (weizenbrötchen). Demo: symbolic run on seed 0 by this report (default parameters; a smoke test, not a validation).

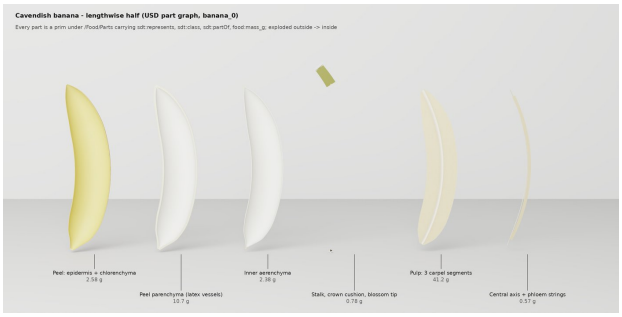
Designator	Goal	Twin model	Demo
PickRoll	roll held, not dented	RollTruth.grasp_corridor	ok
SliceRollOpen	cut_state = sliced_open (or hinged), crumb not squashed	RollTruth.saw	ok
ReheatRoll	crust_texture = crisp, temperature = warm	RollTruth.reheat	ok
ToastRollHalves	toast_level = golden on the cut faces	RollTruth.toast_faces	ok

Table 20: Roll squeeze, incremental stiffness 1–3 N in N/mm (test_reports/roll_squeeze_summary.json).

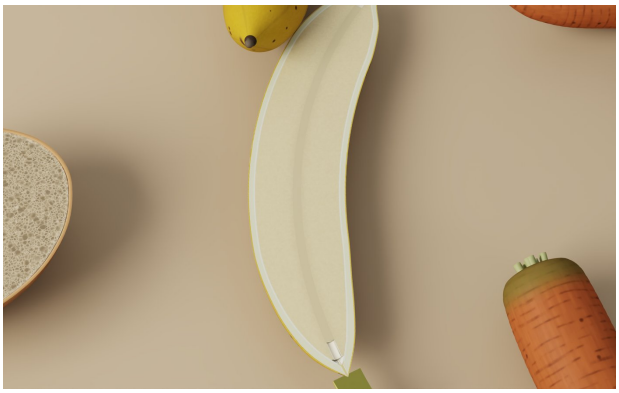
E (kPa)	age (h)	k 1000 Hz	k 2000 Hz	change %	k model (N/mm)
20	5	0.276	0.289	+4.5	0.29
70	48	0.497	0.631	+21.2	1.01
125	96	0.546	0.797	+31.5	1.81
157	168	0.535	0.876	+38.9	2.28

4.2 Cavendish banana

A banana finger is a parthenocarpic berry of three carpels. The peel (exocarp and outer mesocarp) is four tissue layers with latex vessels; the pulp three carpel segments round the placental axis with ovule remnants and the 'strings'. Four finger instances are built. Built instance: `banana_0`, 117.1 g; states: whole, half_a (58.3 g), half_b (58.8 g), tip_half (59.1 g), stalk_half (58.0 g). Research notes: `docs/wave1/BANANA.md`.



anatomy plate



lengthwise half

Figure 15: Banana: exploded anatomy plate and lengthwise half (existing renders).

Table 21: Part tree of banana_0.

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
Peel	`food:BananaPeel`	visible		
Epidermis	`food:Epidermis`	visible		0.58
Chlorenchyma	`food:Chlorenchyma`	hidden		4.58
Parenchyma	`food:PeelParenchyma`	visible after peeling		21.43
InnerAerenchyma	`food:Aerenchyma`	visible after peeling		4.76
Laticifers	`food:Laticifer`	hidden		1,500 points
Stalk	`food:Pedicel`	visible		1.34
CrownCushion	`food:CrownRemnant`	visible		0.21
BlossomTip	`food:PerianthRemnant`	visible		0.01
Pulp	`food:BananaPulp`	visible after peeling		
Segment_1..3	`food:PulpSegment`	visible after peeling	3	83.00
CentralAxis	`food:Placenta`	visible after cutting		1.05
Ovules	`food:AbortedOvule`	visible after cutting		258 points
PhloemStrings	`food:PhloemBundle`	visible after peeling		
String_1..3	`food:PhloemBundle`	visible after peeling	3	0.12

Table 22: Key register entries of the banana.

Register key	Value / distribution	Status	Source and note
`length_outer`	N(210, 15) in [170, 250] mm	SOURCED	convex-face length incl. pedicel (eu1333_2011 definition): 190 (mal22), 233 (ciro05)
`calibre`	N(35, 1.5) in [30, 40] mm	SOURCED	ciro05 34/36/34 mm; eu1333 min 27 mm
`peel_thickness`	N(2.6, 0.25) in [2, 3.2] mm	SOURCED	mal22 2.56 mm (Grand Naine); amn12 1.1-3.8 mm, thinning on ripening
`ripeness_stage`	U(2 .. 7) 1..7	STANDARD	markon / ma22 seven-stage colour scale
`peel_hue_by_stage`	[118, 98.7, 97.3, 92.7, 89.4, 87.5, 82] deg	SOURCED-VERIFY	ringer21 R2..R7; stage 1 from col20 Gros Michel day 0 (h ~122) → 118
`pulp_E_compression`	1.75 (day 1) → 1.09 (day 12) MPa	SOURCED	ciro05 cylinders 20 x 40 mm
`peel_fracture`	53.0 → 2.1 on ripening N	SOURCED-VERIFY	khanal23 (test geometry not in abstract)
`finger_drop_force`	~49 on day 1, > 2x lower by day 3 N	SOURCED-VERIFY	imsabai; mbeguie09
`peel_bond_ratios`	(0.45, 0.3, 1.5)	ILLUSTRATIVE	peel-to-pulp adhesion / ridge seam / strip tear strength as multiples of the peel fracture force (khanal23 53 → 2.1 N, SOURCED-VERIFY); adhesion > seam so the peel splits along the ridges (khanal23) instead of sliding off as a sleeve; GAP: no peeling-force study
`stage_rate_20C`	N(0.6, 0.15) in [0.3, 1] stage/d	SOURCED-VERIFY	hou15: 1 → 7 in 7 d with ethylene (0.86/d); ma22 day bands ~0.4/d; 18 C RS6 at 9 d
`ridges`	one of 4, 5	ILLUSTRATIVE	'slightly ridged' (mal22), 'nearly 4-sided'; count GAP

States and dynamics. Schema attributes: ripeness_stage, edible, peel_state, neck, attached_to_hand, bruised, chilling_injury, pulp_browning, integrity. Ripening on the seven-stage colour scale (STANDARD) at a SOURCED-VERIFY rate; peel hue per stage SOURCED-VERIFY; the peeling bonds are ILLUSTRATIVE ratios of the peel fracture force, because no published peeling force was found (GAP).

Table 23: Task designators for the cavendish banana. Demo: symbolic run on seed 0 by this report (default parameters; a smoke test, not a validation).

Designator	Goal	Twin model	Demo
SeparateBanana	one finger free, peel at the pedicel intact	BananaTruth.separate_from_hand	ok
PeelBanana	peel_state = off, pulp unbruised, strings mostly on the peel	BananaTruth.snap_neck, peel_strip	ok
SliceBananaRounds	integrity = rounds of ~6 mm	BananaTruth.cut_rounds, browning_class	ok

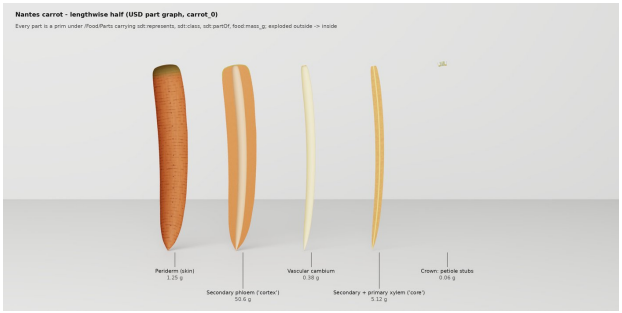
Bench: peeling (Isaac Sim). `scripts/sim_banana_peel.py` models the peel as 4 strips of 12 rigid segments joined by breakable adhesion, seam and hinge joints, and pulls strip 1 from the snapped neck with a spring finger. The force level inherits the ILLUSTRATIVE bond ratios, so only the trend is meaningful: green fruit needs higher peaks and its strip may tear short; ripe fruit peels cleanly at a fraction of a newton. The stage-4 strip tore below its nominal strength (joint forces of light segments are spiky), and the outcome flips between runs. The rendered filmstrip shows the strips only faintly.

Table 24: Banana peel bench (test_reports/banana_peel.json).

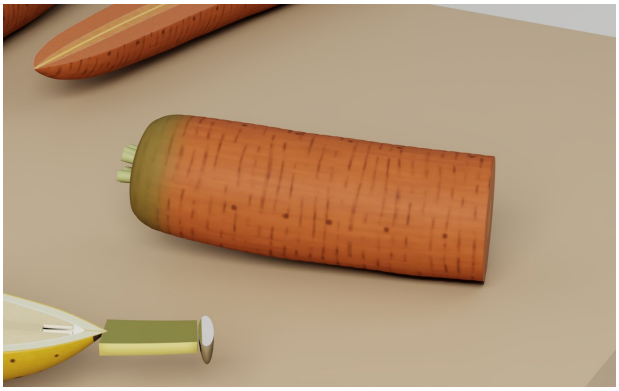
stage	adh. / seam / tear (N)	peak (N)	median (N)	strip 1	others	outcome
2	13.9 / 9.3 / 46.4	14.2	2.67	11/12	0	strip peeled off
4	4.7 / 3.2 / 15.8	7.1	1.26	5/12	0	strip torn short
6	1.6 / 1.1 / 5.4	2.5	0.43	12/12	2	strip peeled off

4.3 Nantes carrot

A carrot is a storage taproot: periderm, orange secondary phloem, vascular cambium, paler secondary xylem (the core) and the primary xylem; the crown carries petiole stubs, lenticel bands and lateral-root scars are point sets. Built instance: `carrot_0`, 115.0 g; states: whole, half_a (57.4 g), half_b (57.6 g), tip_piece (52.4 g), crown_piece (62.6 g). Research notes: `docs/wave1/CARROT.md`.



anatomy plate



crown piece

Figure 16: Carrot: exploded anatomy plate and crown piece (existing renders).

States and dynamics. Schema attributes: firmness_class, peel_state, trim_state, cut_shape, cooked_state, surface_blush, green_shoulder. Boiling uses the Peng 2014 softening kinetics (SOURCED) with a conduction lag and CALIBRATED doneness thresholds; limpness follows turgor and relative water content (SOURCED-VERIFY).

Bench: why cooks plank a carrot (Isaac Sim). `scripts/sim_carrot_plank.py` presses a knife onto carrot_0 up to the cutting force (44 N, the jah18 blade force scaled to this root, SOURCED-VERIFY), straight down or with a slicing stroke, on the whole root and on a planked half. A straight press acts through the axis and hardly moves the round root; a stroke rolls it away at once. The planked half holds during strokes up to about 10 N, then the blade friction yaws it – a claw-grip hand is needed. Escape criterion: roll > 10 deg or slide > 5 mm before F_cut.

Table 25: Part tree of carrot_0.

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
Periderm	`food:Periderm`	visible		2.51
SecondaryPhloem	`food:SecondaryPhloem`	visible after peeling		101.41
VascularCambium	`food:VascularCambium`	visible after cutting		0.76
SecondaryXylem	`food:SecondaryXylem`	visible after cutting		10.09
PrimaryXylem	`food:PrimaryXylem`	visible after cutting		0.14
Crown	`food:Crown`	visible		
Shoulder				
PetioleStubs	`food:PetioleStubs`	visible		
Stub_1..7	`food:PetioleStub`	visible	7	0.09
RootTip				
LateralRootScars	`food:LateralRootScar`	visible		48 points
LenticelBands	`food:LenticelBand`	visible		843 points

Table 26: Key register entries of the carrot.

Register key	Value / distribution	Status	Source and note
`length`	N(190, 15) in [150, 230] mm	SOURCED	con22: 182.7 / 210.8 mm (Nantes)
`shoulder_diameter`	N(31, 2) in [25, 38] mm	SOURCED	con22: 30.3 / 31.4 mm
`xylem_radius_fraction`	N(0.33, 0.04) in [0.25, 0.42]	SOURCED-VERIFY	singh17 ~0.38 over mixed genotypes; Nantes bred for a small core; vob24 trend crown → tip; peng14 4-7 mm core
`E_phloem`	N(4.6, 0.15) in [4.33, 4.85] MPa	SOURCED-VERIFY	vob24 after Stopa 2017
`poisson`	0.475	SOURCED-VERIFY	vob24
`bend_Fmax`	48.6 N	SOURCED-VERIFY	jah18 three-point bending (span: see source)
`blade_shear`	41.1 N	SOURCED-VERIFY	jah18 straight blade 1.4 mm, 30 deg
`limp_rwc`	0.95	SOURCED-VERIFY	herp99_calbo00: firmness from turgor ~0 below RWC 0.95
`softening_kinetics`	2nd-order fractional conversion; k(80 C) 0.035, k(110 C) 1.453; Ea 138.9 kJ/mol; Finf/F0 0.02 1/min	SOURCED	peng14
`peel_depth`	2 mm	SOURCED	con22 hand peeler; mass 12.1-14.6 %
`boil_rounds_6mm`	4-5 crisp-tender, +1-2 soft min	SOURCED-VERIFY	carrot_boil

Table 27: Task designators for the nantes carrot. Demo: symbolic run on seed 0 by this report (default parameters; a smoke test, not a validation).

Designator	Goal	Twin model	Demo
PeelCarrot	peel_state = peeled, ~2 mm removed	CarrotTruth.peel_stroke	ok
CutCarrotRounds	cut_shape = rounds (5-6 mm)	CarrotTruth.cut	ok
BoilCarrot	cooked_state = crisp_tender (or tender)	carrot_models.boil (peng14 + conduction lag)	ok

Table 28: Carrot plank bench (test_reports/carrot_plank.json).

carrot	knife	roll at F_{cut} (deg)	slide (mm)	escapes at (N)
whole	press	5.1	1.5	–
whole	stroke	26.1	7.2	1
planked	press	0.1	0.0	–
planked	stroke	105.1	389.5	10

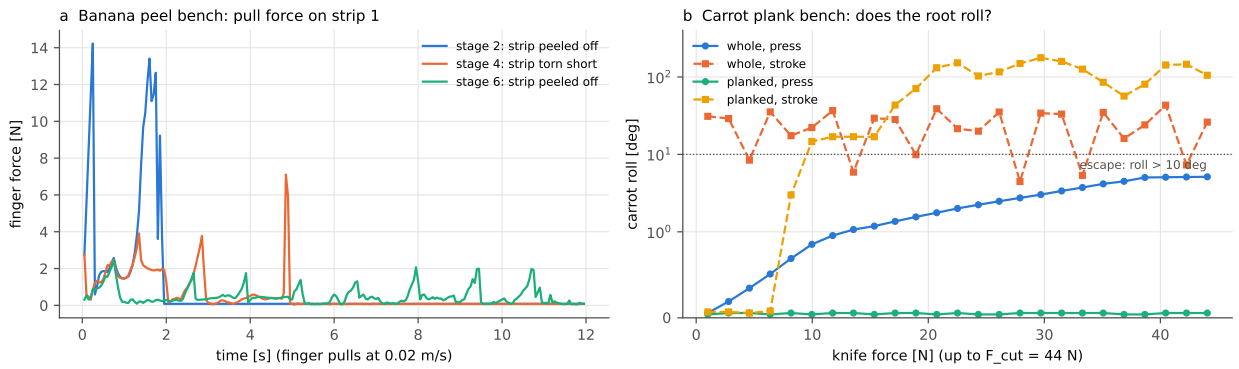


Figure 17: Banana and carrot bench traces drawn by this report from the stored logs: (a) finger force during peeling per ripeness stage; (b) roll angle of the carrot over knife force (log scale above 1 deg).

5 Wave 2: loaf, onion, orange, potato, tomato

The five wave-2 items are built as USD assets with cut states and have exploded anatomy plates, research notes in `docs/wave2/` and ten task designators (`food/wave2_tasks.py`). Path-traced renders of each item, whole and cut, were added on 20 October. They have no entries in `food/schema.py` (their discrete state comes from the truth classes' `discrete()`), and only the tomato has a physics bench.

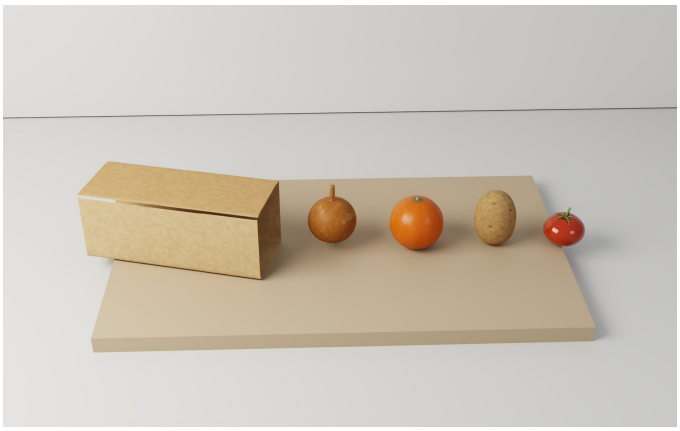
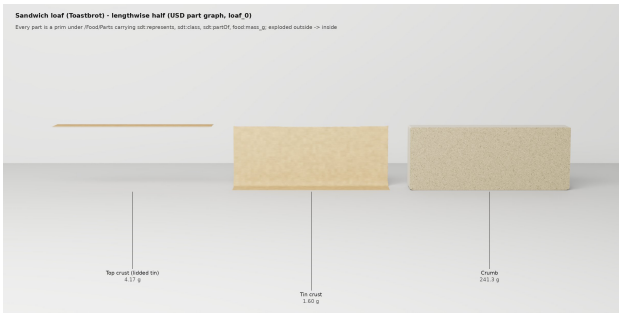


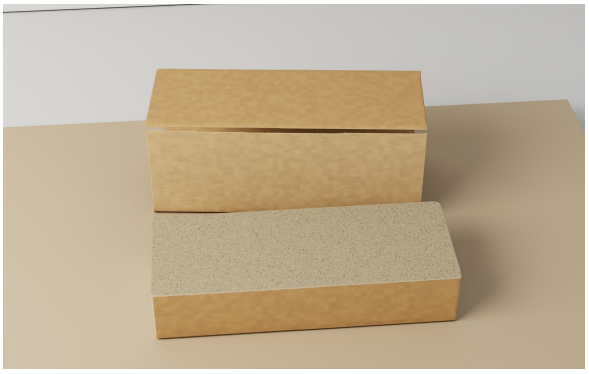
Figure 18: Wave 2 items, path-traced: sandwich loaf, onion, orange, potato and tomato.

5.1 Sandwich loaf (Toastbrot)

Industrial German Toastbrot, baked in a lidded tin: thin soft crust on the tin faces and the flat top, square cut ends (most packs have no heel slices), pre-sliced into 20 slices. Discrete attributes exposed by the truth class: freshness, mould_risk, slices_left. Built instance: `loaf_0`, 494.6 g; states: whole, half_a (247.1 g), half_b (247.5 g), slices (494.6 g). Research notes: `docs/wave2/LOAF.md`.



exploded anatomy plate



whole and cut half on the worktop

Figure 19: loaf_0: anatomy plate and path-traced render (Isaac Sim 6.2).

Table 29: Part tree of loaf_0 (indexed siblings merged).

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
Crust	`food:BreadCrust`	visible		
TopCrust	`food:TopCrust`	visible		8.70
TinCrust	`food:TinCrust`	visible		8.20
Crumb	`food:BreadCrumb`	visible after cutting		484.62
GasCells	`food:GasCell`	visible after cutting		6,217,490 points

Table 30: Key register entries of the sandwich loaf (toastbrot).

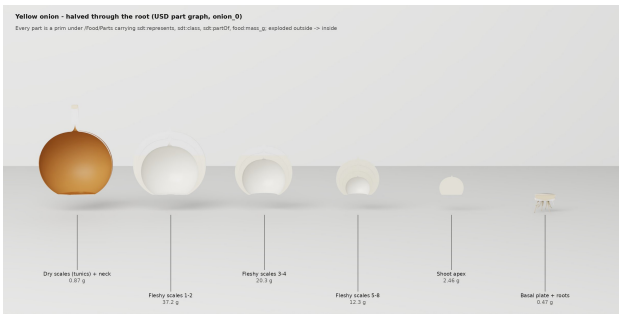
Register key	Value / distribution	Status	Source and note
`length`	N(250, 6) in [235, 265] mm	SOURCED-VERIFY	toast_pack ~0.25 m
`crumb_density`	N(190, 10) in [170, 215] kg/m3	SOURCED	nuss90 0.19 g/cm3
`slices`	20	SOURCED-VERIFY	toast_pack
`heels`	one of 0, 2	SOURCED	test25_toast: most packs no heels; organic 2
`crust_thickness_top`	U(0.8 .. 1.3) mm	ILLUSTRATIVE	thin soft lidded crust; GAP
`firmness_by_day`	1.98 / 3.62 / 4.18 / 5.64 at day 1 / 5 / 9 / 13 (25 C, bagged) N	SOURCED	sehn20 lab white pan loaf
`tensile_failure`	0.31-0.39 (12 mm neck) at 9.2-9.7 mm N	SOURCED	nuss90
`toast_time`	80 .. > 150 to 'best' (per toaster) s	SOURCED	toaster26
`fridge_staling_factor`	6	TYPICAL	toast_misc (popular, weak)

Table 31: Task designators for the sandwich loaf (toastbrot). Demo: symbolic run on seed 0 by this report (default parameters; a smoke test, not a validation).

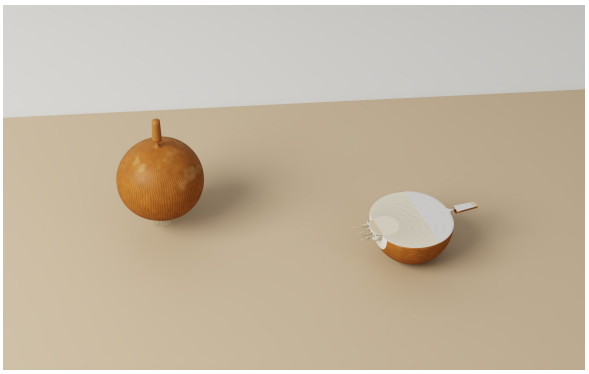
Designator	Goal	Twin model	Demo
TakeToastSlice	one slice out of the bag, untorn; bag closed again	LoafTruth.take_slice; Isaac: loaf 'slices' state (one rigid body per slice)	ok
ToastSlice	toast_level = golden	LoafTruth.toast	ok

5.2 Yellow onion

Swollen leaf bases (fleshy scales) on a compressed stem, the basal plate, under dry papery outer skins; each scale is a revolved region converging on the plate. Discrete attributes exposed by the truth class: cut_state, peel_state, sprouting, trim_state. Built instance: `onion_0`, 146.5 g; states: whole, half_a (73.5 g), half_b (73.0 g), top_piece (56.1 g), root_piece (90.4 g). Research notes: `docs/wave2/ONION.md`.



exploded anatomy plate



whole and cut half on the worktop

Figure 20: onion_0: anatomy plate and path-traced render (Isaac Sim 6.2).

Table 32: Part tree of onion_0 (indexed siblings merged).

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
DryScales	food:DryScaleGroup	visible		
Tunic_1..2	food:DryScale	visible after peeling	2	1.49
FleshyScales	food:FleshyScaleGroup	visible after peeling		
Scale_1..7	food:FleshyScale	visible after cutting	7	138.95
ShootApex	food:ShootApex	visible after cutting		4.91
BasalPlate	food:BasalPlate	visible		0.88
Roots	food:RootCrown	visible		
Root_1..25	food:Root	visible	25	0.03
Neck	food:Neck	visible		0.23

Table 33: Key register entries of the yellow onion.

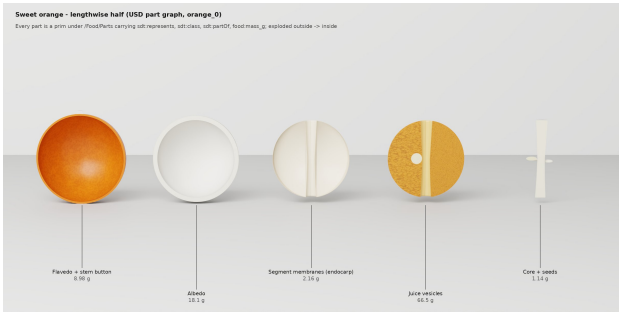
Register key	Value / distribution	Status	Source and note
diameter	N(65, 7) in [50, 85] mm	SOURCED	hb66_onion medium 51-83 mm; bah04 51.7
n_fleshy	one of 7, 8	SOURCED-VERIFY	kim_heb20 7.6 ± 0.2
scale_thickness_outer	U(3.5 .. 4.5) mm	ILLUSTRATIVE	kim_heb20 5.5 mm (bulk mean), wu25 ~3 mm; constrained by count x radius
n_dry	one of 2, 3	ILLUSTRATIVE	GAP (ordinal skin-thickness classes only)
flesh_density	1040 kg/m3	SOURCED-VERIFY	bah04 bulb 1.04 g/cm3
bulb_E	2.03-5.45, nu 0.26-0.45 MPa	SOURCED-VERIFY	jafari16
tissue_E	2.2 skin on / 1.4 off MPa	SOURCED	wu25
crushing_load	400-780 N	SOURCED-VERIFY	bah04
lf_window	peak ~1, gone ~10 min	SOURCED	lokke12
dice_spacing	mince ≤3, fine 3-6, medium 6-13, coarse 13-19 mm	SOURCED	onion_howto

Table 34: Task designators for the yellow onion. Demo: symbolic run on seed 0 by this report (default parameters; a smoke test, not a validation).

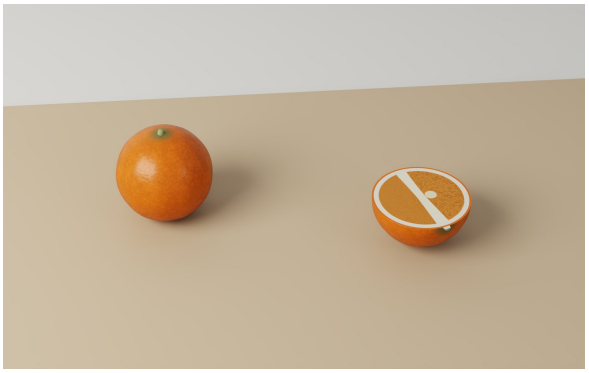
Designator	Goal	Twin model	Demo
DiceOnion	even dice of the target size, root end kept until last	OnionTruth.dice, onion_models.lf_dose	ok
SauteOnion	translucent | golden | caramelised as asked	onion_models.saute_state	ok

5.3 Sweet orange

A hesperidium: flavedo with oil glands, spongy albedo, segment membranes (endocarp) with juice vesicles, three vascular bundles per segment and the central axis; navel or Valencia. Discrete attributes exposed by the truth class: juice, peel_state, segments. Built instance: `orange_0`, 193.6 g; states: whole, stem_half (88.7 g), navel_half (104.9 g), half_a (96.9 g), half_b (96.7 g). Research notes: `docs/wave2/ORANGE.md`. The hand-juicing model gives 63 mL for



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whole and cut half on the worktop

Figure 21: orange_0: anatomy plate and path-traced render (Isaac Sim 6.2).

Table 35: Part tree of orange_0 (indexed siblings merged).

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
Peel	`food:CitrusPeel`	visible		
Flavedo	`food:Flavedo`	visible		17.88
Albedo	`food:Albedo`	visible after peeling		36.09
OilGlands	`food:OilGland`	visible		8,666 points
StemButton	`food:Calyx`	visible		0.07
Core	`food:Columella`	visible after cutting		2.13
Segments	`food:SegmentGroup`	visible after peeling		
Segment_1..10	`food:Segment`	visible after peeling	10	
Membrane	`food:SegmentMembrane`	visible after peeling	10	4.32
Vesicles	`food:JuiceVesicles`	visible after cutting	10	133.01
Seed_1..4	`food:CitrusSeed`	visible after cutting	4	0.13

Table 36: Key register entries of the sweet orange.

Register key	Value / distribution	Status	Source and note
`cultivar`	one of navel, valencia	ILLUSTRATIVE	German supermarket: navel (winter) / Valencia (summer)
`diameter`	N(75, 6) in [61, 88] mm	SOURCED	ucr_citrus Washington navel 61-88 mm
`peel_thickness`	N(5, 0.6) in [3.5, 6.4] mm	SOURCED	ucr_citrus rind 4.1-6.4 / 3.5-6.2 mm
`segments`	one of 9, 10, 11, 12	SOURCED-VERIFY	citrus_patents 9.6-11
`flavedo_E`	103 MPa	SOURCED	smith18 (30-300 across species)
`albedo_E`	0.79 MPa	SOURCED	smith18 (±0.58)
`gland_burst_pressure`	30-140 kPa	SOURCED	smith18
`juice_fraction`	U(45 .. 54) %	SOURCED	ucr_citrus navel 45-54 (Valencia 37-56)
`peel_strip_force`	navel 4, Valencia 8 per 16 mm strip N	ILLUSTRATIVE	ranking from ritenour13; GAP
`bitter_juice_h`	3 h	SOURCED-VERIFY	edis891 'within several hours' (navel)

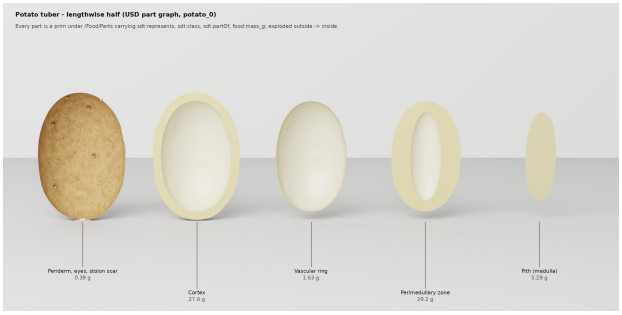
orange_0 (notes; source 60–80 mL by hand, TYPICAL). Peeling force in newtons is a GAP; the register's strip forces are ILLUSTRATIVE.

Table 37: Task designators for the sweet orange. Demo: symbolic run on seed 0 by this report (default parameters; a smoke test, not a validation).

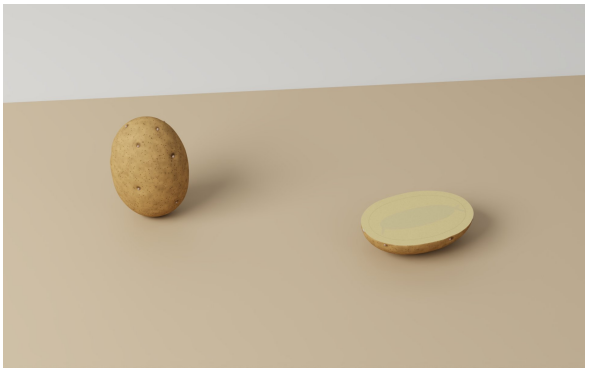
Designator	Goal	Twin model	Demo
JuiceOrange	juice in the glass, few seeds	OrangeTruth.juice	ok
PeelOrange	peel off in few pieces, segments intact	OrangeTruth.peel, bend_peel	ok

5.4 Potato

The tuber is a stem: periderm with lenticels, eyes on a spiral (denser at the rose end), cortex, vascular ring, perimedullary starch store and a branched pith. Discrete attributes exposed by the truth class: cooked_state, greening, peel_state, sweetened. Built instance: `potato_0`, 126.4 g; states: whole, half_a (63.5 g), half_b (63.0 g), rose_piece (56.1 g), heel_piece (70.3 g). Research notes: `docs/wave2/POTATO.md`.



exploded anatomy plate



whole and cut half on the worktop

Figure 22: potato_0: anatomy plate and path-traced render (Isaac Sim 6.2).

Table 38: Part tree of potato_0 (indexed siblings merged).

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
Periderm	`food:Periderm`	visible		0.69
Lenticels	`food:Lenticel`	visible		1,415 points
Eyes	`food:EyeGroup`	visible		
Eye_1..12	`food:Eye`	visible	12	0.05
StolonScar	`food:StolonScar`	visible		0.02
Cortex	`food:Cortex`	visible after peeling		53.75
VascularRing	`food:VascularRing`	visible after cutting		3.24
PerimedullaryZone	`food:PerimedullaryZone`	visible after cutting		58.14
Pith	`food:Pith`	visible after cutting		10.53

Model checks reported in the notes. `docs/wave2/POTATO.md` reports the boiling model (conduction lag plus first-order softening, k100 and Ea SOURCED) calibrated to 2 cm cubes done at 17.5 min (sources 15–20) and checked on 3 cm cubes (22 min) and a large 75 mm tuber (50 min; sources up to 60); a whole 55 mm tuber takes 34 min against 20–25 in German guides, so the model runs long there. The register's 1.1 mm peel removes 10.0 % of the twin's mass against about 9 % measured – an independent check of the geometry.

Table 39: Key register entries of the potato.

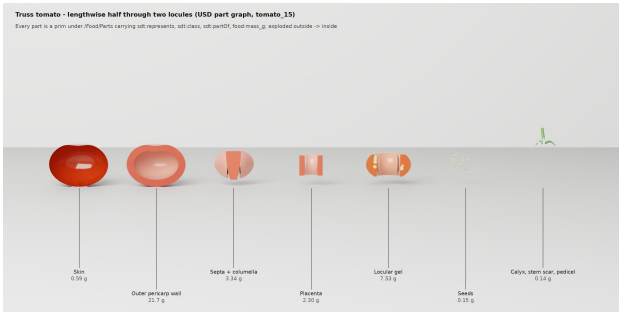
Register key	Value / distribution	Status	Source and note
`cultivar`	one of Linda, Belana, Gala	STANDARD	bsl21
`length`	N(85, 10) in [60, 110] mm	SOURCED-VERIFY	75 / 65 / 99 mm (Astrix / Diamont / Santana)
`width`	N(55, 4) in [45, 65] mm	SOURCED-VERIFY	48 / 45 / 58 mm
`eyes`	integer 8 .. 14	SOURCED-VERIFY	Norgold 10-12 buds; 6-9 apical (mag25)
`phyllotaxis`	0.4 turn	ILLUSTRATIVE	spiral arrangement sourced; the 2/5 fraction is a GAP
`periderm_thickness`	N(65, 10) in [46, 84] um	SOURCED	mc20 phellem
`density`	N(1070, 12) in [1050, 1100] kg/m3	SOURCED-VERIFY	1053 (kk18); SG 1.070-1.103
`E`	U(3.1 .. 5.6) MPa	SOURCED-VERIFY	potato_mech
`softening`	first order, k(100 C) 0.28-0.40, Ea 103-145 1/min, kJ/mol	SOURCED	potato_cook (Harada)
`boil_times`	diced 15-20, large whole up to 60, whole 20-25 (cold start) min	SOURCED	gritzer
`peel_depth`	1.1 mm	SOURCED-VERIFY	potato_peel (~9 % waste)

Table 40: Task designators for the potato. Demo: symbolic run on seed 0 by this report (default parameters; a smoke test, not a validation).

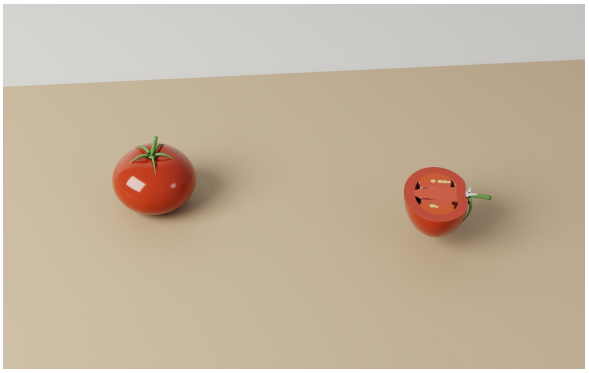
Designator	Goal	Twin model	Demo
BoilPotatoes	done: a skewer passes without resistance	potato_models.boil (conduction + first-order softening, calibrated to 2 cm cubes)	ok
PeelPotato	peeled, eyes and green parts removed	potato_models.peel_loss_fraction	ok

5.5 Round / truss tomato

Skin, outer pericarp wall, septa, columella with placenta, locules filled with gel round the seeds; 2–4 locules; truss tomatoes keep calyx and pedicel. Discrete attributes exposed by the truth class: bruised, cut_state, flavour, ripeness_class, skin. Built instance: `tomato_15`, 71.7 g; states: whole, stem_half (35.6 g), blossom_half (36.0 g), half_a (35.7 g), half_b (35.9 g). Research notes: `docs/wave2/TOMATO.md`.



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whole and cut half on the worktop

Figure 23: tomato_15: anatomy plate and path-traced render (Isaac Sim 6.2).

Bench: plate firmness (Isaac Sim FEM). `scripts/sim_tomato_squeeze.py` is an independent consistency check of the register: lanes take the whole-fruit Hertz modulus of one study (tomato_mech, 0.58 MPa small red .. 1.97 MPa small green) and compare the FEM plate stiffness with Hertz theory and with the plate firmness measured by another study (ince16). Analytically the red fruit matches; the green one comes out lower than measured. The FEM converges for the soft lane only at 4000 Hz / 250 iterations; at 1000 Hz all lanes gave about 1.8 N/mm whatever the modulus.

Table 41: Part tree of tomato_15 (indexed siblings merged).

Part (below /Food/Parts)	sdt:class	observability	n	mass (g)
Skin	`food:TomatoSkin`	visible		1.18
Pericarp	`food:Pericarp`	visible after cutting		
OuterWall	`food:OuterPericarp`	visible after cutting		43.54
Septa	`food:SeptumGroup`	visible after cutting		
Septum_1..3	`food:Septum`	visible after cutting	3	2.60
Columella	`food:Columella`	visible after cutting		3.23
Locules	`food:LoculeGroup`	visible after cutting		
Locule_1..3	`food:Locule`	visible after cutting	3	
Placenta	`food:Placenta`	visible after cutting	3	4.82
Gel	`food:LocularGel`	visible after cutting	3	15.73
Seeds	`food:SeedGroup`	visible after cutting	3	
Seed_1..90	`food:TomatoSeed`	visible after cutting	90	0.34
Calyx	`food:Calyx`	visible		
Sepal_1..5	`food:Sepal`	visible	5	0.11
StemScar	`food:StemScar`	visible		0.03
Pedicel	`food:Pedicel`	visible		0.08

Table 42: Key register entries of the round / truss tomato.

Register key	Value / distribution	Status	Source and note
`diameter`	N(63, 4) in [52, 76] mm	SOURCED	fdc_tomato 61-66 mm for 91-123 g
`locules`	one of 2, 3, 4	SOURCED	ucd_rost: small types 2, globe 3-5
`pericarp_thickness`	N(6.5, 0.8) in [5, 9] mm	SOURCED-VERIFY	meta19 5.3 mm (small); 7-12 germplasm
`plate_firmness`	initial G/P/R 15.1 / 6.0 / 4.5; average 13.2 / 4.7 / 3.1 N/mm	SOURCED-VERIFY	ince16
`skin_rupture_2mm`	G/P/R 6.92 / 3.99 / 2.74 N	SOURCED-VERIFY	ince16
`skin_E`	252-293 (red); failure ≤ 25-28 MPa at 9-10 % MPa	SOURCED	bn05
`damage_strain`	0.03	SOURCED-VERIFY	tomato_mech (Li 2017 FEM)
`skin_hue_by_stage`	[118, 110, 96, 71, 52, 43] deg	DERIVED	h = atan2(1, a*/b*) at the band centres of ince16; ripe ~40 deg (Radzevicius 2009, SV)
`chilling_threshold`	12.5 C	SOURCED	wflo_tomato 12.5-12.8
`flesh_density`	1000 kg/m3	ILLUSTRATIVE	GAP (cherry 1194, suspect)

Table 43: Task designators for the round / truss tomato. Demo: symbolic run on seed 0 by this report (default parameters; a smoke test, not a validation).

Designator	Goal	Twin model	Demo
SliceTomato	slices 6-13 mm, not squashed, gel kept	TomatoTruth.cut	ok
PeelTomato	skin off, flesh firm	TomatoTruth.blanch	ok

Table 44: Tomato plate stiffness 4–20 N in N/mm (test_reports/tomato_squeeze*.json). Measured values ince16, SOURCED-VERIFY.

E (MPa)	FEM 1000 Hz	FEM 4000 Hz	Hertz	FEM/Hertz	measured initial / average
0.58	1.76	4.10	4.53	0.90	red: 4.5 / 3.1
1	1.81	5.23	6.51	0.80	–
1.4	1.76	5.63	8.15	0.69	–
1.97	1.79	6.00	10.23	0.59	green: 15.1 / 13.2

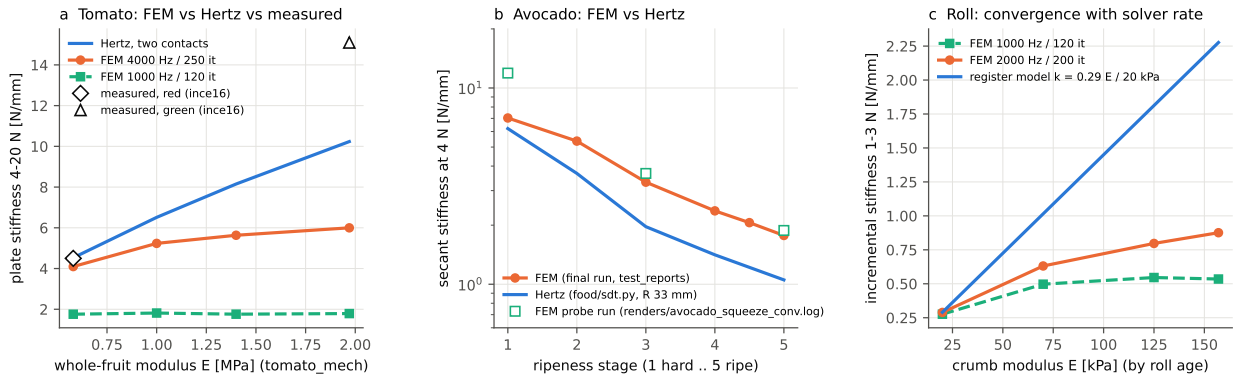


Figure 24: Squeeze benches compared by this report: (a) tomato FEM at two solver settings vs Hertz and the measured plate firmness; (b) avocado FEM vs the project's Hertz model and an unrecorded-settings probe run; (c) roll FEM at two solver settings vs the register's linear model. Common finding: soft lanes converge, stiff lanes need a higher solver rate than used.

6 Wave 3: lemon, cucumber, croissant, baguette, pepper

The five wave-3 items exist as code (`food/<item>.py` with geometry, part tree and asset spec; `food/<item>_models.py` with hidden models and beliefs) and as registers. On 20 October they were added to the asset builder and built (seed 0: USD part graph, cut states, belief and truth layers), rendered on the worktop and as exploded anatomy plates. They still have no task designators and no benches. Their module docstrings refer to research notes in `docs/wave3/`, which **does not exist** in the repository. The part trees below come from instantiating each item (seed 0) on the CPU.



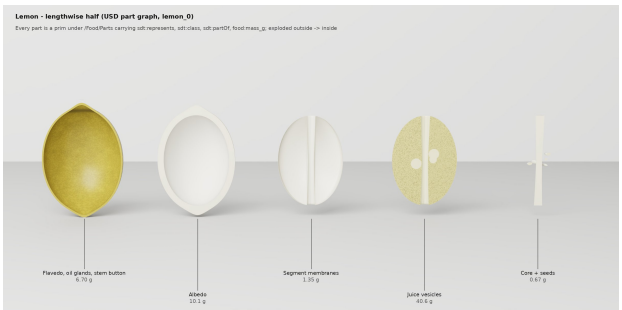
Figure 25: Wave 3 items (seed 0), path-traced: lemon, cucumber, croissant, baguette and pepper.

Table 45: Wave-3 status (checked by this report; cache in report/cache/wave3_parts.json).

Item	Module	Reg.	% ILL.	instantiates	USD states	images
Lemon	`food/lemon.py`	31	29	yes, 43 parts	stem_half, blossom_half, half_a, half_b	plate + render
Greenhouse cucumber	`food/cucumber.py`	28	43	yes, 8 parts	half_a, half_b, rounds	plate + render
Butter croissant	`food/croissant.py`	21	62	yes, 3 parts	top_half, bottom_half, half_a, half_b	plate + render
Baguette	`food/baguette.py`	42	45	yes, 9 parts	half_a, half_b, top_half, bottom_half, crostini	plate + render
Bell pepper	`food/pepper.py`	27	37	yes, 12 parts	half_a, half_b, top_piece, bottom_piece	plate + render

6.1 Lemon

A hesperidium like the orange (the class derives from it) with a prolate outline, the stylar nipple at the blossom end and a short neck at the stem end; thicker, paler albedo.



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Figure 26: lemon_0: anatomy plate and path-traced render (Isaac Sim 6.2).

Table 46: Part tree of lemon seed 0 from `parts()`.

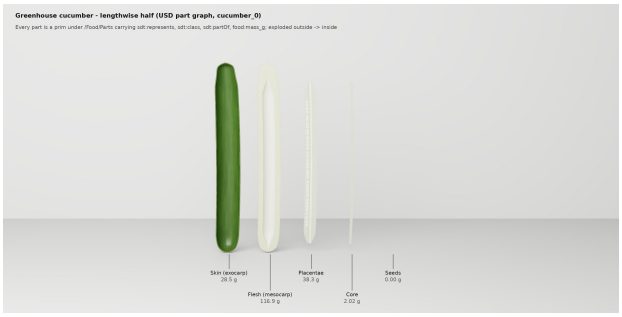
Part (below /Food/Parts)	class	observability	n
Peel	`food:CitrusPeel`	visible	
Flavedo	`food:Flavedo`	visible	
Albedo	`food:Albedo`	visible after peeling	
OilGlands	`food:OilGland`	visible	
StemButton	`food:Calyx`	visible	
Core	`food:Columella`	visible after cutting	
Segments	`food:SegmentGroup`	visible after peeling	
Segment_1..10	`food:Segment`	visible after peeling	10
Membrane	`food:SegmentMembrane`	visible after peeling	10
Vesicles	`food:JuiceVesicles`	visible after cutting	10
Seed_1..6	`food:CitrusSeed`	visible after cutting	6

Table 47: Key register entries of the lemon.

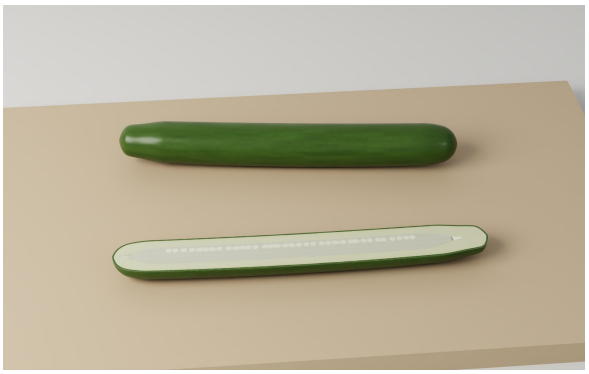
Register key	Value / distribution	Status	Source and note
`diameter`	N(61, 3) in [55, 67] mm	SOURCED	agu20 55-67 mm
`shape_index`	U(1.3 .. 1.55)	SOURCED	length / diameter from agu20 ED/PD 0.6-0.8 (body without nipple)
`peel_thickness`	N(5.6, 0.5) in [4.9, 7.1] mm	SOURCED	agu20, jen22_lemon
`juice_fraction`	U(30 .. 45) %	SOURCED	agu20 31.7-44.3 (EU min 20)
`puncture_force`	41.2 N	SOURCED	ben26 5 mm probe
`cutting_force`	36.3 N	SOURCED	ben26 craft knife
`compression_force`	165.8 at 16 mm N	SOURCED	ben26
`nipple_height`	U(3 .. 7) mm	ILLUSTRATIVE	stylar mammilla; GAP

6.2 Greenhouse cucumber

A pepo of three fused carpels: skin, firm mesocarp, a seed cavity with three placentae carrying immature seeds round a small core; bent onto a gentle arc.



exploded anatomy plate



whole and cut half on the worktop

Figure 27: cucumber_0: anatomy plate and path-traced render (Isaac Sim 6.2).

Table 48: Part tree of cucumber seed 0 from `parts()`.

Part (below /Food/Parts)	class	observability	n
Skin	`food:CucumberSkin`	visible	
Flesh	`food:Mesocarp`	visible after cutting	
SeedCavity	`food:SeedCavity`	visible after cutting	
Placenta_1..3	`food:Placenta`	visible after cutting	3
Core	`food:CentralCore`	visible after cutting	
Seeds	`food:CucumberSeed`	visible after cutting	

Table 49: Key register entries of the greenhouse cucumber.

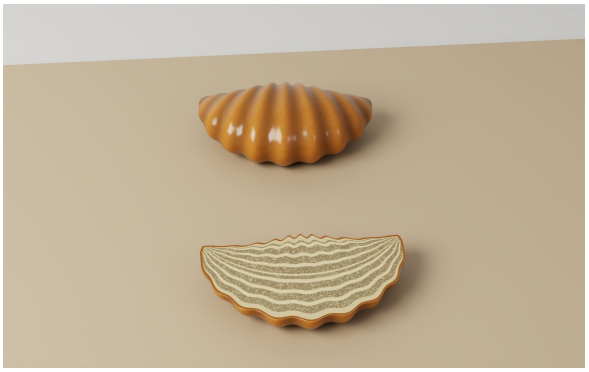
Register key	Value / distribution	Status	Source and note
`length`	N(320, 25) in [250, 380] mm	SOURCED	unece_cuc 25-30 cm minimum; sakata11
`diameter`	N(42, 3) in [36, 50] mm	TYPICAL	derived: 400 g / 32 cm / 950 kg/m3 → ~41 mm
`skin_thickness`	U(1 .. 1.6) mm	SOURCED-VERIFY	cuc_misc peelable exocarp 1-2 mm
`cavity_fraction`	U(0.45 .. 0.55)	ILLUSTRATIVE	seed cavity / fruit radius; GAP
`puncture_3mm`	flesh 8.7, placenta 2.8, skin 9.7 N	SOURCED	sakata11
`E_apparent`	0.85-1.39 (slice) | 2.5 (DiSECt) MPa	SOURCED-VERIFY	jah16; disect_cuc
`firmness_loss_9d`	49 %	SOURCED-VERIFY	jah16
`chilling`	< 10 C after 1-3 d C, d	SOURCED	ucd_cuc

6.3 Butter croissant

A rolled triangle of laminated dough: a glazed crust, a laminated crumb whose cut faces show the spiral of dough sheets and voids, and loose flakes.



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whole and cut half on the worktop

Figure 28: croissant_0: anatomy plate and path-traced render (Isaac Sim 6.2).

Table 50: Part tree of croissant seed 0 from `parts()`.

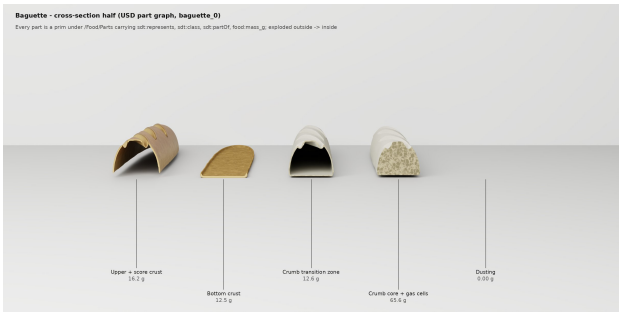
Part (below /Food/Parts)	class	observability	n
Crust	`food:GlazedCrust`	visible	
LaminatedCrumb	`food:LaminatedCrumb`	visible after cutting	
Flakes	`food:Flake`	visible	

Table 51: Key register entries of the butter croissant.

Register key	Value / distribution	Status	Source and note
`length`	N(135, 8) in [115, 160] mm	ILLUSTRATIVE	finished dimensions GAP; chosen for ~180 cm3 = 57 g x 3.13 cm3/g (ste25, USDA medium)
`mass_check`	42 / 57 / 67 (small / medium / large) g	SOURCED	croissant_misc (USDA)
`rolling_turns`	U(3.5 .. 4)	SOURCED	croissant_misc
`butter_layers`	27	SOURCED	ste25: 3 single folds
`crumb_density`	300 kg/m3	DERIVED	1 / 3.13 cm3/g (ste25) = 0.32 g/cm3 whole
`crust_thickness`	U(0.8 .. 1.4) mm	ILLUSTRATIVE	GAP
`firmness`	5.16 (1 h) → 19.2 (24 h), TPA 25 mm 50 % N	SOURCED	ste25
`flakes`	integer 40 .. 120	ILLUSTRATIVE	loose flakes; GAP

6.4 Baguette

The roll's baked-foam model on a long pointed footprint with five to seven overlapping diagonal scores (grignes) and a thicker crust.



exploded anatomy plate



whole and cut half on the worktop

Figure 29: baguette_0: anatomy plate and path-traced render (Isaac Sim 6.2).

Table 52: Part tree of baguette seed 0 from `parts()`.

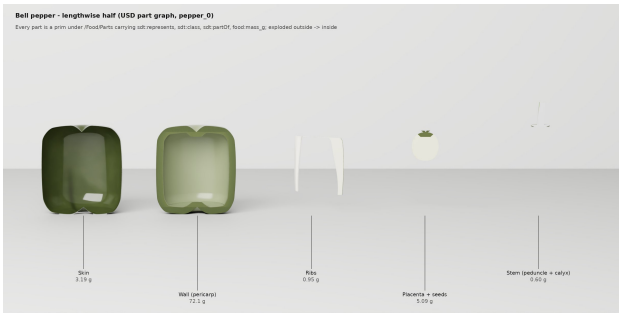
Part (below /Food/Parts)	class	observability	n
Crust	`food:BreadCrust`	visible	
UpperCrust	`food:UpperCrust`	visible	
ScoreCrust	`food:ScoreCrust`	visible	
BottomCrust	`food:BottomCrust`	visible	
Crumb	`food:BreadCrumb`	visible after cutting	
TransitionZone	`food:CrustCrumbTransition`	visible after cutting	
CrumbCore	`food:CrumbCore`	visible after cutting	
GasCells	`food:GasCell`	visible after cutting	
Dusting	`food:FlourDusting`	visible	

Table 53: Key register entries of the baguette.

Register key	Value / distribution	Status	Source and note
`length`	N(600, 30) in [540, 660] mm	SOURCED	wp_baguette 55-65 cm
`width`	N(55, 4) in [45, 62] mm	SOURCED	wp_baguette 4-6 cm
`n_scores`	one of 5, 6, 7	SOURCED	wp_grigne: 5 classic; 5-7 typical for 55-65 cm
`crust_thickness_top`	N(1.9, 0.3) in [1.4, 3.5] mm	SOURCED	dv2012 1.7 mean (zou2024: 3-4, SV)
`crumb_density`	N(200, 20) in [160, 250] kg/m3	DERIVED	~250 g / ~0.9 L bulk (dimensions, wp_baguette) minus a denser crust
`crumb_porosity`	U(0.75 .. 0.85)	ILLUSTRATIVE	open irregular crumb; tomography GAP
`cell_diameter_median`	U(1.3 .. 1.9) mm	ILLUSTRATIVE	'irregular voids of various sizes'
`crust_L`	N(52, 3) in [44, 58] L*	SOURCED-VERIFY	zou2024 powder 57.5; intact darker

6.5 Bell pepper

A hollow berry: pericarp wall round an air cavity, 2–4 lobes, ribs (septa), a placenta hanging from the stem end with the seeds, peduncle and calyx.



exploded anatomy plate



whole and cut half on the worktop

Figure 30: pepper_0: anatomy plate and path-traced render (Isaac Sim 6.2).

Table 54: Part tree of pepper seed 0 from `parts()`.

Part (below /Food/Parts)	class	observability	n
Pericarp	`food:Pericarp`	visible	
Skin	`food:PepperSkin`	visible	
Wall	`food:PepperWall`	visible after cutting	
Ribs	`food:SeptumGroup`	visible after cutting	
Rib_1..3	`food:Septum`	visible after cutting	3
Placenta	`food:Placenta`	visible after cutting	
Seeds	`food:PepperSeed`	visible after cutting	
Stem	`food:StemGroup`	visible	
Peduncle	`food:Peduncle`	visible	
Calyx	`food:Calyx`	visible	

Table 55: Key register entries of the bell pepper.

Register key	Value / distribution	Status	Source and note
`diameter`	N(78, 4) in [68, 90] mm	SOURCED	he22 77.6 mm
`length`	N(83, 6) in [70, 100] mm	SOURCED	he22 81.3-85.3 mm
`lobes`	one of 3, 4	SOURCED	sha26: 2-4, mostly 3
`wall_thickness`	N(7.2, 0.6) in [5.5, 8.5] mm	SOURCED	he22 7.5 red / 6.9 orange
`seeds`	integer 40 .. 140	SOURCED-VERIFY	pepper_misc 38-142
`rib_depth`	U(6 .. 12) mm	ILLUSTRATIVE	GAP
`firmness_whole`	43.5 N	SOURCED	pepper_misc (8 mm probe)
`weight_loss`	2.9-4.8 at 17 C / 55 % RH; shrivel from 6-8 % loss g/d	SOURCED	alt20
`red_LCh`	(38.8, 41.9, 41.6) L*,C*,h	SOURCED	bar20

7 An avocado as the robot's belief state: from perception to action

The sections before describe food items as part graphs with beliefs and truths. This section follows one item, the Hass avocado, through the whole loop a robot needs: perceive it, keep a probabilistic belief about its hidden properties, ask and answer questions about it, predict what an action will do, act, and check the outcome against the prediction. Everything here is code in this repository and runs on the twin (`avocado_4` as the hidden truth); nothing is validated on a real fruit or a real robot.

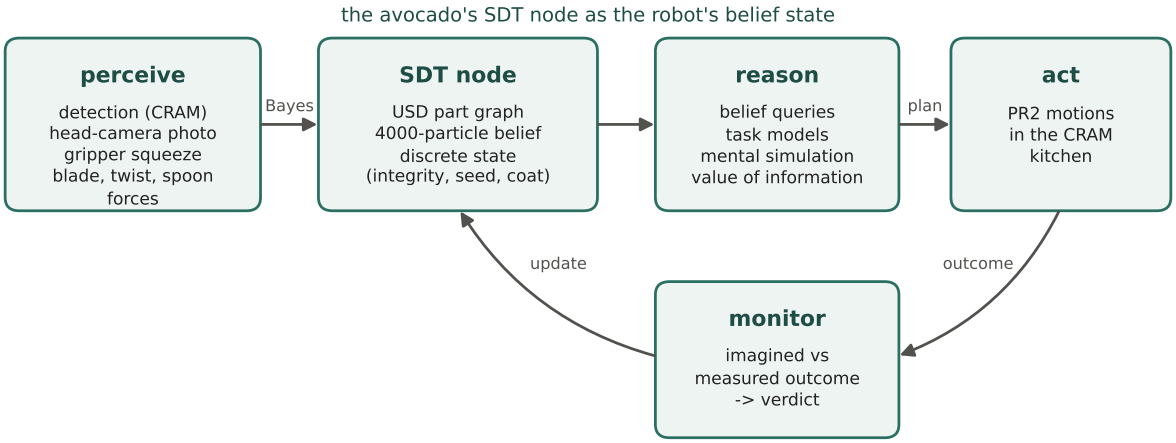


Figure 31: The loop of this section. The SDT node (USD part graph, particle belief, discrete state) is the robot's belief state about the avocado; perception updates it by Bayes' rule, reasoning queries it, action outcomes are compared with what the node predicted and fed back.

7.1 From a photo to an SDT node

`scripts/avocado_from_image.py` (`food/avocado_vision.py`) estimates an avocado's node from one photo: colour clustering picks the fruit-like blob, GrabCut refines it and its convex hull restores the shaded flank; the white balance comes from the fruit's own specular highlights (dichromatic reflection: highlights carry the illuminant's colour); the silhouette gives the aspect ratio and, by fitting the twin's meridian, the neck taper; the skin's CIELAB lightness and chroma enter a ripeness likelihood (hue only as green / not green, because the register's hue chart for ripe Hass is doubtful); skin relief and the stem button are read off as well. Every estimate is labelled OBSERVED, ESTIMATED or PRIOR, and the observations go into the belief of the next subsection, from whose posterior the instance is then built. On the rendered camera view the shape estimates are close (aspect ratio and length within a few per cent of the



analysis sheet of a real photo

the PR2's head-camera view (Isaac Sim render of avocado_4)

Figure 32: Photo input. Left: the analysis of a user photo (segmentation, white balance, profile fit, skin colour). Right: the image the PR2 scenario uses as its camera view, rendered from the hidden-truth instance.

instance), but the ripeness reading is biased: the rendered skin is darker than the colour chart expects for the true stage, and the photo moves the belief to stage 3.97 ± 0.53 while the truth is 2.96. Skin colour is a weak ripeness cue

Table 56: Estimates from the burlap photo (assets/food/avocado_photo_burlap/observation.json).

Estimate	value	sd	unit	status	method
aspect_L_over_D	1.56	0.047	1	OBSERVED	silhouette length / maximum width along the principal axis (side view assumed)
neck	0.025	0.03	1	OBSERVED	least-squares fit of the twin's meridian to the silhouette profile (rms 0.029)
length_L	98	6	mm	PRIOR	register prior (no scale reference in the image); pass –length-mm to measure it
diameter_D	62.9	4.3	mm	ESTIMATED	length / observed aspect ratio
white_balance_gains_BGR	1.06, 1.02, 0.93	–	-	OBSERVED	from 840 specular-highlight pixels on the skin (dichromatic model)
skin_Lab_L	12.7	–	L*	OBSERVED	median of the skin (highlights / shadow out)
skin_chroma	7.8	–	C*	OBSERVED	median chroma of the skin
skin_hue	47.9	–	deg	OBSERVED	median hue angle of the skin
ripeness_stage	3.93	0.73	1..5	ESTIMATED	posterior over the Avocados Australia chart: skin L*, C* against the twin's colour model, hue as green / not green (likelihoods ILLUSTRATIVE)
skin_relief_index	0.596	–	1	OBSERVED	std of high-passed luminance / mean luminance on the skin
pebble_scale	1.2	0.2	mm	ESTIMATED	half the dominant texture wavelength (6.0 px) at the image scale, clipped to the register range 1.2-2.0 mm (ILLUSTRATIVE mapping)
stem_button	present	–	-	OBSERVED	lighter, browner region at the narrow end near the axis
stem_button_length	6.2	1	mm	ESTIMATED	visible length along the axis / 2 (side view of a clipped button, ILLUSTRATIVE)
pose	standing on the blossom end	–	-	OBSERVED	direction of the principal axis from the wide to the narrow end in the image
interior	not observable	–	-	PRIOR	seed, flesh bands, bruising: register priors until the fruit is cut or squeezed

(avo_au_stages: firmness is the better index); the likelihood is not calibrated on renders. Section 7.6 shows how the robot recovers from this.

7.2 The node as a probability distribution: Bayesian filtering

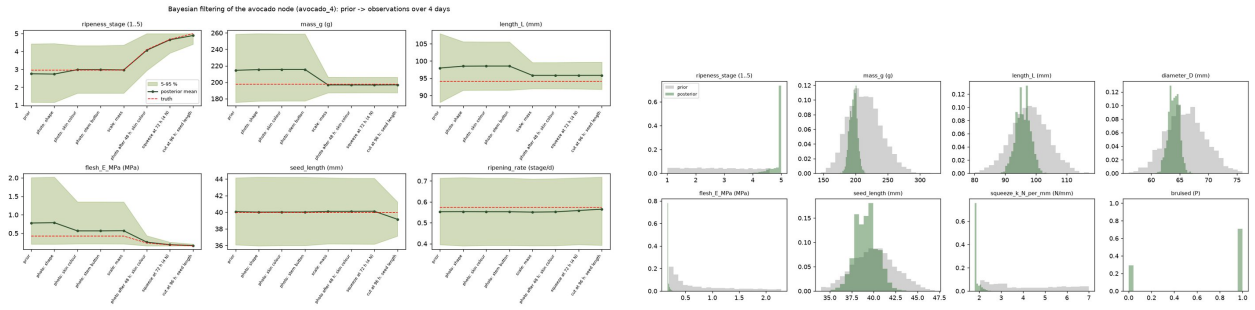
The avocado node is a particle filter (`food/avocado_belief.py`): 4000 particles, each a complete candidate avocado drawn from the register (length, diameter, neck, skin thickness, tubercle size, seed length, width and offset, ripeness stage and ripening rate, skin friction, stem button, bruise flag, which side the seed coat sticks to). Derived per particle: mass (body of revolution of the twin's meridian plus the seed, calibrated on the built twins), aspect ratio, penetrometer force and flesh modulus (register ripeness model), squeeze stiffness (FEM table of the previous section), seed and flesh mass. *Prediction* is ripening at each particle's own rate; a squeeze above a particle's bruise threshold flips its bruise flag. *Measurement models* (noise ILLUSTRATIVE unless stated): photo shape and skin colour, stem button, calliper, scale, gripper squeeze, the knife's stop depths round the seed, the twist outcome and torque, the spoon's lever force, the extracted seed's length, the seed coat seen in the cavity, a bruise seen after cutting. Resampling is systematic with a regularising jitter when the effective sample size halves. The node is written into USD as `sdt_belief.usda`: per variable mean, sd, 5/50/95 % quantiles, histogram and support as attributes of the part prims, plus the observation history and the particle file.

Table 57: Validation of the filter on avocado_4 with simulated noisy observations over four days (scripts/avocado_belief_demo.py; truth: mass 197 g, seed 40.0 mm).

Step	t (h)	ripeness belief	truth	mass (g)	seed length (mm)
prior	0	2.75 ± 1.04	2.96	215 ± 26	40.1 ± 2.4
photo: shape	0	2.74 ± 1.05	2.96	215 ± 26	40.0 ± 2.4
photo: skin colour	0	2.99 ± 0.80	2.96	216 ± 25	40.0 ± 2.4
photo: stem button	0	2.99 ± 0.80	2.96	216 ± 25	40.0 ± 2.4
scale: mass	0	2.97 ± 0.79	2.96	197 ± 6	40.1 ± 2.4
photo after 48 h: skin colour	48	4.07 ± 0.65	4.11	197 ± 6	40.1 ± 2.4
squeeze at 72 h (4 N)	72	4.66 ± 0.38	4.68	197 ± 6	40.1 ± 2.4
cut at 96 h: seed length	96	4.89 ± 0.21	5.00	197 ± 6	39.2 ± 1.3

7.3 Watching the robot peel: a vision-based peeling monitor

`food/peel_monitor.py` judges a peeled avocado half (lying cut face down, one oblique RGB-D view) against the twin's *expectation*: the ideal peeled surface of this instance, registered to the observed depth (trimmed ICP and a height refinement over x, y and yaw). Every visible surface point is classed as peeled well, skin remaining (pebbled texture, darker, or standing proud of the expected surface), flesh missing (deeper than expected, with a slope-dependent threshold and hysteresis over the torn area) or not judged. From the classes come the measures (peeled share, skin area and largest patch, missing area, lost volume, deepest gouge, coverage), a score from 0 to 100 and a verdict (success / rework / not judged) with reasons, written into the node. The test set is synthetic: `food/avocado_peel.py` turns



belief over time posteriors

Figure 33: The avocado node's belief while it ripens and is observed (assets/food/avocado_4/belief/).

peeling outcomes into USD states (skin patches kept on the flesh, gouges cut out of it by mesh booleans, ground truth recorded) and Isaac Sim renders them with depth and segmentation, the expectation at a perturbed pose (2–4 mm, 2–4°) so that the monitor must register it itself. The thresholds are calibrated on these renders only. Depth carries

Table 58: Peeling monitor on 24 synthetic peels (6 clean, 6 with skin patches, 6 with gouges, 6 mixed; peel_set/eval/summary.json). IoU, precision and recall are means over the samples that contain the defect; err: mean absolute area error.

Input	verdicts	reworks found	false alarms	IoU skin	P / R skin	err (mm2)	IoU missing	P / R missing	err (mm2)
RGB-D	22/24	14/15	1	0.46	0.53 / 0.69	15	0.53	0.83 / 0.59	40
RGB only	20/24	13/15	2	0.32	0.35 / 0.75	38	0.00	0.00 / 0.03	220

the gouges: from colour alone the missing flesh is not found at all, while the skin patches, being darker and pebbled, are found from colour too. While building the set, three rendering faults of edited meshes had to be fixed before the evaluation meant anything: per-corner normals that were not trimmed with the faces, texture coordinates lost in the mesh booleans, and material subsets that became empty – each made the renderer drop the mesh's material, so the skin patches came out pale.

7.4 Halving and pitting: an emulator

The emulator (`food/avocado_emulator.py`, mechanics in `food/avocado_process.py`) is a state machine on the node: whole → scored / partly scored → halved / halved with torn flesh → seed out, the seed coat on the seed or left in the cavity. Each action checks its precondition, runs quasi-static mechanics on the hidden instance and writes the new USD states, and returns what a robot would sense (*measured*) apart from the truth:

Table 59: Emulator actions.

Action	Mechanics (constants ILLUSTRATIVE)	USD state
`cut_round_seed`	puncture at the stem end; force = skin resistance × skin thickness + flesh cutting toughness(stage) × depth; the blade stops on the seed coat (stop-depth profile); a force limit can stall the blade	kerf carved from skin and flesh down to the seed coat; cut faces labelled
`twist_apart`	torque = uncut flesh bridges (polar moment × flesh shear strength) + seed adhesion over half the seed; ripe fruit releases at the seed, unripe tears (p_{clean}(stage))	two halves; torn flesh as holes in the cavity wall
`remove_seed`	spoon lever (0.3 × adhesion × area), knife twist (flagged hazardous for a hand-held half, farley2019), squeeze-out (ripe only); the spoon may gouge; the seed coat stays in the cavity when it adheres to the flesh	pitted half with seed-coat sheets and gouges; the seed

7.5 Competence questions and mental simulation

With the node as its belief state, the robot answers competence questions from three kinds of knowledge (`cram/avocado_sdt.py`): the *belief* (what is it, how ripe and when edible, where is the seed and how deep will the blade go), the *task models* (what can I do now, what am I doing and what can go wrong), and *mental simulation*: the halving and pitting mechanics run on avocados drawn from the belief. An imagined avocado needs only the geometry the mechanics use, which takes 1 ms to build instead of 1 s for the full twin, so a question can roll out 150 imagined avocados in seconds. Answers carry the evidence they rest on.

Should it squeeze first? Value of information. The goal is "guacamole now, or store the avocado two days". Utility = quality of the pitted halves (penalties for torn flesh, seed coat left, gouges) × how well the flesh mashes at preparation time (0 below stage 2.5, 1 from stage 4; ILLUSTRATIVE), waiting costs 0.05. The robot first picks the

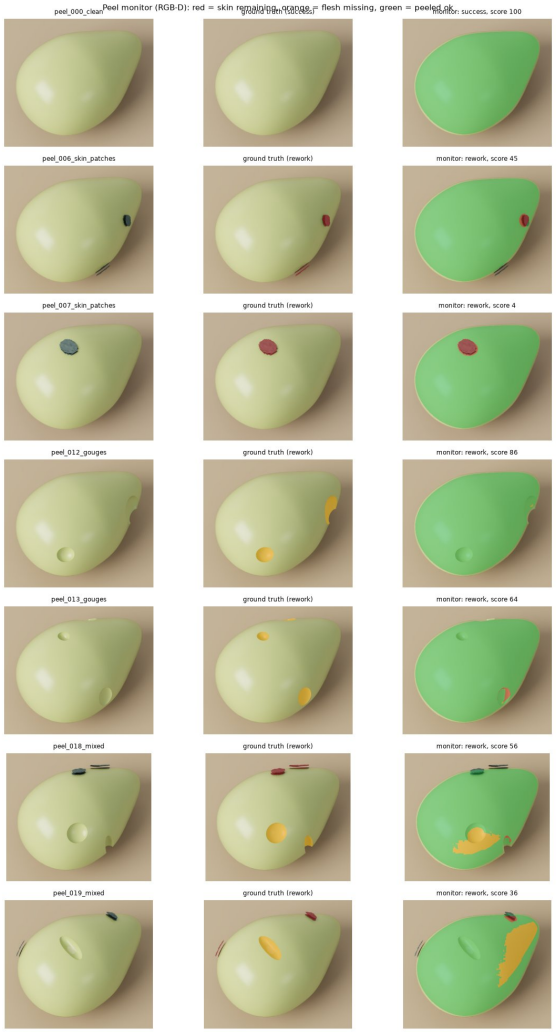


Figure 34: Peeling monitor examples (left: render; middle: ground truth; right: the monitor's classes, red skin remaining, orange flesh missing, green peeled well).

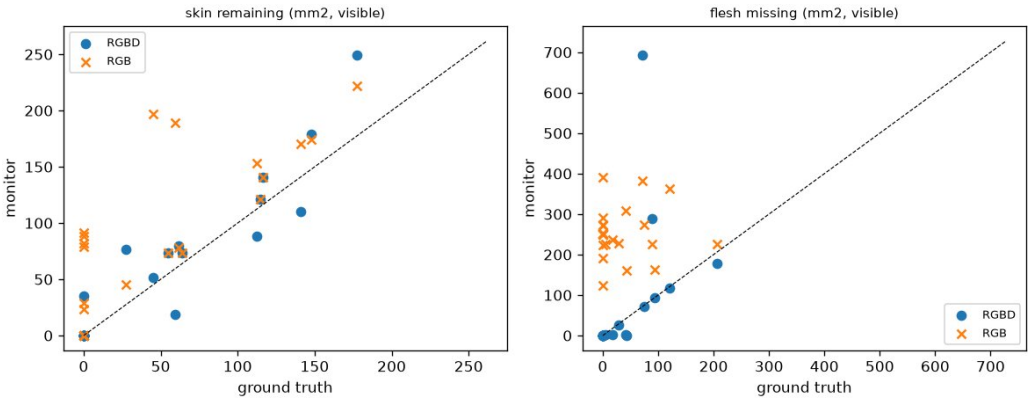


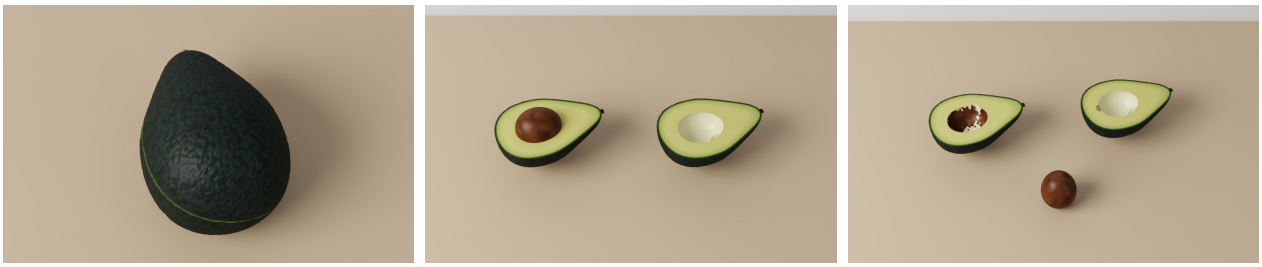
Figure 35: Skin and missing areas: monitor against ground truth, RGB-D and RGB only.

Table 60: Avocado task designators (food/avocado_tasks.py); CutRoundSeed is coraplex's CuttingAction with the technique HALVING.

Designator	Goal	Failures (cue, recovery)
`CutRoundSeed`	integrity = scored: a lengthwise cut through skin and flesh all round the seed	partial_cut, no_entry
`TwistApart`	integrity = halved; the seed stays in one half	stuck, torn
`RemoveSeed`	seed = out; cavity clean (no seed coat, no gouge)	stuck, testa_left, gouge

Table 61: Three emulator runs on avocado_4 (process/summary.json; truth: stage 2.96, forced to 1.8 in the unripe run; seed 40.0 mm; seed coat adheres to the flesh). The belief starts from the register prior and is updated only from the measured part of each outcome.

Scenario	Step	Outcome	ripeness belief	seed (mm)	P(coat on flesh)
ripe	`CutRoundSeed`	scored	2.81 ± 1.03	40.5 ± 2.1	0.49
ripe	`TwistApart`	halved_torn	1.98 ± 0.60	40.6 ± 2.1	0.47
ripe	`RemoveSeed`	pitted_testa_left	2.92 ± 0.36	41.4 ± 0.9	0.97
unripe	`CutRoundSeed`	partly_scored	2.80 ± 1.03	40.5 ± 2.1	0.49
unripe	`TwistApart`	stuck	2.80 ± 1.03	40.5 ± 2.1	0.49
unripe	`CutRoundSeed`	scored	2.80 ± 1.02	40.6 ± 2.0	0.49
unripe	`TwistApart`	halved_torn	1.98 ± 0.59	40.6 ± 2.0	0.45
unripe	`RemoveSeed`	pitted_testa_left	1.91 ± 0.45	38.2 ± 0.8	0.96
knife	`CutRoundSeed`	scored	2.81 ± 1.03	40.5 ± 2.1	0.49
knife	`TwistApart`	halved_torn	1.98 ± 0.60	40.6 ± 2.1	0.47
knife	`RemoveSeed`	pitted_testa_left	2.93 ± 0.36	39.8 ± 0.9	0.97



cut round the seed | halves (torn twist) | pitted, seed, other half

Figure 36: Emulator states, path-traced in Isaac Sim from the USD states the emulator writes.

squeeze force: the strongest of 4, 3 and 2 N whose bruise risk under the belief is below 2 % (bruise threshold 0.6 × penetrometer force, register). Then a preposterior analysis imagines squeeze readings for the imagined avocados (the belief's own measurement model; sensor error grows as the force falls), updates on each and asks whether the decision changes and by how much the expected utility rises.

Table 62: The robot's competence answers in the PR2 scenario (cram/results/pr2_run.json, recorded 2026-10-20 19:40).

Question	Answered from	Answer
What is the object?	belief	a Hass avocado (Avocado < Fruit < Produce < Food), 217 ± 26 g, 95 x 67 mm, ripeness stage 4.0 ± 0.5 (nearly ripe); parts: exocarp_skin: tubercles, stem_button; mesocarp_flesh: fibre_strands; seed: seed_coat (testa), cotyledons x2, embryo_axis
How ripe is it, and when can it be eaten?	belief	P(ready to eat, stage ≥ 4.5) = 0.13; P(still hard, stage < 2) = 0.00; ready in about 0.7 days (median over the belief)
Where is the seed, how deep will the knife go?	belief	blade stops on the seed coat at a depth (90 % interval) of side 13..19 mm, top 32..41 mm, bottom 15..22 mm
What can I do with it now?	task model	state integrity=whole, seed=in: CutRoundSeed: integrity = scored: a lengthwise cut through skin and flesh all round the seed
Should I squeeze it before deciding?	mental simulation	yes: without squeezing I would choose 'store 2 days, then guacamole' (expected utility 0.77; guacamole now (halve, spoon out the seed): 0.68, store 2 days, then guacamole: 0.77); a 3 N squeeze (P(bruise) 0.000) changes that choice in 8% of the imagined readings and is worth +0.012
What did the squeeze tell me?	execution monitoring	as expected: 3.40 N/mm (imagined 2.05..3.97); ripeness belief ripeness_stage 3.97 ± 0.53; penetrometer_N 9.32 ± 5.28 → ripeness_stage 3.79 ± 0.52; penetrometer_N 10.12 ± 3.89
Guacamole now, or store it?	mental simulation	store 2 days, then guacamole (expected utility guacamole now: 0.64, store 2 days, then guacamole: 0.77)
What happens if I halve and pit it now?	mental simulation	halve now, spoon out the seed: P(success) 1.00, P(clean halves) 0.92, P(torn) 0.08, P(seed coat left) 0.47, P(gouge) 0.19, quality 0.80 | halve now, knife-twist the seed out: P(success) 1.00, P(clean halves) 0.92, P(torn) 0.08, P(seed coat left) 0.47, P(gouge) 0.00, quality 0.85, HAZARD | wait 2 days, halve, spoon: P(success) 1. ...
Should I rather wait?	mental simulation	waiting 2 days raises the expected quality from 0.80 to 0.86 (P(clean halves) 0.92 → 1.00)
How should I do it (parameters)?	mental simulation	halve now, spoon out the seed; twist torque limit 0.30 N m (imagined p95 0.20), seed removal limit 6.5 (imagined p95 4.3); keep the blade on the seed coat
What am I doing with it?	task model	CutRoundSeed: integrity = scored: a lengthwise cut through skin and flesh all round the seed; steps: place the fruit on its side on the board → pierce at the stem end until the blade meets the seed → keep the edge on the seed coat → turn the fruit once round under the blade → withdraw; possible failures: partial_cut: watch f ...

Table 63: Imagined outcomes of the plans the robot compared (after the photo and the squeeze).

Plan	wait (h)	n	P(clean)	P(torn)	P(coat left)	P(gouge)	hazard	quality
halve now, spoon out the seed	0	150	0.92	0.08	0.47	0.19	no	0.80
halve now, knife-twist the seed out	0	150	0.92	0.08	0.47	0.00	yes	0.85
wait 2 days, halve, spoon	48	150	1.00	0.00	0.43	0.13	no	0.86

7.6 The PR2 in the CRAM kitchen

`cram/avocado_pr2_demo.py` puts the node into the CRAM framework (`~/WORK`, imported, not changed): the apartment kitchen of the CRAM resources with the PR2, an avocado on a 4 cm cutting board on the counter, a knife in the PR2's right gripper and a spoon on the counter. `Avocado(Fruit)` is a semantic annotation of the semantic digital twin that carries the robot's node; the twin's meshes are exported as coloured part bodies (skin, three flesh

Table 64: Execution monitoring in the PR2 scenario (cram/results/pr2_run.json).

Step	Imagined	Measured	Verdict	Belief after
`SqueezeTest`	stiffness 2.05..3.97 N/mm (95 %), mean 2.51	stiffness 3.40 N/mm at 3 N	as expected	ripeness_stage 3.79 ± 0.52; penetrometer_N 10.12 ± 3.89
`CutRoundSeed`	P(scored) = 1.00; F_puncture_N imagined 6.24..14.65	scored, F_puncture_N 13.10	as expected	ripeness_stage 3.80 ± 0.53; seed_length 40.45 ± 2.18; testa_adheres_to_flesh 0.48 ± 0.50
`TwistApart`	P(halved_torn) = 0.07; T_peak_Nm imagined 0.07..0.15	halved_torn, T_peak_Nm 0.18; flesh tears and stays on the seed	unexpected	ripeness_stage 3.00 ± 0.38; seed_length 40.43 ± 2.09; testa_adheres_to_flesh 0.44 ± 0.50
`RemoveSeed`	P(pitted_testa_left) = 0.49; F_peak_N imagined 2.65..5.71	pitted_testa_left, F_peak_N 3.37; spoon took flesh from the cavity rim	as expected	ripeness_stage 3.16 ± 0.35; seed_length 41.45 ± 0.85; testa_adheres_to_flesh 0.96 ± 0.19

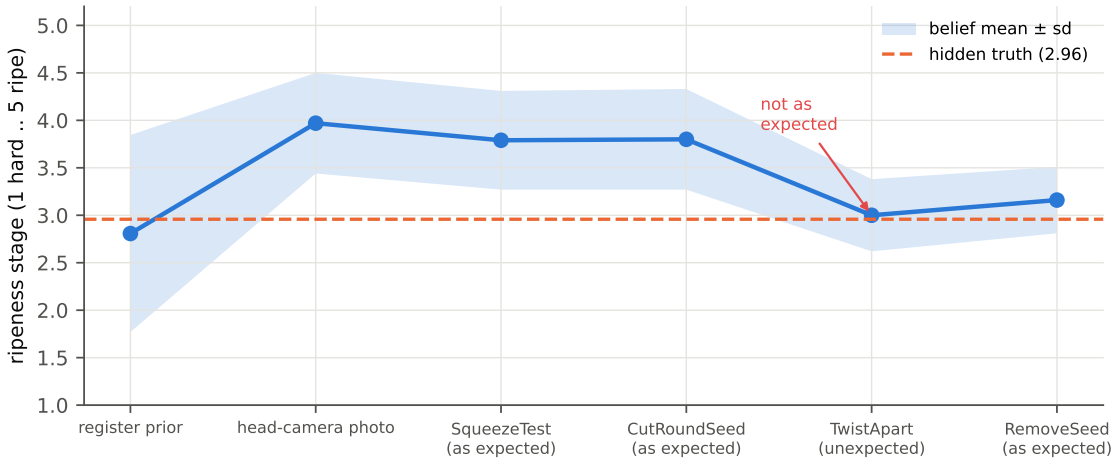


Figure 37: The robot's ripeness belief through the PR2 scenario: the photo makes the avocado look riper than it is, the squeeze (a weak cue) corrects little, the plan predicts clean halves, the twist tears the flesh – not as expected – and the belief falls towards the truth.

so it proceeds and expects torn flesh with low probability – which is exactly where it is wrong, because the photo misled it.

7.7 An interactive kitchen in CRAMERA

`cram/avocado_kitchen.py` keeps the same kitchen live in CRAMERA, the CRAM viewer (3D scene, EQL query panel, plan graph), and lets an operator drive it from the query panel. Commands and questions are ready-made queries (buttons); typed requests are matched to them: A command is checked against the robot's and the avocado's

Table 65: The operator's controls in the CRAMERA query panel.

Kind	Entries
commands (queued, executed with PR2 motions)	Detect the avocado; Squeeze the avocado; Wait two days; Cut round the seed; Twist the halves apart; Spoon out the seed; Reset the kitchen
questions answered at once	What is the object? How ripe is it? Where is the seed? What can you do now? What are you doing?
questions answered by mental simulation	Should you squeeze it first? Guacamole now, or store it? What happens if you halve it now?
views	the robot's answers, its belief, what it imagined, how the execution went, the command log, the commands with their availability
typed phrasings (examples)	squeeze it; halve it; open the avocado; remove the seed; is it ripe; what can you do; start over

state and refused with the reason ("the halves have to be apart first", "I have not detected an avocado yet"), otherwise queued and executed by a worker while the viewer follows the robot. "Wait two days" ripens the hidden avocado and the belief; a twist after waiting usually comes out clean. Two details of CRAMERA shaped the design: the viewer builds every preset's code to word it in English, so a command fires only when the query is evaluated (the command sits in an attribute read by the query's filter); and the world records every annotation it adds or removes as JSON, so the node is kept beside the annotation, not in it. Running beside another CRAMERA session needs its own bridge port, data directory and viewer. No screenshot of the 3D view is included: the headless browser available here has no WebGL.

What this section does not show. No real avocado, no real robot. The kinematic simulation has no contacts: knife, fingers and spoon pass through the fruit, and the mechanics supply forces and outcomes. The mechanics' constants, the utility and the photo's colour likelihood are ILLUSTRATIVE; the peeling monitor's thresholds are calibrated on the twin's own renders. The robot's seed-centre estimate is the twin's nominal position (no sideways offset), and torn flesh is not shown on the seed piece.

8 From reasoning to acting, and from acting to theories

Section 7 gave the robot a belief about one avocado and the means to ask questions about it. This section takes three further steps. First, the robot becomes an *agent* that knows which effects its task must bring about and which it must not cause, predicts both over the fruit's lifetime, and chooses, executes and audits plans accordingly. Second, the knowledge it acts on is written down as a *microtheory* – the egg microtheory of the AICOR working report transferred to the avocado – and tested like a scientific theory against held-out simulated fruit, with ablations and misspecified plants. Third, the robot *learns* that theory: it grows from a deliberately weak seed by its own failures, and reinforcement-learning problems are generated from its successive versions. The full microtheory is a separate document, `report/avocado_microtheory.pdf` (15 pages); this section summarises it. Everything is simulation evidence on partly ILLUSTRATIVE laws.

8.1 An agent that enforces and prevents effects

`cram/avocado_agent.py` turns the node's knowledge into decisions. A *mission* names what is to be served when (guacamole now; half now and half for tomorrow's breakfast; guacamole in two days) and bounds every effect: a goal must hold with at least its probability, a prevented effect may occur with at most its probability, and a bound of zero is a safety rule that no utility can trade against (Table 66). The plan space combines when to cut (now or at the serving time), where the whole fruit waits (counter or fridge), how the seed comes out (spoon or knife twist), whether seed-coat sheets are lifted, and how a kept half is stored (six treatments from nothing to vacuum-packed in the fridge). For each plan the agent runs mental simulations over 80 avocados drawn from its belief: the lifetime model (`food/avocado_lifetime.py`: ripening at the fruit's rate with a Q_{10} of 2.5 between counter and fridge, a ripe window of 24–48 h, decline, spoilage, browning of a cut face as $\mathrm{d}E = k\sqrt{t\, f_{\text{treatment}}}$) advances each one to the cut, the processing mechanics halve and pit it with the repairs the agent would make, and the servings are scored. A geometric check (`plan_hold`) places the holding gripper at least 10 mm from the blade's plane, pinching the fruit's top or, if that fails, pressing it from above.

Table 66: Effects and bounds of the standard missions (cram/avocado_agent.py).

Effect	Kind	Bound	Meaning
halves_apart	enforce	≥ 0.95	the two halves are separated
seed_out	enforce	≥ 0.95	the seed is out of the fruit
cavity_clean	enforce	≥ 0.90	no seed-coat sheets left in the cavity
edible_at_service	enforce	≥ 0.70	every serving mashes well and is not past its best (eating quality ≥ 0.6)
fresh_at_service	enforce	≥ 0.80	no serving's cut face is visibly brown
flesh_intact	enforce	≥ 0.30	no torn flesh, no gouges
hazard	prevent (safety rule)	≤ 0.00	a knife strike into the seed of a held half (pitting injuries, farley2019)
blade_near_hand	prevent (safety rule)	≤ 0.00	the holding gripper within 10 mm of the blade's plane
bruise	prevent	≤ 0.02	a squeeze bruises the flesh
spoiled	prevent	≤ 0.05	the fruit spoils before it is eaten
browned	prevent	≤ 0.20	a cut face is visibly brown when served
testa_left	prevent	≤ 0.10	seed-coat sheets stay in the cavity
torn	prevent	≤ 0.60	the twist tears the flesh (it clings to the seed)
gouge	prevent	≤ 0.60	the spoon takes flesh

Table 67: The agent's decisions for the avocado the PR2 sees on the counter (belief after the head-camera photo: ripeness_stage 3.97 ± 0.53; seed_length 40.03 ± 2.46; testa_adheres_to_flesh 0.51 ± 0.50). Every knife-twist plan is rejected by the safety rule (P(hazard) = 1).

Mission	Chosen plan	feasible	Why
Guacamole now	cut now; seed out with the spoon; peel off seed-coat sheets	1/4	meets every goal and stays within every risk bound; expected quality 0.75 (best of 1 feasible plans out of 4)
Half now, half for tomorrow's breakfast	cut now; seed out with the spoon; peel off seed-coat sheets; keep the other half: vacuum+fridge	3/24	meets every goal and stays within every risk bound; expected quality 0.68 (best of 3 feasible plans out of 24)
Guacamole in two days	keep it whole in the counter for 48 h, then cut; seed out with the spoon; peel off seed-coat sheets	2/12	meets every goal and stays within every risk bound; expected quality 0.89 (best of 2 feasible plans out of 12)

Lifetime-based decisions. The same fruit leads to different plans for different missions. Its lifetime under the belief: counter: edible from 0.0 d to 3.8 d, P(spoiled) > 5 % from 4.5 d; fridge: edible from 0.0 d to 12.0 d, no spoilage risk within 12 d. For guacamole in two days the agent keeps the fruit whole on the counter and cuts at the serving time (expected quality higher than cutting now, because the fruit ripens into its best window). For half now and half tomorrow the kept half is what decides: only treatments that slow browning enough keep its cut face fresh after 15 hours (Table 68).

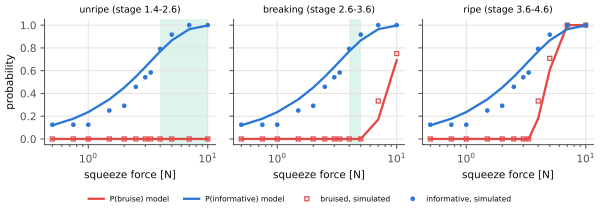
Table 68: Half now, half tomorrow: the plans that cut now, use the spoon and lift the seed coat, by storage of the kept half.

Kept half stored as	browning factor	P(browned)	quality	Feasible / violated bound
none	1.00	1.00	0.37	fresh_at_service 0.00 < 0.80; browned 1.00 > 0.20
lemon	0.45	1.00	0.46	fresh_at_service 0.00 < 0.80; browned 1.00 > 0.20
wrap	0.35	1.00	0.50	fresh_at_service 0.00 < 0.80; browned 1.00 > 0.20
wrap+fridge	0.12	0.00	0.60	yes
lemon+wrap+fridge	0.06	0.00	0.65	yes
vacuum+fridge	0.03	0.00	0.68	yes

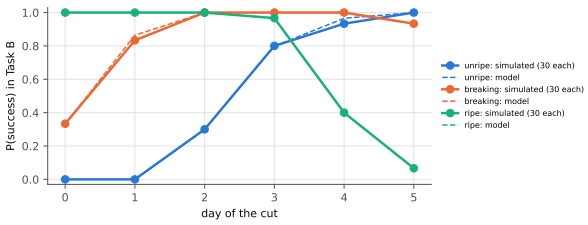
Execution with a goal watch and an effect audit. In the CRAMERA kitchen (`cram/avocado_kitchen.py`) a mission is a command: the agent decides, explains the choice and the rejected alternatives, and the PR2 executes the plan with the motions of Section 7.6 (with new ones for holding, lifting the seed coat with the spoon tip and squeezing lemon on the kept half). After every step a goal watch compares the observed effects with the predicted ones and reassesses the remaining risk (e.g. a torn twist raises the gouge risk of the spoon); after the mission an effect audit lists each goal and prevented effect as achieved, violated or still pending (the kept half until breakfast). Advancing the kitchen's clock ripens, spoils and browns the hidden fruit, so a decision to wait can be checked against what happens.

8.2 An avocado manipulation microtheory

The egg microtheory distinguishes grasping an egg intact from cracking it open: the same fracture is a failure in one task and the goal in the other. For the avocado the event is the separation of the flesh: forbidden while the fruit is squeezed to judge its ripeness (Task A), required in the cut plane and at the twist when it is halved and pitted (Task B), and forbidden everywhere else (bruise, torn cavity wall, gouge). The microtheory (`theory/`) has the egg's parts: a research SDT whose hidden parameters are drawn once per avocado and whose events are patches over relational facts; reduced-order laws for squeezing, bruising, cutting, twisting, seed removal, peeling, ripening and browning; a probabilistic hybrid model whose jump probabilities are marginals over the hidden parameters; an observation model and a particle belief; and phase-indexed requirements over the patch trace. Its two central predictions are a squeeze *corridor* between an informative and a bruising force, which is empty for ripe fruit, and a ripeness *band in time* for the cut, between torn flesh and overripeness (Figure 38).



Task A: squeeze corridor (model lines, simulation markers; green: safe and informative)



Task B: success by day of the cut (simulated, model dashed)

Figure 38: The two central predictions of the avocado microtheory, checked against held-out simulated avocados (details in avocado_microtheory.pdf).

Table 69: Held-out evaluation: the same 90 avocados (30 unripe, 30 breaking, 30 ripe) for every policy; discordant pairs and exact McNemar tests in the microtheory document.

Task	Policy	success	Paired against (only policy : only baseline)
A: assess ripeness by touch, intact	closed-loop squeeze under a 1 \% bruise chance constraint	82/90	thumb 10 N 46:0; photo only 14:2; open-loop prior plan 3:1
B: halve and pit within 5 days	belief band: daily squeeze, cut on the planned day, repairs, seed-coat lifting	88/90	cut now 48:0; wait 2 d 21:2; wait 4 d 20:2; photo-only band 2:2
B, ablations	remove one distinction from the robot's model	–	no seed-coat side 49:0; ready / not ready 1:2; no repair 0:0; population ripening rate 0:0

What it shows. The belief regulates better than fixed policies (the thumb press bruises every ripe fruit; cutting at once tears unripe fruit; waiting four days lets ripe fruit go past its best). Only one removed distinction is load-bearing on these avocados – which side the seed coat sticks to; the others make no detectable difference, which needs a discriminating test, not a conclusion. An error that makes the flesh weaker than believed cannot be corrected in time (the closed loop then succeeds on 4/30 ripe fruit: the bruise is the first evidence), while a camera bias is corrected by touch. A replay audit of 1,350 episodes (8,699 steps) finds no mismatch between the relational action axioms and the

simulator's patches; 22 pairs of steps from the same relational state reach different branches – the hidden attributes (ripeness, the seed-coat side) decide, so a proof needs them.

8.3 Learning the theory from failures

Following *Manipulation as Theory Construction*, the robot's competence is a theory $\mathcal{T}_k = \langle \Sigma_k, A_k, M_k, C_k, E_k \rangle$: concepts that are perceivable through the SDT, actionable through designators and traceable; assertions; body-world-system families (scopes); scoped claims $\Gamma = \text{Competent}(F_{\text{task}}, F_{\text{BWS}}, A_{\text{assess}})$; and evidence (`theory/package.py`). The seed $\mathcal{T}_0$ knows only how to halve and pit a ripe Hass avocado. Five requests expand the task or the family; each runs the executive derived from the current theory, the failures become a localized gap, competing hypotheses are tested by discriminating experiments in the research SDT on development avocados, a patch is fitted, and the integrator admits, restricts or versions it; the revised claim is assessed on the held-out avocados (`theory/learning.py`, Table 70). The

Table 70: The revision path of the avocado theory (simulation evidence).

	Request	Gap: theory before	Patch	Held-out after
T0	halve and pit a ripe Hass	–	–	30/30 verified
T1	Halve this firm avocado.	RipenessDependentSeparation: 10/60	admit; +RipenessStage, TornFlesh	88/90 verified
T2	Pit it; the cavity must be clean.	SeedCoatUnrepresented: 39/90	admit; +SeedCoatSide	88/90 verified
T3	Use the new camera (lighter images).	ColourRipenessMisspecified: 27/30	restrict + version; +TactileRipeness	30/30 verified
T4	Keep the other half for tomorrow's breakfast.	BrowningUnrepresented: 0/90	admit; +Browning	88/90 verified
T5	Halve and pit this other cultivar.	CultivarOutsideFamily: 41/60	restrict	41/60 refuted

patches recover the plant's laws from the experiments: the clean-release law's centre at stage 2.72 (plant 2.8; its width is overestimated on a coarse grid), the share of fruit whose seed coat sticks to the flesh 0.53, the peel repair's success 0.82 per pass (plant 0.85) and the new camera's lightness bias +10.4 (plant +10). The last request – a cultivar whose seed clings ten times harder – is answered by *restriction*: the only remedy, the knife twist, is forbidden, so the claim is retracted for that family; the evidence grows while the warranted scope contracts. At runtime a *commitment gate* separates warranted commitments from uncertain, unsupported, counterevidenced and out-of-scope ones:

Table 71: The commitment gate of the final theory.

Situation	State	Reason
ripe fruit, photo + touch	warranted	claim Γ2 clean cavity verified (88/90), instance in scope (P = 0.97)
photo only	uncertain	the scope requires tactile ripeness evidence; only photo
knife twist proposed	unsupported	forbidden effect predicted: hazard P = 1.00
sticky cultivar	counterevidenced	claim Γ2 on a sticky cultivar retracted
not an avocado?	out-of-scope	P(instance in Hass, any ripeness) = 0.30
browning not represented	unsupported	concepts not operationally covered: Bruising

8.4 Reinforcement learning programmed by the theory

Reinforcement learning lives inside the theory (`theory/prl.py`): the learning problem is generated from a theory version. Its state features are the theory's perceivable concepts (believed stage and spread, day, a torn twist, the seed coat seen), its actions the designators' parameters (cut or wait, spoon or knife twist, peel passes only if the theory has the seed-coat concept), its reward the claim's KPIs, and its hard constraints the forbidden effects, as a shield that masks every action whose forbidden effect the theory predicts. A tabular Monte-Carlo learner trains on 5,600 research-SDT episodes and is evaluated greedily on the held-out avocados (Table 72). The learner inherits the theory's distinctions

Table 72: Programmable RL on the 90 held-out avocados of Task B.

Program / policy	success	hazards
RL from T1 (no seed-coat concept), shielded	38/90	0
RL from T3, shielded	88/90	0
RL from T3, hazard as a soft penalty 0.05	0/90	90
hand-coded executive of the same theory (belief band)	88/90	0

and its safety rules: from $\mathcal{T}_3$ with the shield it learns the band and matches the hand-coded executive; from $\mathcal{T}_1$, which has no seed-coat concept, no amount of training cleans a cavity whose dirt it cannot represent; and with the hazard as a small penalty instead of a shield it buys the forbidden knife strike in every episode. What the theory does not represent the learner cannot learn, and what the theory forbids a reward must not be allowed to buy.

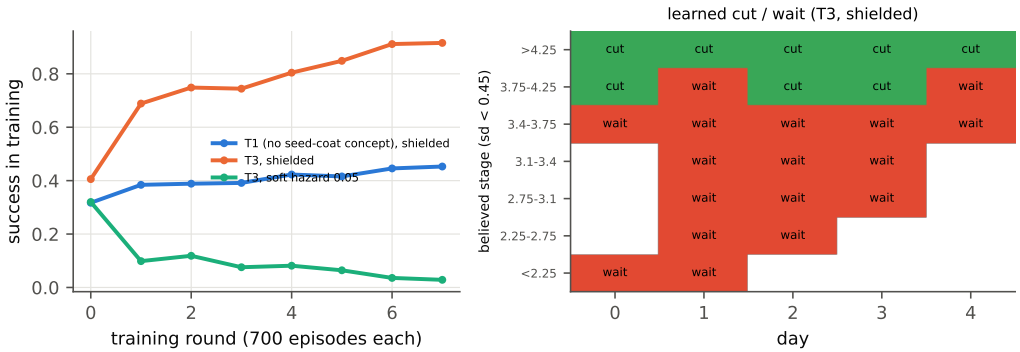


Figure 39: Left: success during training. Right: the cut / wait rule learned from $\mathcal{T}_3$ with the shield, by believed stage and day – the ripeness band, learned.

What this section does not show. No physical validation: the plant and the robot share the laws except where a misspecification is injected; contact is quasi-static mechanics, not contact simulation; the photo in the microtheory is a symbolic observation from the colour law; the learning loop's agents are scripted functions, and the RL is tabular on a handful of features. The 30 held-out avocados per class are reused across tables (paired, not independent).

9 The kitchens: EASE and GARMI apartments

`scripts/import_apartments.py` imports the EASE and GARMI apartments and the GARMI robot from the CRAM viz scene bundles (read in place from `~/ResearchSpace`, which stays read-only) into USD with Isaac Sim's URDF importer: doors and drawers become joints, collisions are convex decompositions. The GARMI URDF needed its gazebo, ros2_control and transmission blocks stripped before the importer would accept it.

Table 73: Imported scenes, counted in the source URDFs by this report.

Scene	links	joints	movable joints
EASE (IAI) apartment	118	117	42
GARMI (TUM MIRMI) apartment	79	78	11
GARMI robot (cleaned URDF)	64	63	26

The GARMI rig. `household/garmi_rig.py` drives the imported GARMI (two FR3 arms with Franka hands, a two-stage lift, a pan-tilt head, a wheeled base) as one PhysX articulation with the base fixed for manipulation (the probe found 26 degrees of freedom and 53 links): damped-least-squares Cartesian IK per arm on the PhysX Jacobian with a null-space posture bias, force-limited grasping (finger drives in force mode, targets ramped – an instantly set stiff target slams the fingers shut, a lesson from the egg bench), and the Franka TCP 0.1034 m along the hand axis.

Food in the kitchen. `scripts/render_apartment_kitchen.py` places GARMI and a cutting board on the EASE kitchen island with seven food assets: apple_3 whole and half, avocado_4 whole and half, and three eggs. Only these egg-set items have been placed; none of the wave-1, wave-2 or wave-3 items is in a kitchen yet, and the GARMI apartment has been imported but not rendered or furnished with food.

End-to-end task: not yet working. `scripts/sim_garmi_egg.py` is the first end-to-end task (GARMI picks an egg, cracks it on the bowl rim and opens it with both hands). Its last run (`renders/garmi_egg.log`) stopped during the approach: the arm's joint state and TCP became NaN and the controller raised *cannot convert float NaN to integer*. No GARMI task log exists yet.

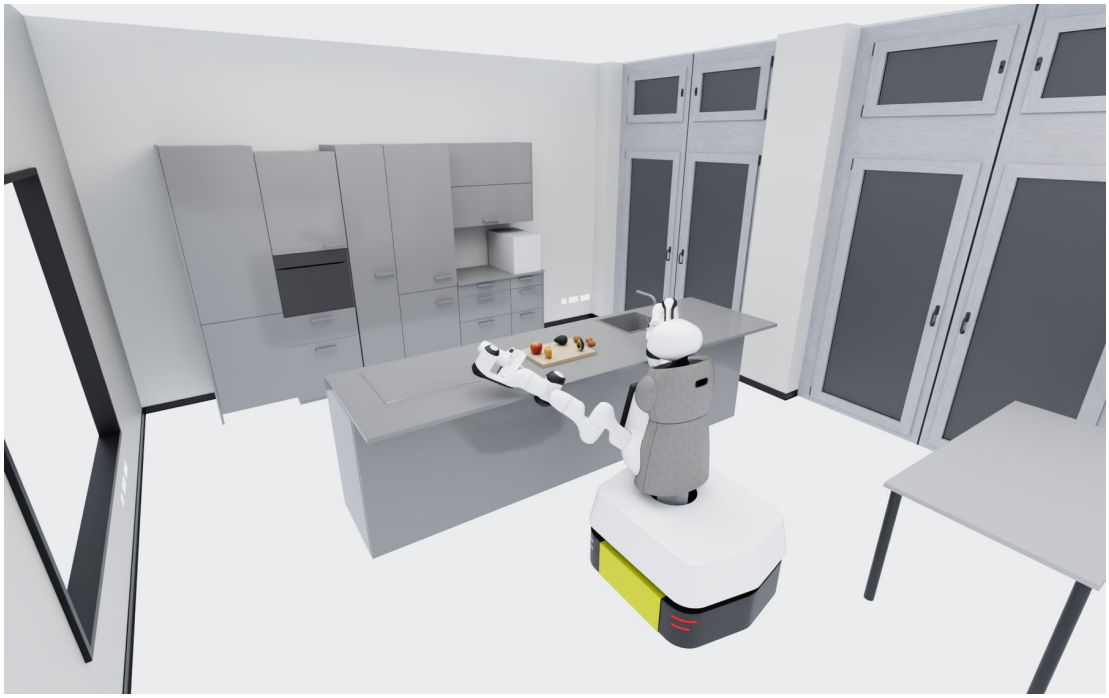
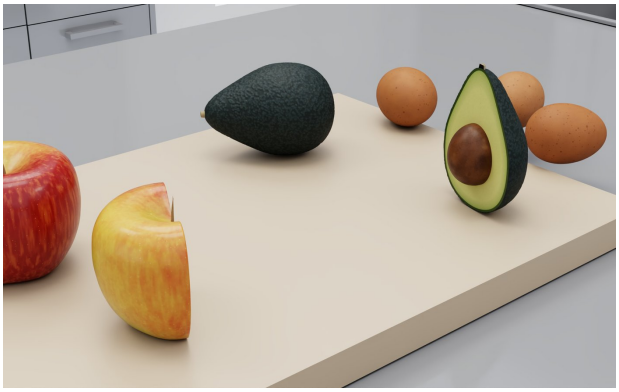


Figure 40: The EASE apartment kitchen with GARMI (existing render, path-traced in Isaac Sim 6.2).



island with board and food



close-up: apple, avocado, eggs

Figure 41: Food twins on the EASE kitchen island (existing renders).

10 Data quality

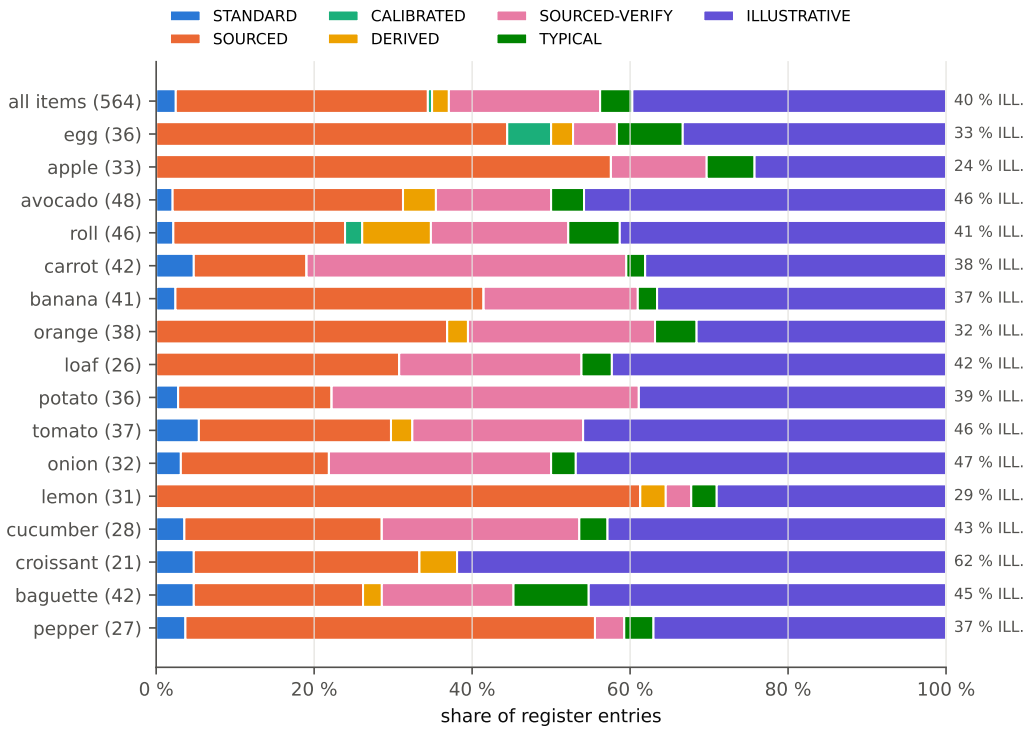


Figure 42: Register entries by status, per item and overall (drawn by this report from food/params.py). Numbers in brackets: entries per item.

Of 564 register entries, 180 (32 %) are SOURCED, 108 (19 %) SOURCED-VERIFY, 14 STANDARD, 12 DERIVED, 3 CALIBRATED, 23 TYPICAL and 224 (40 %) ILLUSTRATIVE. The ILLUSTRATIVE values concentrate where sources end: colours (L*C*h of flesh and skin), internal dimensions (membrane, rib and core sizes), friction coefficients, the peeling and cutting forces, and the outcome constants of the strike and peel models. The sampled instance state (age, storage) is ILLUSTRATIVE by design: it is the scenario, not a property of food.

Table 74: Register status counts per item (food/params.py).

Item	STANDARD	SOURCED	CALIBRATED	DERIVED	SOURCED-VERIFY	TYPICAL	ILLUSTRATIVE	total
egg	0	16	2	1	2	3	12	36
apple	0	19	0	0	4	2	8	33
avocado	1	14	0	2	7	2	22	48
roll	1	10	1	4	8	3	19	46
carrot	2	6	0	0	17	1	16	42
banana	1	16	0	0	8	1	15	41
orange	0	14	0	1	9	2	12	38
loaf	0	8	0	0	6	1	11	26
potato	1	7	0	0	14	0	14	36
tomato	2	9	0	1	8	0	17	37
onion	1	6	0	0	9	1	15	32
lemon	0	19	0	1	1	1	9	31
cucumber	1	7	0	0	7	1	12	28
croissant	1	6	0	1	0	0	13	21
baguette	2	9	0	1	7	4	19	42
pepper	1	14	0	0	1	1	10	27
all	**14**	**180**	**3**	**12**	**108**	**23**	**224**	**564**

Consistency checks run by this report. The checks below are computed when the report is built; they test the repository, not the food. Further known inconsistencies: `docs/wave1/ROLL.md` gives roll_0 as 48.1 g (the part sum) while the asset's whole body is 47.4 g; the `food/dynamics.py` docstring still calls both impact constants ILLUSTRATIVE although one is now CALIBRATED; the drop-surface stiffnesses and the strike-model constants live in code (`dynamics.SURFACES`, `egg_crack.P`) and are mirrored only as a text note in the register.

Table 75: Repository consistency checks.

Check	Result
part masses sum to the whole	24/27 instances within 0.5 %; roll and loaf +1.4 %
egg crack study reproducible with the current register	yes
SOURCED / SOURCED-VERIFY / CALIBRATED entries naming a source key of params.SOURCES	264/291
source keys in params.SOURCES that the register cites	166/205
egg assets built with the current egg register	yes
egg_1 truth.usda impact_effective_mass_fraction vs register	0.53 in the asset vs 0.53 (CALIBRATED) in the register
docs/ANATOMY.md up to date with the register	yes
research notes for wave 3 (docs/wave3)	missing
wave-1 and wave-2 designator demos (seed 0)	20/20 reach their goal

11 Limitations and next steps

What is validated, and how far.

- *Against published measurements, independently:* the tomato Hertz stiffness from one study's modulus matches another study's measured red-fruit plate firmness (the green fruit does not); the potato peel loss (10.0 vs about 9 %); part and cut masses against the register's mass checks.
- *Fits:* the egg boiling thresholds, the egg impact constant, the roll crust-water curve and the carrot boiling thresholds are CALIBRATED – they reproduce the data they were fitted to.
- *Solver-limited:* the FEM squeezes converge only for soft lanes (ripe avocado, fresh roll, red tomato at 4000 Hz); stiff lanes are lower bounds.
- *Mechanism only:* banana peeling, carrot planking and the egg crack study show plausible trends, but their forces or outcome constants are ILLUSTRATIVE.
- *Not validated at all:* appearance (no comparison with reference photographs yet), wave-3 behaviour, any real robot run.

Gaps in the build.

- **Anatomy review gate.** The exploded plates were sent for the user's review; the gate is still pending for every item, so all part trees are provisional.
- **Wave 3 incomplete:** built and rendered (20 October), but the `docs/wave3/` research notes, task designators, schema entries and benches are missing.
- **Wave 2** has no schema entries; only the tomato has a bench.
- **Apple without designators or bench.** The avocado has designators, an emulator and the PR2 scenario (kinematic: no contact forces; Section 7).
- **Kitchens:** only the egg set has been placed (EASE); the GARMI apartment is imported but unused; the GARMI egg task fails with NaN in the arm controller.
- **Bodies are simplified in physics:** homogeneous FEM bodies (no skin, seed or crust), rigid peel segments, kinematic shell fragments, a rigid yolk.
- **Stale artefacts:** rebuild the egg assets and `docs/ANATOMY.md` with the current register; fix the bread part-mass overlap.

Next steps. (1) Pass the anatomy review; (2) rerun the stiff FEM lanes at higher solver rates (the memory note: 4000 Hz / 250 iterations was enough for soft lanes only) or replace them by the analytic models; (3) fix the GARMI IK failure and log the first end-to-end egg task in the EASE kitchen; (4) build wave 3 and render wave 2; (5) for the avocado: calibrate the photo ripeness likelihood and the peeling monitor on real images, give the PR2 contact physics for the knife, the twist and the spoon (breakable joints carrying the bridge and adhesion strengths), and estimate the seed's sideways offset; (6) verify the 108 SOURCED-VERIFY entries and calibrate the ILLUSTRATIVE ones that the benches depend on (peel bonds, strike constants, cutting forces, friction); (7) test the avocado microtheory on real fruit (penetrometer, squeeze stiffness and photos at known stages to calibrate the bruise ratio and the colour and stiffness likelihoods) and take a second fruit through the theory-construction loop.

A Parameter register

Table 76: Register of the hen egg (params.EGG, 36 entries).

Key	Value / distribution	Status	Source and note
length_L	N(57, 1.8) in [52, 63] mm	SOURCED	kop: 58.05 mm; turjaf: 56.12 mm
shape_index_B_over_L	N(0.765, 0.02) in [0.7, 0.81]	SOURCED	kop: 78.25 %; alt: normal 72-76
asymmetry_w	U(1.6 .. 3.4) mm	ILLUSTRATIVE	narushin2021 parameter w; range assumed
shell_thickness	N(0.35, 0.03) in [0.28, 0.42] mm	SOURCED	kop: 0.28-0.41 mm; sb: 0.35 mm
cuticle_thickness	N(7.85, 1.5) in [5, 12.5] um	SOURCED	ir: 5.6-12.5 um, mean 7.85
mammillary_thickness	100 um	SOURCED	fr: ~100 um (palisade ~300, vertical crystal 3-8)
membranes_thickness	N(70, 8) in [50, 90] um	SOURCED	fr/mem: ~70 um combined
outer_membrane_share	0.75	SOURCED	ubc: outer membrane ~3x thicker than inner
pore_count	integer 10000 .. 20000	SOURCED	por: 10,000-20,000; ~2x denser at blunt pole
shell_E	U(15 .. 43) GPa	SOURCED	sev: 41-43 GPa nanoindentation; sb model 15 GPa
shell_poisson	0.3	TYPICAL	
shell_density	2300 kg/m3	ILLUSTRATIVE	checked: shell 8-11 % of mass (br)
albumen_density	1035 kg/m3	SOURCED-VERIFY	cs: liquid white ~1.03 g/ml (weak source)
yolk_density	1030 kg/m3	TYPICAL	
whole_specific_gravity	1.078-1.097 fresh, 1.040-1.059 after >30 d	SOURCED	ro
yolk_mass_fraction	N(0.3, 0.015) in [0.27, 0.32]	SOURCED	br: 27-32 %; usda 31 %
albumen_layer_shares	outer thin 0.23, thick 0.57, inner thin 0.17, chalaziferous 0.03	SOURCED	usda egg grading manual
germinal_disc_diameter	U(2 .. 3) mm	SOURCED	ab: infertile blastodisc 2-3 mm
yolk_colour_fan	U(9 .. 14) DSM	SOURCED	dsm YolkFan 1-16; consumers prefer ≥ 12
storage_temp_C	one of 6, 22 C	ILLUSTRATIVE	fridge or counter
age_days	U(1 .. 21) d	ILLUSTRATIVE	days since lay
air_cell_height	1.5 + k(T) * age; k(6 C) = 0.08, k(22 C) = 0.19 mm/d mm	SOURCED-VERIFY	gr: < 4 mm after 28 d at 6 C; > 6 mm after 25 d at 22 C; graded vs eu589/usda
F_break_equatorial	N(35, 8) in [12, 59] N	SOURCED	kop: Hisex 36.43 N short axis, 10-59 N; icf11 34±6 N
F_break_axial	N(37, 9) in [12, 62] N	SOURCED	kop 34.92 N long axis; icf11 41±8 N; alt: highest along the polar axis at the sharp end
F_break_thickness_scaling	F ~ (t / 0.35 mm)^2	ILLUSTRATIVE	thin-shell scaling assumed
shell_contact_stiffness	Reissner k = 4 E t^2 / (R_local sqrt(3(1-nu^2))) N/mm	DERIVED	reissner1946 + shell_E + shell_thickness
microcrack_band	0.85	ILLUSTRATIVE	F/F_break in [0.85, 1) may leave an invisible micro-crack
microcrack_strength_loss	0.4	ILLUSTRATIVE	F_break of a micro-cracked shell x (1 - 0.40)
membrane_tear_ratio	1.6	ILLUSTRATIVE	check → leaker when the load exceeds 1.6 x crack force
edge_concentration	F_crack_edge = F_break * sqrt(r_edge / 5 mm), r_edge ≥ 0.2 mm	ILLUSTRATIVE	sharp tool parts crack at lower force (Morgenstern crackforce depends on parts)
friction_shell_rubber	U(0.45 .. 0.75)	ILLUSTRATIVE	dry shell vs gripper pad
crack_strike_model	food/egg_crack.py P: coupled mass 0.05 kg; membrane tears at F/Fc 1.8 (flat), 1.05 (rim), 3.0 (egg-on-egg); shatter at 2.6 / 2.2 / 2.4; shards 0.15 / 0.6 / 0 per F/Fc; yolk puncture 0.06 per shard	ILLUSTRATIVE	structure from atk_egg, egginfo_crack, atk_egg_on_egg; numbers to calibrate
cook_thermal_props	albumen 0.55/1035/3700, yolk 0.34/1030/2800, shell 0.45/2300/900 W/mK, kg/m3, J/kgK	TYPICAL	egg thermophysical properties, to cite (e.g. Denys et al. 2003)
cook_yolk_thresholds	f65 < 0.15 runny, < 0.45 jammy, < 0.85 creamy, else firm	CALIBRATED	alphafoodie timing table (hot start, medium eggs) + Lopez-Alt fridge-egg timings
impact_effective_mass_fraction	0.53	CALIBRATED	c = sqrt(m_eff)/rate fitted to cohen2025: side-on drops onto a hard surface crack ~45 % at 10 mm (model 45 %), vertical 55 % at 8 mm (model ~57 %); the model population is more heterogeneous than one MIT batch (8 mm side-on: model ~28 % vs 5 %)
impact_rate_factor	1.3	ILLUSTRATIVE	dynamic/static crack force; strength rises with loading rate (only sqrt(m_eff)/rate is identified by the calibration)

Table 77: Register of the apple (gala-like) (params.APPLE, 33 entries).

Key	Value / distribution	Status	Source and note
cultivar	Gala-like	ILLUSTRATIVE	shape/colour archetype (striped orange-red over yellow)
diameter_D	N(75, 3.5) in [66, 86] mm	SOURCED	wiki_gala: 78 % of Gala fruit 70-80 mm
height_over_D	N(0.88, 0.03) in [0.8, 0.96]	SOURCED	uspto_pp20028: 71.3 x 81.4 mm
lobe_amplitude	U(0.008 .. 0.03)	ILLUSTRATIVE	five-fold relief at the calyx end
lopsidedness	U(0 .. 0.03)	ILLUSTRATIVE	
skin_thickness	N(55, 10) in [33, 73] um	SOURCED	konarska2006: 33.3-73.1, mean 55.2 um
skin_tensile_strength	2.56 MPa	SOURCED-VERIFY	jets_skin_tensile
lenticel_density	U(4.3 .. 12.6) 1/cm2	SOURCED	ishs_1099_97: mean 7.11 /cm2
flesh_density	N(860, 40) in [760, 970] kg/m3	SOURCED	wiki_gala 0.86 g/cm3; fuji 955-968 kg/m3 (degruyter)
cortex_porosity	U(0.17 .. 0.3)	SOURCED	fmach_porosity: Jazz 17 % .. Golden Del. 29.8 %

Table 77: Register of the apple (gala-like) (params.APPLE, 33 entries). (continued)

Key	Value / distribution	Status	Source and note
core_density_excess	60 kg/m3	ILLUSTRATIVE	core porosity 14.5 % vs cortex 26 % (frontiers2015)
flesh_E	lognormal, median 3, factor 1.7 MPa	SOURCED	fuji 5.0 MPa (consensus f4110d16); quasi-static 4.2-7.8; stored 0.5-1.7 (tjaf2005)
flesh_poisson	0.3	TYPICAL	
storage_days	U(0 .. 120) d	ILLUSTRATIVE	E multiplied by exp(-storage_days / 180)
mt_firmness	N(75, 8) in [50, 95] N	SOURCED	msu2025: Gala ~18 lb (80 N) at harvest
flesh_shear_strength	N(0.4, 0.05) in [0.28, 0.5] MPa	SOURCED	fuji shear 0.45 MPa; failure stress 0.32-0.37 MPa (consensus)
bruise_energy_threshold	U(0.004 .. 0.012) J	SOURCED-VERIFY	no bruise below 0.25 m/s impact; bruises from 0.011 J pendulum (consensus)
bruise_volume_per_J	0.01 * V_fruit / 0.41 J cm3/J	SOURCED-VERIFY	fsct2008 (about 1 % of fruit volume at 410 mJ); linear in energy
cutting_force_peak	N(25, 5) in [12, 40] N	ILLUSTRATIVE	no apple cutting force found
friction_skin_rubber	U(0.7 .. 0.95)	SOURCED-VERIFY	fuji vs TPU rubber 0.88 (consensus)
core_line_radius	N(14, 1.5) in [11, 18] mm	SOURCED	uspto: core diameter 24.7-36.7 mm
core_length	N(41, 3) in [34, 48] mm	SOURCED	uspto_pp29089: 41.1 mm
locule_length	N(16, 1) in [13, 19] mm	SOURCED	uspto: 15.2 x 7.9, 16.6 x 9.9 mm
locule_width	N(8.5, 0.8) in [7, 10.5] mm	SOURCED	uspto (radial width in the long. section)
endocarp_thickness	0.35 mm	ILLUSTRATIVE	leathery/papery carpel wall
seeds_per_carpel	P(0,1,2) = 0.1, 0.4, 0.5	SOURCED	wiki_apple: 1-2 seeds per carpel
seed_length	N(8.5, 0.6) in [7.5, 9.6] mm	SOURCED	sciRep2021: 7.69-9.43 mm
seed_width	N(4.5, 0.3) in [3.9, 5] mm	SOURCED	sciRep2021: 4.0-4.9 mm
seed_thickness	N(2.7, 0.2) in [2.2, 3.2] mm	TYPICAL	
stem_length	U(10 .. 28) mm	SOURCED	uspto: 10.5, 12-14, 28.2 mm
stem_diameter	N(2.8, 0.15) in [2.5, 3.1] mm	SOURCED	uspto: 2.7-3.0 mm
vascular_bundle_radius	0.45 mm	ILLUSTRATIVE	10 primary bundles (frontiers2015)
browning_dE	2.38 (0 min), 2.68 (30 min), 4.05 (60 min) dE	SOURCED	pmc7022590 (Golden Delicious)

Table 78: Register of the hass avocado (params.AVOCADO, 48 entries).

Key	Value / distribution	Status	Source and note
cultivar	Hass	SOURCED	avo_au_stages; uspto_pp14239
length_L	N(98, 6) in [80, 120] mm	SOURCED-VERIFY	specialty: 7-15 cm; no Hass mm data held
diameter_D	N(66, 3.5) in [56, 76] mm	TYPICAL	chosen so mass ~ 200 g (uspto mean 204 g)
fruit_mass_check	142-340 (CAC), 200-300 (wiki), mean 204 (uspto_pp14239) g	SOURCED	
neck	U(0.08 .. 0.2)	ILLUSTRATIVE	obovate/pyriform taper of the stem end
skin_thickness	N(1.6, 0.15) in [1.2, 2] mm	SOURCED	uspto_pp14239: Hass skin 1.6 mm
skin_layers	cuticle, epidermis (1 layer), hypodermis (1-3 cells), parenchyma, stone-cell layer	SOURCED	cs42
pebble_scale	U(1.2 .. 2) mm	ILLUSTRATIVE	no source for tubercle size
seed_length	N(40, 2.5) in [34, 47] mm	DERIVED	seed mass 28.4-30.6 g (uspto) at 1.1 g/cm3
seed_width_ratio	N(0.86, 0.04) in [0.78, 0.95]	SOURCED	uspto: 'variable, mostly ellipsoid'
seed_offset	U(0.5 .. 2.5) mm	SOURCED-VERIFY	cs42: seed slightly off-centre
seed_coat_thickness	0.25 mm	ILLUSTRATIVE	two papery testa layers (cs42); 'paper-thin' (ufl)
endocarp_thickness	0.15 mm	ILLUSTRATIVE	a few rows of flattened cells (cs42)
testa_adheres_to	one of seed, flesh	SOURCED-VERIFY	cs42: endocarp may adhere to the seed coat; side not settled
flesh_density	990 kg/m3	ILLUSTRATIVE	whole fruit 970-1018 (clark2007)
seed_density	1100 kg/m3	ILLUSTRATIVE	
skin_density	1050 kg/m3	ILLUSTRATIVE	
mass_ratio_seed_skin_flesh	16:12:72 (avocadosource) / 14:13:73 (uspto) %	SOURCED	check
ripeness_stage	U(1 .. 4.6) 1..5	STANDARD	avo_au_stages; continuous hidden value
ripening_rate	N(0.55, 0.1) in [0.3, 0.8] stage/d	SOURCED-VERIFY	ashton2006: firm-ripe ~day 8, soft-ripe ~9.5 at 20 C; landahl: 304 → 2 N in 6 d at 18 C
penetrometer_force	stage 1..5 = 118, 46, 15, 8.3, 4.9 N (geometric band centres), log-linear N	SOURCED	avo_au_stages: hard >10, rubbery 2.2-10, breaking 1.1-2.1, firm-ripe 0.7-1.0, ripe 0.4-0.6 kgf (11.1 mm tip, 8 mm)
flesh_E	E = 2.29 (F_pen / 118 N)^0.837 → 2.29 .. 0.16 MPa MPa	SOURCED	farinas2021: 2.29±0.75 → 0.16±0.08 MPa over ripening (endpoints); mapping ILLUSTRATIVE
flesh_poisson	0.45	ILLUSTRATIVE	near-incompressible soft tissue
bruise_force_ratio	0.6	SOURCED-VERIFY	hia2018: a 10 N thumb press bruises firm-ripe fruit (visible within 48 h); ratio of thumb pressure to penetrometer pressure at stage 4
bruise_visible_after	24 h	SOURCED	hia2018: bruises take ~24 h to show
friction_skin_rubber	U(0.6 .. 0.9)	ILLUSTRATIVE	pebbled skin
skin_colour_Lab	L 40→27, C 15→3, h 125→250 over ripening (whole fruit) L*,C*,h	SOURCED	ashton2006
flesh_bands_Lab	three equal bands: dark green L 74→64 h 113; pale green L 83→75 h 105; yellow L 83→75 h 101; C 43→35 L*,h	SOURCED	ashton2006
vascular_strands	6	SOURCED	cs42: pedicel cylinder splits into 6 strands, join near the apex and enter the seed coat as one
squeeze_stiffness_model	Isaac FEM table (test_reports/avocado_squeeze.json): k(4 N) 7.0 / 5.4 / 3.3 / 2.4 / 2.1 / 1.8 for stages 1 / 2 / 3 / 4 / 4.5 / 5 N/mm	DERIVED	convergence (1000 Hz/120 it vs 2000 Hz/200 it): stages ≥ 3 within 10 %; stages 1-2 under-resolved (stage 1: 7.0 vs 11.9 N/mm) → unripe fruit is stiffer than the table

Table 78: Register of the hass avocado (params.AVOCADO, 48 entries). (continued)

Key	Value / distribution	Status	Source and note
stem_button_length	U(2 .. 4) mm	SOURCED-VERIFY	alder: ~3 mm clipped button
process_protocol	cut lengthwise round the seed, twist the halves apart, remove the seed with a spoon (knife strike discouraged)	SOURCED	afm_howto; farley2019 (knife injuries when pitting)
skin_puncture_force	U(6 .. 15) N	ILLUSTRATIVE	GAP: no knife-puncture data for Hass skin held; sharp chef's knife at the stem end
skin_cut_resistance	U(1.5 .. 3) N/mm	ILLUSTRATIVE	force per mm of skin thickness while slicing along the perimeter; GAP
flesh_cut_toughness_unripe	300 J/m2	ILLUSTRATIVE	analogue: fruit-flesh cutting toughness 0.1-0.4 kJ/m2 (apple, potato); scaled with sqrt(E / E_unripe) over ripening
seed_adhesion_unripe	20 kPa	ILLUSTRATIVE	seed-flesh interface shear strength, scaled with (E / E_unripe)^0.7; GAP
clean_twist_stage	2.8 1..5	ILLUSTRATIVE	centre of the clean-separation transition: unripe flesh clings to the seed (twin AvocadoTruth.halve: clean from ~3; afm_howto: ripe fruit twists apart)
testa_remnant_fraction	U(0.4 .. 1)	ILLUSTRATIVE	share of the seed coat left in the cavity when the testa adheres to the flesh (cs42)
knife_blade_thickness	1.6 mm	TYPICAL	chef's knife spine 2-2.5 mm, edge region ~1-2 mm
storage_temperatures	counter 20, fridge 4 (ripe fruit 2-4 C; softening fruit 5-8 C up to 10 d) C	SOURCED	ucd_avocado; marques2011
ripening_q10	2.5	ILLUSTRATIVE	temperature factor of the ripening rate per 10 C (respiration-type Q10); no avocado-specific value held
ripe_window_counter	U(24 .. 48) h	SOURCED-VERIFY	ripe_window_guides: stage 5 lasts ~1-2 d at room temperature
decline_per_day_counter	0.6 1/d	ILLUSTRATIVE	eating-quality loss per day past the ripe window (overripe: soft, stringy, brown spots), slowed by the fridge with ripening_q10
spoil_days_past_window	U(1 .. 3) d	ILLUSTRATIVE	days past the ripe window at 20 C until the fruit is unusable (rot, flesh browning)
browning_rate_cut	3 dE/sqrt(h)	ILLUSTRATIVE	enzymatic browning of exposed flesh at 20 C, dE ~ k sqrt(t) (PPO, oxygen-limited); apple slices dE 2.4-4 within 1 h (pmc7022590) as an analogue
browning_visible_dE	5 dE	ILLUSTRATIVE	colour difference at which the cut face looks brown
browning_treatment_factors	none 1.0, lemon juice 0.45, plastic wrap 0.35, wrap + fridge 0.12, vacuum + fridge 0.03; seed left in: no effect on the cut face	ILLUSTRATIVE	ordering from atk_browning; factors are assumptions
peel_coat_success	0.85	ILLUSTRATIVE	chance that one pass with the spoon tip lifts the seed-coat sheets left in the cavity

Table 79: Register of the wheat roll (weizenbrötchen) (params.ROLL, 46 entries).

Key	Value / distribution	Status	Source and note
product	Weizenbroetchen, oval, one lengthwise score ('Schnittbroetchen')	STANDARD	leitsaetze_1993: ≥ 90 % wheat flour; testde_2021: the score raises the crust share
length	N(102, 5) in [90, 115] mm	ILLUSTRATIVE	oval single-score roll; round crispy roll ~90 mm (hirte_2012), round buns 86-92 mm (musiienko_2025)
width	N(80, 4) in [70, 90] mm	ILLUSTRATIVE	chosen with length/height for ~50 g at ~0.2 g/cm3
height	N(52, 3) in [44, 60] mm	SOURCED	hirte_2012 ~50 mm; musiienko_2025 50-55 mm
mass_check	50-90 (bake-at-home rolls, testde_2021); RMT ~55 g dough per roll (ireks_rmt) g	SOURCED	
base_edge_height	U(6 .. 10) mm	ILLUSTRATIVE	rounded base edge
lumpiness	U(0.01 .. 0.03)	ILLUSTRATIVE	hand-moulded irregularity
score_length_frac	U(0.7 .. 0.85)	ILLUSTRATIVE	share of the long axis
score_width	U(14 .. 22) mm	ILLUSTRATIVE	opened 'Ausbund' width
score_depth	U(4 .. 8) mm	ILLUSTRATIVE	
crust_thickness_top	N(1.3, 0.2) in [0.9, 2] mm	SOURCED	hirte_2012: cut-off crust 1.0-1.5 mm
crust_thickness_score	U(0.6 .. 0.9) mm	ILLUSTRATIVE	thinner, paler Ausbund crust
crust_thickness_bottom	U(1.2 .. 1.8) mm	ILLUSTRATIVE	GAP: no bottom-crust measurement held
transition_thickness	U(1 .. 2) mm	ILLUSTRATIVE	soleimani_2014 distinguishes an inner-crust colour band; thickness GAP
crumb_density	N(180, 15) in [150, 220] kg/m3	DERIVED	roll specific volume 5.9-6.4 ml/g (kurek_2016) → whole ~0.16-0.17 g/cm3, minus the crust share; crumb 0.088-0.252 (zghal_1999, SOURCED-VERIFY)
crust_density	N(320, 30) in [260, 400] kg/m3	TYPICAL	crust 50-100 % denser than crumb
transition_density	240 kg/m3	ILLUSTRATIVE	between crust and crumb
crumb_porosity	U(0.73 .. 0.81)	SOURCED	hirte_2012 uCT
cell_diameter_median	N(0.7, 0.08) in [0.5, 0.9] mm	DERIVED	lagrain_2012 mean 2D cell area 0.32 mm2 → equivalent diameter 0.64 mm; 2D underestimates (x1.5 volume, rathnayake_2018)
cell_diameter_gsd_log	0.55	ILLUSTRATIVE	lognormal spread of cell size (2-2189e3 um3 range, hirte)

Table 79: Register of the wheat roll (weizenbrötchen) (params.ROLL, 46 entries). (continued)

Key	Value / distribution	Status	Source and note
cell_number_density	1200 1/cm3	DERIVED	83 cells/cm2 (lagrain_2012) / ~0.07 cm mean diameter
cell_elongation	U(1.2 .. 1.5)	ILLUSTRATIVE	cells stretched in the rise direction
crust_L	N(52, 4) in [42, 62] L*	SOURCED-VERIFY	pan-bread crust 47.6 (chaple_2023); roll browning index 50.4 → L* ~50 (kurek_2016)
crust_C	36 C*	SOURCED-VERIFY	chaple_2023 a* 17.05, b* 32.2 → C* 36.4
crust_h	62 deg	SOURCED-VERIFY	chaple_2023 → h 62
crumb_L	74 L*	SOURCED-VERIFY	pan-bread crumb 72.06 (chaple_2023); white roll slightly lighter
crumb_C	15 C*	SOURCED-VERIFY	chaple_2023 a* 0.94, b* 15.37
crumb_h	86 deg	SOURCED-VERIFY	chaple_2023
variant	one of plain, dusted	ILLUSTRATIVE	flour-dusted or plain
slice_height_frac	0.45	TYPICAL	wikihow_slice, tastingtable_bagel: horizontally through the middle
friction_crust_rubber	U(0.5 .. 0.8)	ILLUSTRATIVE	
age_h	one of 0.5, 2, 6, 20, 30 h	ILLUSTRATIVE	time since baking (fresh .. 'vom Vortag')
storage	one of paper_bag, plastic_bag, open	ILLUSTRATIVE	oekotest_storage: paper bag recommended; plastic softens the crust
crumb_E_kPa_by_age	20 @ 5, 70 @ 48, 125 @ 96, 157 @ 168 kPa @ h	SOURCED-VERIFY	lagrain_2012 pan bread, read off figures; log-time interpolation
crumb_poisson	0.103	SOURCED	lagrain_2012
crust_water_model	w(t) = 9.73 t_h^0.112 (40 % RH, 22 C) %	CALIBRATED	power law through hirte_2012 9.0 % @ 0.5 h and 13.9 % @ 24 h; predicts 10.6 % @ 2.1 h (measured: 10.1 % critical reached at 2.1 h) → residual +0.5 points
crust_water_critical	10.1 %	SOURCED	hirte_2012: half the sensory crispness lost
plastic_bag_factor	4	ILLUSTRATIVE	crust softening clock runs x4 in a closed bag (oekotest: 'plastic softens the crust'; no rate held)
crust_break_force	3 N	SOURCED	hirte_2012: ≥ 3 N with a 1 x 10 mm blade
avrami_firmness	F = 66.01 - 59.53 exp(-0.186 t_d) N	SOURCED	cardone_2020 loaf (n=1, k per day)
reheat_oven	150-170, 5-10 C, min	TYPICAL	utopia_aufbacken
toaster_rack	2-3 + flip + 1-2 min	SOURCED-VERIFY	utopia_aufbacken
toast_times	3 light .. 4.5 'starting to burn' (sliced bread, home toaster) min	SOURCED	kaur_2026
saw_setpoint	-30 N, 0.02 m/s, ±0.02 m strokes (baguette) N, m/s, m	SOURCED	luo_2026 (robot)
crumb_squash_strain	0.3	ILLUSTRATIVE	permanent crumb compression when pressing instead of sawing
squeeze_k_model	k = 0.29 E / 20 kPa (30 mm pads across the width, 1 → 3 N) N/mm	DERIVED	Isaac FEM (test_reports/roll_squeeze_summary.json): 20 kPa lane converged (0.276 → 0.29 N/mm, 1000 → 2000 Hz); stiffer lanes under-converged → linear-elastic scaling; crust excluded (GAP)

Table 80: Register of the nantes carrot (params.CARROT, 42 entries).

Key	Value / distribution	Status	Source and note
type	Nantes (cylindrical, blunt tip, small core), loose, topped	STANDARD	eu730_1999; con22
length	N(190, 15) in [150, 230] mm	SOURCED	con22: 182.7 / 210.8 mm (Nantes)
shoulder_diameter	N(31, 2) in [25, 38] mm	SOURCED	con22: 30.3 / 31.4 mm
mass_check	110-135 (con22 Nantes); ≥ 50 g main crop (eu730_1999) g	SOURCED	
tip_bluntness	U(0.3 .. 0.9)	ILLUSTRATIVE	Nantes 'blunt tip' (catalogues)
bend_radius	U(600 .. 3000) mm	ILLUSTRATIVE	slight curvature
periderm_thickness	150 um	ILLUSTRATIVE	GAP (knowles & flore 1983 to obtain); ~3 cells of ~50 um
xylem_radius_fraction	N(0.33, 0.04) in [0.25, 0.42]	SOURCED-VERIFY	singh17 ~0.38 over mixed genotypes; Nantes bred for a small core; vob24 trend crown → tip; peng14 4-7 mm core
primary_xylem_diameter	1 mm	ILLUSTRATIVE	
petiole_stubs	integer 4 .. 9	ILLUSTRATIVE	
petiole_stub_length	U(2 .. 6) mm	ILLUSTRATIVE	topped carrots
green_shoulder_length	one of 0, 6, 12 mm	STANDARD	eu730_1999 Class I ≤ 20 mm on long roots (instances drawn inside Class I)
lateral_root_rows	4	SOURCED-VERIFY	diarch_rows
lateral_root_scar_spacing	U(8 .. 15) mm	ILLUSTRATIVE	
lenticel_band_spacing	U(2 .. 4) mm	ILLUSTRATIVE	GAP: no spacing found
lenticel_band_relief	U(0.05 .. 0.12) mm	ILLUSTRATIVE	
phloem_LCh	(55, 49, 54) L*,C*,h	SOURCED-VERIFY	pac16_dry raw flesh 55.35/28.9/40.05
xylem_LCh	(63, 45, 64) L*,C*,h	ILLUSTRATIVE	paler, yellower core (lutein; aub22) – no L*a*b*
skin_LCh	(50, 44, 52) L*,C*,h	ILLUSTRATIVE	duller than the flesh
soil_residue	U(0 .. 0.25)	ILLUSTRATIVE	washed supermarket carrots
friction_skin_rubber	U(0.5 .. 0.8)	ILLUSTRATIVE	moist skin
density	1040 kg/m3	SOURCED-VERIFY	jah18 1.04 g/cm3 (1.00-1.17)
E_phloem	N(4.6, 0.15) in [4.33, 4.85] MPa	SOURCED-VERIFY	vob24 after Stopa 2017
E_xylem	N(4.75, 0.08) in [4.62, 4.88] MPa	SOURCED-VERIFY	vob24 after Stopa 2017
poisson	0.475	SOURCED-VERIFY	vob24
E_decay_7d	1.32	SOURCED	vob24: E / 1.32 after 7 d in the fridge (zip bag)

Table 80: Register of the nantes carrot (params.CARROT, 42 entries). (continued)

Key	Value / distribution	Status	Source and note
bend_Fmax	48.6 N	SOURCED-VERIFY	jah18 three-point bending (span: see source)
bend_deflection	21.9 mm	SOURCED-VERIFY	jah18
blade_shear	41.1 N	SOURCED-VERIFY	jah18 straight blade 1.4 mm, 30 deg
impact_damage_force	43-45 N	SOURCED-VERIFY	xia21
fracture_energy	40-400 (mixed produce) J/m2	SOURCED-VERIFY	alv00
turgor_fresh	0.7 MPa	SOURCED-VERIFY	herp99_calbo00
limp_rwc	0.95	SOURCED-VERIFY	herp99_calbo00: firmness from turgor ~0 below RWC 0.95
softening_kinetics	2nd-order fractional conversion; k(80 C) 0.035, k(110 C) 1.453; Ea 138.9 kJ/mol; Finf/F0 0.02 1/min	SOURCED	peng14
peel_depth	2 mm	SOURCED	con22 hand peeler; mass 12.1-14.6 %
water_loss_rate	fridge in bag 0.3, room open 2.0 %/d	ILLUSTRATIVE	storage_loss: unwrapped 20 C unacceptable after ~3 d at 5-8 % loss → ~2 %/d; fridge GAP
age_d	U(1 .. 14) d	ILLUSTRATIVE	since purchase; hidden
storage	one of fridge_bag, fridge_open, room_open	ILLUSTRATIVE	
blush_reversible_h	1.25 h	SOURCED-VERIFY	sim09_cz95
blush_irreversible_h	15 h	SOURCED-VERIFY	sim09_cz95
boil_rounds_6mm	4-5 crisp-tender, +1-2 soft min	SOURCED-VERIFY	carrot_boil
cut_dims	rounds 5-6; julienne 3x3x45; brunoise 3; batonnet 6x6x55; dice 6/12/19 mm	TYPICAL	carrot_cuts

Table 81: Register of the cavendish banana (params.BANANA, 41 entries).

Key	Value / distribution	Status	Source and note
cultivar	Cavendish (Musa acuminata AAA; Grand Naine / Valery)	SOURCED	ciro05; mal22
length_outer	N(210, 15) in [170, 250] mm	SOURCED	convex-face length incl. pedicel (eu1333_2011 definition): 190 (mal22), 233 (ciro05)
calibre	N(35, 1.5) in [30, 40] mm	SOURCED	ciro05 34/36/34 mm; eu1333 min 27 mm
mass_check	102 (bant24), 105 (chiw23), 126-148 (polat19) g	SOURCED	
curvature_radius	U(150 .. 260) mm	ILLUSTRATIVE	'slightly curved' (mal22); GAP
ridges	one of 4, 5	ILLUSTRATIVE	'slightly ridged' (mal22), 'nearly 4-sided'; count GAP
angularity	U(0.04 .. 0.08)	ILLUSTRATIVE	ridges round out as the finger matures (fao)
peel_thickness	N(2.6, 0.25) in [2, 3.2] mm	SOURCED	mal22 2.56 mm (Grand Naine); amn12 1.1-3.8 mm, thinning on ripening
peel_thinning	0.2	ILLUSTRATIVE	fractional thinning stage 1 → 7 (amn12: thinning, no % held)
peel_layer_fractions	(0.15, 0.6, 0.25)	ILLUSTRATIVE	chlorenchyma / parenchyma / inner aerenchyma share of the peel (amn12 layer order)
pulp_peel_ratio_check	1.69-2.69 (chiw23, bant24, mal22); rises with ripening	SOURCED	
central_axis_diameter	3 mm	ILLUSTRATIVE	placental axis
stalk_length	N(17.3, 1.5) in [13, 22] mm	SOURCED	mal22 pedicel
stalk_width	N(10.15, 0.8) in [8, 12.5] mm	SOURCED	mal22 pedicel
phloem_strings	integer 3 .. 6	ILLUSTRATIVE	'strings' (guapo22); count GAP
ovules_total	integer 200 .. 300	SOURCED	promusa: 200-300 ovules
ovule_diameter	0.6 mm	ILLUSTRATIVE	tiny black specks
ripeness_stage	U(2 .. 7) 1..7	STANDARD	markon / ma22 seven-stage colour scale
peel_hue_by_stage	[118, 98.7, 97.3, 92.7, 89.4, 87.5, 82] deg	SOURCED-VERIFY	ringer21 R2..R7; stage 1 from col20 Gros Michel day 0 (h ~122) → 118
peel_L_by_stage	[48, 55, 62, 68, 72, 72, 64] L*	SOURCED-VERIFY	shape: hou15 L* peaks at stage 5 then falls; values: hega26 45-57 → 55-73 → 52-66, shahir14 max 74
peel_C_by_stage	[42, 50, 56, 60, 62, 62, 58] C*	SOURCED-VERIFY	ringer21 b* 56-72 (a* ~0 at R5 → C* ~ b*); col20 green C* ~48
pulp_LCh	(86, 24, 92) L*,C*,h	ILLUSTRATIVE	'cream' (mal22); numeric GAP
sugar_spot_density	1	ILLUSTRATIVE	stage 7 = brown flecks (markon); kinetics GAP
friction_peel_rubber	U(0.5 .. 0.8)	ILLUSTRATIVE	waxy peel
peel_density	950 kg/m3	ILLUSTRATIVE	inner aerenchyma; whole fruit SG 1.023 (arcc, SV)
pulp_density	1050 kg/m3	ILLUSTRATIVE	whole fruit SG 1.023 (arcc, SV)
pulp_E_compression	1.75 (day 1) → 1.09 (day 12) MPa	SOURCED	ciro05 cylinders 20 x 40 mm
pulp_E_bending	5.16 → 2.65 MPa	SOURCED	ciro05
pulp_firmness	38.0 → 14.0 over 10 d (2 mm probe) N	SOURCED	jaiswal12; chiw23 44.8 → 22.1
peel_puncture	13.6-15.2 (2 mm probe) N	SOURCED	ciro05
peel_fracture	53.0 → 2.1 on ripening N	SOURCED-VERIFY	khanal23 (test geometry not in abstract)
whole_fracture	129 bending (14 cm span), 121 compression N	SOURCED	ciro05
finger_drop_force	~49 on day 1, > 2x lower by day 3 N	SOURCED-VERIFY	imsabai; mbeguie09
bruise_threshold	200 mJ	SOURCED-VERIFY	bugaud14: no visible bruise up to 200 mJ (Grande Naine)

Table 81: Register of the cavendish banana (params.BANANA, 41 entries). (continued)

Key	Value / distribution	Status	Source and note
peel_bond_ratios	(0.45, 0.3, 1.5)	ILLUSTRATIVE	peel-to-pulp adhesion / ridge seam / strip tear strength as multiples of the peel fracture force (khanal23 53 → 2.1 N, SOURCED-VERIFY); adhesion > seam so the peel splits along the ridges (khanal23) instead of sliding off as a sleeve; GAP: no peeling-force study
stage_rate_20C	N(0.6, 0.15) in [0.3, 1] stage/d	SOURCED-VERIFY	hou15: 1 → 7 in 7 d with ethylene (0.86/d); ma22 day bands ~0.4/d; 18 C RS6 at 9 d
q10_ripening	2	ILLUSTRATIVE	temperature dependence of the stage rate
chilling_threshold	13 C	SOURCED	wflo
brix_by_day	4.5 / 6.8 / 8 / 16.7 / 23.7 at day 1 / 4 / 6 / 8 / 10 Bx	SOURCED	chiw23
mass_loss_rate	1 %/d	SOURCED-VERIFY	chiw23 ~2.2 %/d (22-27 C, outlier-looking); kojima96 ~0
rounds_thickness	6 mm	TYPICAL	banana_howto

Table 82: Register of the sweet orange (params.ORANGE, 38 entries).

Key	Value / distribution	Status	Source and note
cultivar	one of navel, valencia	ILLUSTRATIVE	German supermarket: navel (winter) / Valencia (summer)
diameter	N(75, 6) in [61, 88] mm	SOURCED	ucr_citrus Washington navel 61-88 mm
shape_index	U(1 .. 1.08)	SOURCED	ucr_citrus L/W 1.0-1.1
mass_check	162-335 (navel), 113-299 (Valencia) g	SOURCED	ucr_citrus
peel_thickness	N(5, 0.6) in [3.5, 6.4] mm	SOURCED	ucr_citrus rind 4.1-6.4 / 3.5-6.2 mm
flavedo_thickness	U(1 .. 1.6) mm	ILLUSTRATIVE	peel minus albedo (2-5 mm, citrus_patents; 3.9 mm smith18)
core_diameter	U(5 .. 9) mm	ILLUSTRATIVE	central axis; GAP
segments	one of 9, 10, 11, 12	SOURCED-VERIFY	citrus_patents 9.6-11
membrane_thickness	0.12 mm	ILLUSTRATIVE	endocarp film; GAP
seeds	integer 1 .. 6	SOURCED	ucr_citrus Valencia 1.2-6.6; navel ~0 (set in the model)
seed_length	9 mm	TYPICAL	
oil_gland_density	50 1/cm2	SOURCED-VERIFY	knight01 8,000-12,000 per fruit → ~40-60 /cm2 (derived); one cultivar 208 /cm2 (citrus_patents)
peel_density	746 kg/m3	SOURCED	jentzsch22
albedo_density	700 kg/m3	ILLUSTRATIVE	spongy with air spaces (sadka19)
pulp_density	1045 kg/m3	DERIVED	juice SG 1.045 (orange_juice)
peel_LCh	(60, 74, 57) L*,C*,h	ILLUSTRATIVE	chosen for CCI ~ +11 (cci market target +10..+20); measured L*a*b* GAP
albedo_LCh	(92, 10, 90) L*,C*,h	ILLUSTRATIVE	white; GAP
pulp_LCh	(66, 62, 66) L*,C*,h	ILLUSTRATIVE	'medium orange' (ucr_citrus); GAP
membrane_LCh	(84, 28, 80) L*,C*,h	ILLUSTRATIVE	translucent pale; GAP
friction_peel_rubber	U(0.5 .. 0.8)	ILLUSTRATIVE	waxy, pitted peel
flavedo_E	103 MPa	SOURCED	smith18 (30-300 across species)
albedo_E	0.79 MPa	SOURCED	smith18 (±0.58)
peel_compressive_E	0.02 MPa	SOURCED-VERIFY	jentzsch22 whole peel, compression
peel_poisson	-0.05	SOURCED	jentzsch22
punch_force	15.5-79 (4.3-14.3 mm punches) N	SOURCED-VERIFY	orange_mech
rupture_force	140-221 (Valencia) N	SOURCED-VERIFY	orange_mech
gland_damage_force	10-250 (sphere; oil release) N	SOURCED-VERIFY	orange_mech
oil_jet	8.5 ± 4.0, 102 ± 20 m/s, um	SOURCED	smith18
gland_burst_pressure	30-140 kPa	SOURCED	smith18
juice_fraction	U(45 .. 54) %	SOURCED	ucr_citrus navel 45-54 (Valencia 37-56)
juice_by_hand	60-80 per medium orange mL	TYPICAL	orange_juice
brix	U(9.3 .. 14.1) Bx	SOURCED	ucr_citrus
juice_viscosity	5 mPa s	SOURCED-VERIFY	orange_juice 4-6 at 8 C
peelability	navel medium-easy; Valencia hard (strips break); cold fruit harder	SOURCED	ritenour13 (no N values)
peel_strip_force	navel 4, Valencia 8 per 16 mm strip N	ILLUSTRATIVE	ranking from ritenour13; GAP
storage	3-8, 90-95, ≤ 3 months; chilling after 3-6 weeks at ~4 C C, %RH	SOURCED-VERIFY	orange_storage
bitter_juice_h	3 h	SOURCED-VERIFY	edis891 'within several hours' (navel)
age_d	U(2 .. 20) d	ILLUSTRATIVE	since purchase

Table 83: Register of the sandwich loaf (toastbrot) (params.LOAF, 26 entries).

Key	Value / distribution	Status	Source and note
product	Toastbrot (Buttertoast), lidded tin, pre-sliced, 500 g	SOURCED	test25_toast; toast_pack
length	N(250, 6) in [235, 265] mm	SOURCED-VERIFY	toast_pack ~0.25 m
width	N(100, 3) in [92, 110] mm	SOURCED-VERIFY	tins 100 / 110 mm
tin_height	N(98, 3) in [88, 105] mm	SOURCED-VERIFY	square slices: 25 g x 0.19 g/cm3 / 12.5 mm → ~100 x 100 mm (derived; tins 80-90 mm deep)
dome_height	0 mm	SOURCED	test25_toast: baked under a lid (flat top)

Table 83: Register of the sandwich loaf (toastbrot) (params.LOAF, 26 entries). (continued)

Key	Value / distribution	Status	Source and note
edge_radius	U(8 .. 12) mm	ILLUSTRATIVE	rounded tin corners
slices	20	SOURCED-VERIFY	toast_pack
heels	one of 0, 2	SOURCED	test25_toast: most packs no heels; organic 2
crust_thickness_top	U(0.8 .. 1.3) mm	ILLUSTRATIVE	thin soft lidded crust; GAP
crust_thickness_tin	U(0.5 .. 0.9) mm	ILLUSTRATIVE	GAP
crumb_density	N(190, 10) in [170, 215] kg/m3	SOURCED	nuss90 0.19 g/cm3
crust_density	300 kg/m3	ILLUSTRATIVE	
cell_diameter_median	0.5 mm	ILLUSTRATIVE	finer, more even than a roll; C-Cell values GAP
cell_number_density	2500 1/cm3	ILLUSTRATIVE	
cell_elongation	1.15	ILLUSTRATIVE	
crumb_LCh	(74.7, 15.5, 88.6) L*,C*,h	SOURCED-VERIFY	ishida14 74.73 / 0.37 / 15.51
crust_LCh	(62, 34, 70) L*,C*,h	ILLUSTRATIVE	pale lidded crust; tin loaves bottom out at L* ~64.5 at 200 C (giovanelli_2021, derived)
friction_bread	U(0.6 .. 0.9)	ILLUSTRATIVE	
firmness_by_day	1.98 / 3.62 / 4.18 / 5.64 at day 1 / 5 / 9 / 13 (25 C, bagged) N	SOURCED	sehn20 lab white pan loaf
tensile_failure	0.31-0.39 (12 mm neck) at 9.2-9.7 mm N	SOURCED	nuss90
best_before_d	12-20 d	SOURCED	test25_toast
fridge_staling_factor	6	TYPICAL	toast_misc (popular, weak)
toast_time	80 .. > 150 to 'best' (per toaster) s	SOURCED	toaster26
butter_load	0.068 g/cm2	SOURCED-VERIFY	toast_misc
age_d	U(0.5 .. 8) d	ILLUSTRATIVE	since baking
storage	one of bag_closed, bag_open, fridge	ILLUSTRATIVE	

Table 84: Register of the potato (params.POTATO, 36 entries).

Key	Value / distribution	Status	Source and note
cultivar	one of Linda, Belana, Gala	STANDARD	bsl21
cooking_type	one of festkochend, vorwiegend festkochend	SOURCED-VERIFY	Belana / Linda festkochend, Gala vorwiegend festkochend
length	N(85, 10) in [60, 110] mm	SOURCED-VERIFY	75 / 65 / 99 mm (Astrix / Diamont / Santana)
width	N(55, 4) in [45, 65] mm	SOURCED-VERIFY	48 / 45 / 58 mm
thickness_ratio	U(0.8 .. 0.92)	SOURCED-VERIFY	thickness / width 0.81-0.89
mass_check	64-167 (examples); min square mesh 35 / 30 mm (repealed HKV) g	SOURCED-VERIFY	
lumpiness	U(0.02 .. 0.05)	ILLUSTRATIVE	
eyes	integer 8 .. 14	SOURCED-VERIFY	Norgold 10-12 buds; 6-9 apical (mag25)
eye_crowding	1.8	ILLUSTRATIVE	apical bud density ~7x lateral (mag25)
phyllotaxis	0.4 turn	ILLUSTRATIVE	spiral arrangement sourced; the 2/5 fraction is a GAP
eye_depth	U(0.6 .. 1.4) mm	ILLUSTRATIVE	bsl21 'flach' (ordinal)
eye_width	U(3 .. 4) mm	ILLUSTRATIVE	
periderm_thickness	N(65, 10) in [46, 84] um	SOURCED	mc20 phellem
lenticel_density	U(5 .. 12) 1/cm2	SOURCED	mag25
vascular_ring_depth	U(3 .. 6) mm	ILLUSTRATIVE	GAP (Reeve 1969 paywalled)
pith_fraction	U(0.35 .. 0.5)	ILLUSTRATIVE	GAP
pith_branch_width	U(1.5 .. 2.5) mm	ILLUSTRATIVE	star of the pith; GAP
density	N(1070, 12) in [1050, 1100] kg/m3	SOURCED-VERIFY	1053 (kk18); SG 1.070-1.103
skin_LCh	(62, 30, 75) L*,C*,h	ILLUSTRATIVE	yellow skin, L* > 55 typical (SV); GAP
flesh_LCh	(80, 30, 92) L*,C*,h	ILLUSTRATIVE	medium-yellow (bsl21), L* > 75 (SV)
soil_residue	U(0 .. 0.3)	ILLUSTRATIVE	
friction_skin_rubber	U(0.5 .. 0.8)	ILLUSTRATIVE	
E	U(3.1 .. 5.6) MPa	SOURCED-VERIFY	potato_mech
poisson	0.45	SOURCED-VERIFY	potato_mech (Zhao 2023)
fracture_stress	1.13-1.40 MPa	SOURCED-VERIFY	potato_mech
thermal	k 0.55, cp 3.43, alpha 0.15e-6 W/mK, kJ/kgK, m2/s	SOURCED	kk18
gelatinisation	onset 61, peak 67.5, end 76-82 C	SOURCED-VERIFY	potato_cook
softening	first order, k(100 C) 0.28-0.40, Ea 103-145 1/min, kJ/mol	SOURCED	potato_cook (Harada)
k100	U(0.28 .. 0.4) 1/min	SOURCED	potato_cook
Ea	133 kJ/mol	SOURCED	potato_cook (Harada 133-145)
boil_times	diced 15-20, large whole up to 60, whole 20-25 (cold start) min	SOURCED	gritzer
peel_depth	1.1 mm	SOURCED-VERIFY	potato_peel (~9 % waste)
storage_temp	7-13 dark; < 8 sweetens C	SOURCED-VERIFY	potato_storage
greening_days_20C	1.5 d	SOURCED-VERIFY	potato_storage 1-2.5 d under retail light
age_d	U(2 .. 30) d	ILLUSTRATIVE	
light_exposure_h	one of 0, 24, 48 h	ILLUSTRATIVE	hours under light

Table 85: Register of the round / truss tomato (params.TOMATO, 37 entries).

Key	Value / distribution	Status	Source and note
type	round / truss tomato, 80-120 g	STANDARD	fdc_tomato
diameter	N(63, 4) in [52, 76] mm	SOURCED	fdc_tomato 61-66 mm for 91-123 g
height_ratio	U(0.75 .. 0.9)	SOURCED-VERIFY	ince16 0.75; GRIN 'slightly flattened'
mass_check	91 / 123 (small / medium) g	SOURCED	fdc_tomato

Table 85: Register of the round / truss tomato (params.TOMATO, 37 entries). (continued)

Key	Value / distribution	Status	Source and note
locules	one of 2, 3, 4	SOURCED	ucd_rost: small types 2, globe 3-5
skin_thickness	U(120 .. 250) um	ILLUSTRATIVE	cuticle 9.9-15.7 um (bn05) + epidermis + collenchymatous hypodermis; exocarp 9.6 % FW (meta19)
pericarp_thickness	N(6.5, 0.8) in [5, 9] mm	SOURCED-VERIFY	meta19 5.3 mm (small); 7-12 germplasm
septum_thickness	U(2.5 .. 3.5) mm	ILLUSTRATIVE	
columella_diameter	U(10 .. 16) mm	ILLUSTRATIVE	columella + placenta 10 % FW (meta19)
placenta_width	U(3 .. 5) mm	ILLUSTRATIVE	
seeds_per_fruit	integer 50 .. 120	SOURCED-VERIFY	68.8 (one cultivar)
seed_length	3.5 mm	ILLUSTRATIVE	GAP
rib_depth	U(0.01 .. 0.03)	ILLUSTRATIVE	
sepals	one of 5, 6	ILLUSTRATIVE	GAP
sepal_length	U(12 .. 20) mm	ILLUSTRATIVE	
pedicel_length	one of 0, 12, 15 mm	ILLUSTRATIVE	truss tomatoes keep calyx and pedicel
ripeness_stage	U(3.5 .. 6) 1..6	STANDARD	cfr_tomato colour classes
skin_hue_by_stage	[118, 110, 96, 71, 52, 43] deg	DERIVED	h = atan2(1, a*/b*) at the band centres of ince16; ripe ~40 deg (Radzevicius 2009, SV)
skin_L_by_stage	[56, 56, 53, 49, 45, 42] L*	ILLUSTRATIVE	L* falls on ripening (rad08); GAP
skin_C_by_stage	[38, 36, 40, 50, 60, 66] C*	ILLUSTRATIVE	GAP
flesh_LCh_green	(70, 30, 110) L*,C*,h	ILLUSTRATIVE	GAP
flesh_LCh_breaker	(80, 22, 85) L*,C*,h	ILLUSTRATIVE	pale interior at breaker; GAP
flesh_LCh_red	(48, 58, 36) L*,C*,h	ILLUSTRATIVE	GAP
flesh_density	1000 kg/m3	ILLUSTRATIVE	GAP (cherry 1194, suspect)
gel_density	1010 kg/m3	ILLUSTRATIVE	
friction_skin_rubber	U(0.4 .. 0.7)	ILLUSTRATIVE	smooth waxy skin
plate_firmness	initial G/P/R 15.1 / 6.0 / 4.5; average 13.2 / 4.7 / 3.1 N/mm	SOURCED-VERIFY	ince16
skin_rupture_2mm	G/P/R 6.92 / 3.99 / 2.74 N	SOURCED-VERIFY	ince16
skin_E	252-293 (red); failure ≤ 25-28 MPa at 9-10 % MPa	SOURCED	bn05
damage_strain	0.03	SOURCED-VERIFY	tomato_mech (Li 2017 FEM)
bio_yield_force	2.57 N	SOURCED-VERIFY	tomato_mech
stage_rate_20C	U(0.35 .. 0.7) stage/d	SOURCED-VERIFY	breaker → full colour 7-14 d (patent)
chilling_threshold	12.5 C	SOURCED	wflo_tomato 12.5-12.8
fridge_volatile_loss	65 after 8 d at 5 C %	SOURCED	zhang16
water	94.5 %	SOURCED	fdc_tomato
slice_thickness	6-13 mm	SOURCED	tomato_howto
blanch_s	30-45 (20-60) s	SOURCED	tomato_howto

Table 86: Register of the yellow onion (params.ONION, 32 entries).

Key	Value / distribution	Status	Source and note
type	yellow storage onion (Haushaltszwiebel), cured	STANDARD	hb66_onion
diameter	N(65, 7) in [50, 85] mm	SOURCED	hb66_onion medium 51-83 mm; bah04 51.7
shape_index	U(0.85 .. 1.1)	SOURCED-VERIFY	height / diameter: bah04 57.1 / 51.7; flat-round German types
mass_check	78.7 ± 24 (51.7 mm) g	SOURCED-VERIFY	bah04
n_fleshy	one of 7, 8	SOURCED-VERIFY	kim_heb20 7.6 ± 0.2
scale_thickness_outer	U(3.5 .. 4.5) mm	ILLUSTRATIVE	kim_heb20 5.5 mm (bulk mean), wu25 ~3 mm; constrained by count x radius
scale_thickness_inner	U(2 .. 2.8) mm	ILLUSTRATIVE	inner scales thinner; GAP
scale_gap	0.05 mm	ILLUSTRATIVE	epidermis interface between scales
n_dry	one of 2, 3	ILLUSTRATIVE	GAP (ordinal skin-thickness classes only)
dry_thickness	U(0.08 .. 0.15) mm	ILLUSTRATIVE	GAP
plate_diameter	U(15 .. 22) mm	ILLUSTRATIVE	GAP
plate_thickness	U(4 .. 6) mm	ILLUSTRATIVE	GAP
n_roots	integer 15 .. 30	ILLUSTRATIVE	
root_length	U(3 .. 10) mm	ILLUSTRATIVE	trimmed root crown
neck_length	U(5 .. 20) mm	ILLUSTRATIVE	
neck_diameter	U(7.4 .. 10.9) mm	SOURCED-VERIFY	onion_misc (cured neck, Indian cultivars)
flesh_density	1040 kg/m3	SOURCED-VERIFY	bah04 bulb 1.04 g/cm3
dry_density	400 kg/m3	ILLUSTRATIVE	papery skin
plate_density	1050 kg/m3	ILLUSTRATIVE	
skin_LCh	(52, 45, 65) L*,C*,h	ILLUSTRATIVE	golden brown; GAP
flesh_LCh	(88, 12, 95) L*,C*,h	ILLUSTRATIVE	white to pale yellow; GAP
friction_skin_rubber	U(0.6 .. 1)	SOURCED-VERIFY	bah04 0.67-1.34
bulb_E	2.03-5.45, nu 0.26-0.45 MPa	SOURCED-VERIFY	jafari16
tissue_E	2.2 skin on / 1.4 off MPa	SOURCED	wu25
epidermis_E	3.5-8.0 MPa	SOURCED	wei01
crushing_load	400-780 N	SOURCED-VERIFY	bah04
rolling_angle	20-31 on base, 14-23 on side deg	SOURCED-VERIFY	bah04
lf_window	peak ~1, gone ~10 min	SOURCED	lokke12
storage	0, 65-75, 6-9 months; household 3-4 weeks C, %RH	SOURCED	hb66_onion
dice_spacing	mince ≤3, fine 3-6, medium 6-13, coarse 13-19 mm	SOURCED	onion_howto
caramelise_min	30-45 min	TYPICAL	onion_misc
age_d	U(3 .. 40) d	ILLUSTRATIVE	

Table 87: Register of the lemon (params.LEMON, 31 entries).

Key	Value / distribution	Status	Source and note
cultivar	one of Primofiori, Verna	SOURCED	agu20
diameter	N(61, 3) in [55, 67] mm	SOURCED	agu20 55-67 mm
shape_index	U(1.3 .. 1.55)	SOURCED	length / diameter from agu20 ED/PD 0.6-0.8 (body without nipple)
nipple_height	U(3 .. 7) mm	ILLUSTRATIVE	stylar mammilla; GAP
neck_height	U(1.5 .. 4) mm	ILLUSTRATIVE	stem-end neck; GAP
mass_check	100-184 (agu20); 166 (jen22_lemon) g	SOURCED	
peel_thickness	N(5.6, 0.5) in [4.9, 7.1] mm	SOURCED	agu20, jen22_lemon
flavedo_thickness	U(1 .. 1.4) mm	SOURCED-VERIFY	jen22_lemon ~1 mm (ben26: ~46 % of peel)
core_diameter	U(4 .. 7) mm	ILLUSTRATIVE	GAP
segments	one of 9, 10	SOURCED	agu20 8.8-10.2
membrane_thickness	0.1 mm	ILLUSTRATIVE	GAP
seeds	integer 1 .. 8	SOURCED	agu20 1.3-11.3, lemon_misc 1.3-2.3
seed_length	8 mm	TYPICAL	
oil_gland_density	125 1/cm2	SOURCED	ben26 1.25 /mm2
peel_density	818 kg/m3	SOURCED	jen22_lemon
albedo_density	390 kg/m3	SOURCED	ben26
pulp_density	1030 kg/m3	DERIVED	juice ~1.03 g/mL from USDA portion weights (lemon_misc)
peel_LCh	(66, 46, 93) L*,C*,h	SOURCED	agu20 L* 64-68, b* 42-49, hue 88-99
albedo_LCh	(92, 12, 95) L*,C*,h	ILLUSTRATIVE	GAP
pulp_LCh	(82, 35, 98) L*,C*,h	ILLUSTRATIVE	pale yellow; GAP
membrane_LCh	(88, 18, 95) L*,C*,h	ILLUSTRATIVE	GAP
friction_peel_rubber	U(0.5 .. 0.8)	ILLUSTRATIVE	
juice_fraction	U(30 .. 45) %	SOURCED	agu20 31.7-44.3 (EU min 20)
juice_of_one_lemon	48 g	SOURCED	lemon_misc (USDA)
acidity	56-75 g/L citric	SOURCED	agu20
puncture_force	41.2 N	SOURCED	ben26 5 mm probe
cutting_force	36.3 N	SOURCED	ben26 craft knife
compression_force	165.8 at 16 mm N	SOURCED	ben26
chilling_threshold	10 C	SOURCED	lemon_misc (Eureka)
wedge_mass	5.9 g	SOURCED	lemon_misc (USDA)
bitter_juice_h	1e+06 h	ILLUSTRATIVE	lemon juice does not turn bitter like navel juice

Table 88: Register of the greenhouse cucumber (params.CUCUMBER, 28 entries).

Key	Value / distribution	Status	Source and note
type	greenhouse long cucumber (Schlangengurke), parthenocarpic	STANDARD	dewiki_sg
length	N(320, 25) in [250, 380] mm	SOURCED	unece_cuc 25-30 cm minimum; sakata11
diameter	N(42, 3) in [36, 50] mm	TYPICAL	derived: 400 g / 32 cm / 950 kg/m3 → ~41 mm
mass_check	300-500 (dewiki_sg), typical 350 (cuc_misc) g	SOURCED	
neck_ratio	U(0.8 .. 0.9)	ILLUSTRATIVE	stem end narrower
ribs	3	ILLUSTRATIVE	faint ribs over the three carpels
rib_depth	U(0.005 .. 0.012)	ILLUSTRATIVE	
bend_radius	U(1000 .. 4000) mm	ILLUSTRATIVE	within Class I (≥ ~130 mm, unece_cuc)
skin_thickness	U(1 .. 1.6) mm	SOURCED-VERIFY	cuc_misc peelable exocarp 1-2 mm
cavity_fraction	U(0.45 .. 0.55)	ILLUSTRATIVE	seed cavity / fruit radius; GAP
core_diameter	U(3 .. 5) mm	ILLUSTRATIVE	
seeds_per_row	integer 25 .. 40	ILLUSTRATIVE	parthenocarpic: soft immature ovules
seed_length	U(4 .. 6) mm	ILLUSTRATIVE	< mature 6.4-9.1 mm
density	950 kg/m3	SOURCED-VERIFY	disect_cuc; water 95.2 % (USDA)
skin_LCh	(38, 34, 127) L*,C*,h	SOURCED-VERIFY	cuc_misc a* -20.6, b* 27.5 → C 34, h 127; L* GAP
flesh_LCh	(85, 20, 112) L*,C*,h	ILLUSTRATIVE	pale whitish green; GAP
friction_skin_rubber	U(0.4 .. 0.7)	ILLUSTRATIVE	glossy waxy skin
round_thickness	5 mm	SOURCED	disect_cuc (SliceIt! 5 mm slices)
puncture_3mm	flesh 8.7, placenta 2.8, skin 9.7 N	SOURCED	sakata11
E_apparent	0.85-1.39 (slice) | 2.5 (DiSECt) MPa	SOURCED-VERIFY	jah16; disect_cuc
failure	0.53-0.85 at 0.45-0.53 strain MPa, 1	SOURCED-VERIFY	jah16
firmness_loss_9d	49 %	SOURCED-VERIFY	jah16
storage_optimum	10-12.5, 95, < 14 d C, %RH	SOURCED	ucd_cuc
chilling	< 10 C after 1-3 d C, d	SOURCED	ucd_cuc
weight_loss_15d	wrapped 0.66, unwrapped 11.1 (12 C) %	SOURCED-VERIFY	cuc_misc
salting_min	30 min	SOURCED	cuc_misc (wikibooks)
age_d	U(1 .. 10) d	ILLUSTRATIVE	
storage	one of fridge_wrapped, counter_wrapped, fridge_unwrapped	ILLUSTRATIVE	home fridge 4 C is in the chilling range

Table 89: Register of the butter croissant (params.CROISSANT, 21 entries).

Key	Value / distribution	Status	Source and note
product	Buttercroissant (butter the only fat)	STANDARD	leitsaetze_fb
length	N(135, 8) in [115, 160] mm	ILLUSTRATIVE	finished dimensions GAP; chosen for ~180 cm3 = 57 g x 3.13 cm3/g (ste25, USDA medium)

Table 89: Register of the butter croissant (params.CROISSANT, 21 entries). (continued)

Key	Value / distribution	Status	Source and note
width	N(68, 5) in [55, 80] mm	ILLUSTRATIVE	see length
height	N(42, 4) in [34, 50] mm	ILLUSTRATIVE	see length
mass_check	42 / 57 / 67 (small / medium / large) g	SOURCED	croissant_misc (USDA)
curvature_radius	one of 70, 90, 100000 mm	ILLUSTRATIVE	crescent or straight (fr: butter = straight, SV)
steps	integer 5 .. 7	ILLUSTRATIVE	visible rolls; rolled 3.5-4 turns (croissant_misc)
step_relief	U(0.05 .. 0.1)	ILLUSTRATIVE	
rolling_turns	U(3.5 .. 4)	SOURCED	croissant_misc
butter_layers	27	SOURCED	ste25: 3 single folds
butter_on_flour	U(50 .. 64) %	SOURCED	ste25, recipes; legal min 30 (leitsaetze_fb)
crust_thickness	U(0.8 .. 1.4) mm	ILLUSTRATIVE	GAP
crust_density	450 kg/m3	ILLUSTRATIVE	
crumb_density	300 kg/m3	DERIVED	1 / 3.13 cm3/g (ste25) = 0.32 g/cm3 whole
crust_LCh	(52, 48, 66) L*,C*,h	ILLUSTRATIVE	deep golden, egg-washed; GAP
crumb_LCh	(84, 26, 88) L*,C*,h	ILLUSTRATIVE	pale yellow (butter); GAP
flakes	integer 40 .. 120	ILLUSTRATIVE	loose flakes; GAP
friction_crust	U(0.4 .. 0.7)	ILLUSTRATIVE	
firmness	5.16 (1 h) → 19.2 (24 h), TPA 25 mm 50 % N	SOURCED	ste25
reheat	180, 8 from frozen; refresh 160-180, 3-5 (typical) C, min	SOURCED	croissant_misc
age_h	one of 1, 3, 8, 24 h	ILLUSTRATIVE	

Table 90: Register of the baguette (params.BAGUETTE, 42 entries).

Key	Value / distribution	Status	Source and note
product	baguette de tradition francaise (flour, water, yeast/levain, salt)	STANDARD	wp_baguette
length	N(600, 30) in [540, 660] mm	SOURCED	wp_baguette 55-65 cm
width	N(55, 4) in [45, 62] mm	SOURCED	wp_baguette 4-6 cm
height	N(42, 4) in [32, 50] mm	SOURCED	wp_baguette 3-5 cm
mass_check	~250 (250-300 tradition) g	SOURCED	wp_baguette
base_edge_height	U(4 .. 7) mm	ILLUSTRATIVE	
lumpiness	U(0.01 .. 0.025)	ILLUSTRATIVE	
n_scores	one of 5, 6, 7	SOURCED	wp_grigne: 5 classic; 5-7 typical for 55-65 cm
score_length_mm	U(100 .. 125) mm	TYPICAL	~10-12 cm, overlapping ~1/3
score_axis_angle	U(10 .. 20) deg	TYPICAL	
score_length_frac	1	ILLUSTRATIVE	per score: set by score_length_mm
score_width	U(9 .. 14) mm	ILLUSTRATIVE	opened grigne
score_depth	U(5 .. 9) mm	SOURCED-VERIFY	wp_grigne ~1 cm cut
crust_thickness_top	N(1.9, 0.3) in [1.4, 3.5] mm	SOURCED	dv2012 1.7 mean (zou2024: 3-4, SV)
crust_thickness_score	U(0.8 .. 1.2) mm	ILLUSTRATIVE	
crust_thickness_bottom	U(1.6 .. 2.4) mm	ILLUSTRATIVE	
transition_thickness	U(1 .. 2) mm	ILLUSTRATIVE	
crumb_density	N(200, 20) in [160, 250] kg/m3	DERIVED	~250 g / ~0.9 L bulk (dimensions, wp_baguette) minus a denser crust
crust_density	380 kg/m3	ILLUSTRATIVE	
transition_density	260 kg/m3	ILLUSTRATIVE	
crumb_porosity	U(0.75 .. 0.85)	ILLUSTRATIVE	open irregular crumb; tomography GAP
cell_diameter_median	U(1.3 .. 1.9) mm	ILLUSTRATIVE	'irregular voids of various sizes'
cell_diameter_gsd_log	0.75	ILLUSTRATIVE	wide spread: open crumb
cell_number_density	260 1/cm3	ILLUSTRATIVE	
cell_elongation	U(1.1 .. 1.3)	ILLUSTRATIVE	
crust_L	N(52, 3) in [44, 58] L*	SOURCED-VERIFY	zou2024 powder 57.5; intact darker
crust_C	36.4 C*	SOURCED-VERIFY	zou2024 a* 10.5, b* 34.9
crust_h	73.3 deg	SOURCED-VERIFY	zou2024
crumb_L	70 L*	SOURCED-VERIFY	zou2024 66.4 (powder)
crumb_C	15.8 C*	SOURCED-VERIFY	zou2024
crumb_h	96 deg	SOURCED-VERIFY	zou2024
variant	one of plain	STANDARD	
slice_height_frac	0.42	TYPICAL	split lengthwise for sandwiches
friction_crust_rubber	U(0.5 .. 0.8)	ILLUSTRATIVE	
age_h	one of 0.5, 2, 6, 20 h	ILLUSTRATIVE	
storage	one of paper_bag, open	ILLUSTRATIVE	baguette_misc
crumb_poisson	0.103	SOURCED	lagrain_2012 (bread)
crust_water_critical	10.1 %	SOURCED	hirte_2012
plastic_bag_factor	4	ILLUSTRATIVE	
crust_break_force	3 N	SOURCED	hirte_2012
crumb_squash_strain	0.3	ILLUSTRATIVE	
bias_slice_mm	12 mm	TYPICAL	crostini 1-1.5 cm at 30-45 deg

Table 91: Register of the bell pepper (params.PEPPER, 27 entries).

Key	Value / distribution	Status	Source and note
type	blocky sweet pepper (Paprika)	STANDARD	pepper_misc (EU 2023/2429)
colour	one of red, yellow, green	ILLUSTRATIVE	retail mix
diameter	N(78, 4) in [68, 90] mm	SOURCED	he22 77.6 mm
length	N(83, 6) in [70, 100] mm	SOURCED	he22 81.3-85.3 mm

Table 91: Register of the bell pepper (params.PEPPER, 27 entries). (continued)

Key	Value / distribution	Status	Source and note
mass_check	237 ± 37 (he22), 230-245 German retail (alt20) g	SOURCED	
lobes	one of 3, 4	SOURCED	sha26: 2-4, mostly 3
lobe_depth	U(0.04 .. 0.08)	ILLUSTRATIVE	
wall_thickness	N(7.2, 0.6) in [5.5, 8.5] mm	SOURCED	he22 7.5 red / 6.9 orange
wall_density	1000 kg/m3	TYPICAL	92-94 % water
rib_depth	U(6 .. 12) mm	ILLUSTRATIVE	GAP
rib_thickness	U(1.2 .. 2) mm	ILLUSTRATIVE	GAP
placenta_diameter	U(20 .. 28) mm	ILLUSTRATIVE	GAP
placenta_height	U(22 .. 32) mm	ILLUSTRATIVE	GAP
seeds	integer 40 .. 140	SOURCED-VERIFY	pepper_misc 38-142
seed_diameter	3.5 mm	ILLUSTRATIVE	1000-seed weight 4-6 g (sha26); size GAP
stem_length	U(10 .. 25) mm	ILLUSTRATIVE	cut peduncle (EU: neatly cut)
red_LCh	(38.8, 41.9, 41.6) L*,C*,h	SOURCED	bar20
yellow_LCh	(61.4, 50.2, 84.3) L*,C*,h	SOURCED	bar20
green_LCh	(33.7, 19, 116.9) L*,C*,h	SOURCED	bar20
friction_skin_rubber	U(0.35 .. 0.6)	ILLUSTRATIVE	glossy waxy skin; robots report slip
firmness_whole	43.5 N	SOURCED	pepper_misc (8 mm probe)
bite_firmness	red 11.6, orange 6.9; -81 % in 15 d at 20 C N	SOURCED	he22
chilling_threshold	7 C	SOURCED	pepper_misc
weight_loss	2.9-4.8 at 17 C / 55 % RH; shrivel from 6-8 % loss g/d	SOURCED	alt20
shelf_life_d	12-28 d	SOURCED	bar20
ring_mass	10 g	SOURCED	pepper_misc (USDA)
age_d	U(1 .. 10) d	ILLUSTRATIVE	

B Sources

Table 92: Sources held in food/params.py (SOURCES).

Key	Source
narushin2021	Narushin V.G., Romanov M.N., Griffin D.K. (2021) Egg and math: introducing a universal formula for egg shape. Ann. N.Y. Acad. Sci. 1505:169-177 (Hügelschäffer egg model).
eu589_2008	Commission Regulation (EC) No 589/2008, marketing standards for eggs: weight grades S<53, M 53-63, L 63-73, XL≥73 g; Class A air space height ≤6 mm, 'extra' ≤4 mm.
icf11_eggshell	ICF11 conference paper on eggshell fracture, gruppofrattura.it/ocs/index.php/ICF/ICF11/paper/download/10244/9618: failure at 34±6 N (equatorial compression) and 41±8 N (axial, across the poles). Species to verify in the full text.
nedomova2004	Hen-egg quasi-static compression, Acta Univ. Agric. Silvic. Mendel. Brun. 2004 (actavia.mendelu.cz/pdfs/acu/2004/05/07.pdf): model with shell thickness 0.35 mm and Young's modulus 15,000 MPa.
reissner1946	Reissner E. (1946) Stresses and small displacements of shallow spherical shells. J. Math. Phys. 25: point-load stiffness of a shallow spherical shell k = 4 E t^2 / (R sqrt(3(1-nu^2))).
apple_E_quasistatic	Quasi-static equivalent Young's modulus of apple flesh: Jonagold 4.24±0.96, Joly Red 5.09±1.27, Kanzi 7.82±0.41 MPa (impact study, found via researchportal.be/en/node/4347597; citation to verify).
tjaf2005	Turkish J. Agric. For. 29(6) 2005 (tar-29-6-6-0411-8): apple flesh modulus falls with storage, 1.68→0.51 MPa (Williams), 1.45→0.88 (Starkspur Golden Delicious), 1.51→1.1 (Starking).
fsct2008	J. Food Sci. Technol. (Iran) 5(18):37-44 (2008): bruise volume at 410 and 1210 mJ impacts; Golden Delicious about 1 % of fruit volume at 410 mJ; bruise volume rises with impact energy.
hertz	Hertz contact theory (Johnson, Contact Mechanics, 1985): sphere-on-flat force, pressure, impact.
avo_au_stages	Avocados Australia, 'Avocado Stages of Ripeness chart for Hass' (2017): stages 1 (hard) .. 5 (ripe/eat now); penetrometer (11.1 mm tip, 8 mm into skinless flesh) >10 kgf at stage 1, 0.4-0.6 kgf at 'eat now'; firmness is a better ripeness index than skin colour.
avo_firmness_12d	Hass flesh firmness falling from 75.43 to 2.63 N over 12 days of ripening (reported in a ripening study surfaced by literature search; citation to verify).
wiki_apple	https://en.wikipedia.org/wiki/Apple and https://en.wikipedia.org/wiki/Pome: flesh from the receptacle/hypanthium; usually 5 carpels with 1-2 seeds; endocarp = leathery core case.
wiki_gala	https://en.wikipedia.org/wiki/Gala_(apple): 70-80 mm, striped orange-red over yellow, density 0.86.
konarska2006	Hort. Sci. (Prague) 2006 (agriculturejournals.cz/pdfs/hor/2006/03/04.pdf): apple skin 33.3-73.1 um.
ishs_1099_97	ISHS Acta Hortic. 1099_97: lenticel density 4.3-12.6 per cm2, mean 7.11.
frontiers2015	Front. Plant Sci. 2015, 6:679 (fpls.2015.00679): porosity 10.6 % centre .. 26.4 % periphery; 10 primary bundles (5 sepal + 5 petal), carpellary 10 ventral + 5 dorsal bundles.
uspto	USPTO apple plant patents PP29089, PP20028, PP21710, PP23862, PP26766: core 24.7-36.7 mm, core length 41.1 mm, locules 15.2 x 7.9 / 16.6 x 9.9 mm, stems 10.5-28.2 x 2.7-3 mm, stem cavity 8-25 mm deep, calyx basin 3-17 mm deep.
sciRep2021	Sci. Rep. 2021 (s41598-021-02143-z): apple seed length 7.69-9.43 mm, width 4.0-4.9 mm.
msu2025	MSU Extension apple maturity report 2025: Gala ~18 lb Magness-Taylor at harvest.
fmach_porosity	openpub.fmach.it 10449/23207: apple porosity 17 % (Jazz) .. 29.8 % (Golden Delicious).
jets_skin_tensile	J. Eng. Technol. Sci. (journals.itb.ac.id/index.php/jets/article/view/1231): skin 2.56 MPa.
pmc7022590	PMC7022590: Golden Delicious slices browning dE 2.38 / 2.68 / 4.05 at 0 / 30 / 60 min.
atk_cut_apple	America's Test Kitchen 'How to cut an apple without coring': four cuts beside the core, slices 6 mm (matchsticks) / 13 mm (dice).
usda_egg	USDA AMS Egg-Grading Manual (ams.usda.gov/.../Egg Grading Manual.pdf): cuticle (not 'bloom'), palisade/'spongy' and mammillary layers, two membranes, air cell grades AA ≤3.2 mm, A ≤4.8 mm; albumen = outer thin 23 %, thick 57 %, inner thin 17 %, chalaziferous 3 %; yolk ~31 %; check vs leaker.
fr	Front. Immunol. 2022, 13:838525: cuticle 0.5-12 um, vertical crystal 3-8 um, palisade ~300 um, mammillary ~100 um, membranes ~70 um; pores ~2x denser at the blunt pole.
ir	ScienceDirect S2666469024000101: cuticle 5.6-12.5 um, mean 7.85 um (search snippet).
por	Rev. Bras. Cienc. Avic. (scielo ZBg3hyp6Swr9HgR9XSQSQsN): 10,000-20,000 pores (snippet).
ubc	UBC open collections 1.0104002: outer shell membrane ~3x thicker than inner.
kop	J. Process. Energy Agric. (aseestant.ceon.rs 32280): 58.05 x 45.46 mm, SI 78.25, 65.89 g, shell 12.76 %; compression 32.41 N mean, Hisex Brown 36.43 N short axis / 34.92 N long axis, range 10.03-58.96 N.
alt	Effect of egg shape index on mechanical properties (unis.gop.edu.tr): SI classes sharp <72, normal 72-76, round >76; highest breaking force along the polar axis at the sharp end.

Table 92: Sources held in food/params.py (SOURCES). (continued)

Key	Source
br	Redalyc 395940112007: shell 8-11 %, albumen 56-61 %, yolk 27-32 % of egg mass.
sev	Acta Univ. Agric. Mendel. 2010/01/19: eggshell E 41±3.9 GPa radial, 43±4.1 GPa tangential.
sb	Acta Univ. Agric. Mendel. 2004/05/07: single egg failed ~49 N, linear to failure, shell 0.35 mm, E 15 GPa.
gr	Grashorn, Lohmann Information 49 (2016) egg storage: air cell < 4 mm after 28 d at 6 C; > 6 mm after 25 d at 22 C; mass loss 1 / 2.58 / 4.54 g at 6 / 15 / 22 C over 28 d.
ro	Analele Univ. Oradea 2011 (Usturoi): fresh egg specific gravity 1.078-1.097; 1.040-1.059 after >30 d.
ab	Alberta Agriculture: infertile blastodisc 2-3 mm, fertile blastoderm 5-9 mm (snippet).
unsw	A Laboratory Manual of Vertebrate Embryology (1947), UNSW embryology: concentric yolk layers, latebra, neck of latebra to the nucleus of Pander under the germinal disc; two twisted chalazae.
eng	Engormix 'Protoporphyrin in shell and cuticle': 13-20 % of brown pigment in the cuticle, ~80 % in shell.
dsm	DSM YolkFan (formerly Roche): 15 shades since 1965, 16th added 2016.
cohen2025	Cohen et al. 2025, Communications Physics (MIT; physics.aps.org/articles/v18/100): 180 drops onto a hard surface from 8-10 mm: side-on 5 % (8 mm) / ~45 % (10 mm) cracked, vertical 55 % / ~80 %; static breaking ~45 N in both orientations.
bps1976	British Poultry Science 1976 (doi 10.1080/00071667608416318): some well-formed eggs crack from drops of 3.3 mm onto a stiff flat surface; typically a single hair crack round the equator or to a pole.
alphafoodie_boil	alphafoodie.com/how-long-to-boil-eggs: hot start, medium eggs: 4.5 min white just set/yolk very runny, 6 jammy, 7 barely runny, 8-9 set but creamy, 10-11 hard, >14 overcooked.
kenji_boil	Lopez-Alt (NYT 2019, via summaries): fridge eggs into boiling water: 6 min liquid yolk, 8.5 fudgy, 11 just firm; hot start peels cleanly ~90 % vs ~50 % cold start.
egginfo_crack	British Lion egginfo.co.uk 'How to crack an egg': rounded corner (bowl or pan edge), light tap; too hard on a flat surface and you get egg everywhere.
atk_egg_on_egg	America's Test Kitchen 'The best surface for cracking an egg is another egg': only the weaker egg cracks, leaving a clean divot; the membrane holds the shards.
atk_egg	America's Test Kitchen 'Egg cracking showdown: bowl vs counter': rim tears the membrane and splits cleanly but leaves shards that can puncture the yolk; counter gives larger pieces, membrane often intact.
wh_separate	wikiHow 'Separate an Egg': shell-to-shell transfer ~3x or hand; cold eggs; third bowl per egg.
cs42	Cummings & Schroeder 1942, Calif. Avocado Soc. Yearbook 27:56-64 (avocadosource CAS_1942_PG_056-064): exocarp (cuticle, epidermis, 1-3 hypodermal cells, parenchyma, stone-cell layer), mesocarp with oil idioblasts and 6 vascular strands, thin endocarp, two-layer papery testa, two cotyledons, embryonic axis.
uspto_pp14239	USPTO PP14239 (Hass comparison): skin 1.6 mm, fruit 204 g, seed 28.4 g, seed:skin:flesh 14:13:73.
clark2007	Clark et al. 2007 Postharvest Biol. Technol. 46:119 (phtnet mb409): fruit density 970-1018 kg/m3.
landahl	Landahl & Terry (Cranfield): Hass whole-fruit compression 304.1 → 2.1 N over 6 d at 18 C.
ashton2006	Ashton et al. 2006 J. Agric. Food Chem. 54:10151: Hass skin L*/C*/h over ripening; flesh in three equal bands dark green / pale green / yellow; cyanidin-3-O-glucoside in ripe skin.
farinas2021	Farinas et al. 2021 J. Food Eng. 289:110266: flesh slice modulus 2.29 → 0.16 MPa with ripening.
hia2018	Hort Innovation 2018 'Avocado squeezing is bad': 10 N thumb press bruises firm-ripe fruit; shoppers 3-30 N; bruises show after ~24 h.
farley2019	Farley et al. 2019 Am. J. Emerg. Med.: ~50,413 US avocado knife injuries 1998-2017, mostly the non-dominant hand while pitting.
ucd_avocado	UC Davis Postharvest Technology Center, avocado produce facts (postharvest.ucdavis.edu/node/7115): mature-green 5-13 C by cultivar; ripe fruit 2-4 C; ethylene at 20 C → ripe in 3-6 d.
marques2011	Marques et al. (2011) VII World Avocado Congress (avocadosource WAC7 MarquesJose2011b): Hass at the softening stage stored up to 10 d at 5-8 C with low risk of quality loss at eating ripe; avoid < 5 C once past the rubbery stage.
ripe_window_guides	Produce guides (axonify produce knowledge; UC Davis-hosted consumer page): a stage-4/5 Hass stays ripe ~24-48 h at room temperature before overripening (weak sources).
atk_browning	America's Test Kitchen 'How to keep avocado from turning brown': vacuum sealing kept halves green 5 d; plastic wrap + fridge minimal discolouration; lemon / lime juice or ascorbic acid slow browning but taste sour; keeping the pit in does not protect the cut face (informal kitchen test).
afm_howto	Avocados From Mexico / freshfruitportal how-to: cut lengthwise around the seed, twist apart; remove seed with a spoon (knife strike discouraged); score to dice; peel from the small end.
usda_handling	USDA ARS egg safety handling (ars.usda.gov .../eggsafety_handlingeggs.pdf): gently remove a stuck egg; cook cracked eggs thoroughly.
masterclass_crack	MasterClass 'How to crack an egg' (snippet): hold firm but gentle.
cookscountry_storage	Cook's Country egg storage (cookscountry.com/how_tos/5900-egg-storage): carton, middle/back shelf 35-40 F; door ~45 F.
eggs_ca	Egg Farmers of Canada (eggs.ca/?p=15718): crack on a flat surface; crack into a separate bowl first.
atk_crack_separate	America's Test Kitchen 'How to crack and separate eggs' (articles/1126): cold eggs, flat surface, three bowls, half shell lifts yolk specks.
egginfo_separate	British Lion egginfo.co.uk 'How to separate egg whites': shell-to-shell, hand, bottle methods.
foodrepublic_separate	Food Republic 'Ways to separate eggs' (1769095): methods and separator tools.
incredibleegg_boil	Incredible Egg 'How to make hard-boiled eggs': cold start, stand 9/12/15 min (M/L/XL); refrigerate eggs 7-10 d before; never microwave in shell.
milkstreet_peel	Milk Street 'How to peel hard-cooked eggs' (2020): steaming 12 min, 5 min ice bath, shake method ~60 % on warm eggs.
atk_scrambled	America's Test Kitchen 'Perfect scrambled eggs' (recipes/6604) and '6 tips' (articles/3118): fork-beat, half-and-half, medium-high then low.
incredibleegg_scrambled	Incredible Egg basic scrambled eggs: medium heat, push curds, off while glossy.
pepin_pbs	Jacques Pepin classic scrambled eggs (PBS American Masters): stop the cooking with reserved egg and cream.
loveandlemons_fried	Love & Lemons 'Fried egg': 1 tsp fat per egg, lid 2-3 min; over easy 15 s second side.
incredibleegg_fried	Incredible Egg basic fried eggs: steam-baste with 1 tsp water and a lid, or butter-baste.
kenji_baste	Lopez-Alt fried eggs via Tasting Table (1346566): baste with shimmering oil ~1 min, crisp lacy edges.
ishs_1119_22	ISHS Acta Hortic. 1119_22: avocado firmness devices (Sinclair iQ, firmometer, analogue meter, hand squeeze) are non-linearly related – readings are device-specific.
hirte_2012	Hirte A. (2012) Extending crispness - controlling water migration in bread. PhD thesis, Wageningen (edepot.wur.nl/221072): crispy roll ~90 x 50 mm; crust 1-3 mm (cut-off crust 1.0-1.5, ~1.3 mm); crumb porosity 0.73-0.81 (uCT 6 um), cell walls 20-61 um; crumb aw ~0.97; crust water 9.0 % at 30 min, 13.9 % at 1 d (40 % RH, 22 C); critical crust water 10.1 % reached after 2.1 h; crust break ≥ 3 N (1 x 10 mm blade); bake-off mass loss 2.4-4.1 %, 4 h storage 1.9-2.9 %.
lagrain_2012	Lagrain et al. (2012) Food Biophysics 7:173 (KU Leuven author manuscript): pan-bread crumb E ~20 kPa (5 h) → ~70 (48 h) → ~125 (96 h) → ~157 kPa (168 h) (read off figures), Poisson 0.103, loss factor 23 → 10 %; mean 2D cell area 0.32 mm2, 83 cells/cm2; open porosity 0.81-0.89; Gibson-Ashby E/Es = (rho/rho_s)^2; crumb moisture 0.46 → 0.42 g/g.
cardone_2020	Cardone et al. (2020) Foods 9:260 (PMC7143436): Avrami crumb firming, control wheat bread, n=1: A0 6.48 N, Ainf 66.01 N, k 0.186 (per day; unit to verify); crust L*~47, a* 9, b* 25.
chaple_2023	Chaple et al. (2023) LWT 174:114326 (PMC9883616): pan bread crust L*a*b* 47.6/17.05/32.2, crumb 72.06/0.94/15.37.
kurek_2016	Kurek et al. (2016) Food Technol. Biotechnol. 54(1) (PMC5105627): wheat rolls from 60 g dough, specific volume 6.4 (3 h) → 5.9 ml/g (48 h); moisture 47.8 %; crust browning index 50.4.
musiienko_2025	Musiienko et al. (2025) Front. Nutr. (PMC12497575): round buns 86-92 x 50-55 mm, 197-263 cm3.
testde_2021	Stiftung Warentest 'Aufbackbroetchen' (test.de 4206027): rolls 50-90 g; target crust 'hellgoldbraun bis dunkelgoldbraun'; crispness lost after 4 h, crumb unchanged; score raises the crust share.
ireks_rmt	IREKS Kompendium, RMT-Backversuch Broetchen: 1000 g T550 flour, 50 g yeast, 15 g salt, 10 sugar, 10 fat → 30 rolls; proof 25 min 32 C; bake 20 min at 230 C with steam.

Table 92: Sources held in food/params.py (SOURCES). (continued)

Key	Source
leitsaetze_1993	Leitsaetze fuer Brot und Kleingebaeck (1993): Kleingebaeck ≤ 250 g; Weizenbroetchen ≥ 90 % wheat flour (2021 revision: verify).
giovanelli_2021	Giovanelli & Cappa (2021) Foods 10:417 (PMC7918450): browning starts > 105-120 C at aw < 0.4-0.7; crust 140-180 C at the end of baking; browning index 35.5 (200 C) / 55.9 (225 C).
soleimani_2014	Soleimani Pour-Damanab et al. (2014) J Food Sci Technol 51:3439 (PMC4571265): crust thickness grows with bake time; colour bands outer crust L* < 51.6, inner crust 51.6-62.3, crumb 62.3-71.2.
kaur_2026	Kaur et al. (2026) Plant Biotechnol. J. (PMC13278543): home toaster 3 / 3.5 / 4 / 4.5 min, 'starting to look burned' at 4.5 min.
utopia_aufbacken	utopia.de 'Broetchen aufbacken': oven 150-170 C for 5-10 min after dampening; toaster rack highest setting 2-3 min, flip, 1-2 min.
oekotest_storage	oekotest.de: keep rolls in a paper bag at room temperature; plastic softens the crust; freeze fresh, do not refreeze.
wikihow_slice	wikiHow 'Slice Bread' (via kiwix mirror): serrated knife, roll flat, hand flat on top, saw from the centre; stop short for a hinge.
tastingtable_bagel	Tasting Table 'safest way to cut a bagel': flat side down, palm on top, serrated knife horizontally; halfway, stand on the uncut edge and finish downwards.
luo_2026	Luo & Yuan (2026) arXiv 2607.04367, robot food cutting: sawing at -30 N down-force, 2 cm/s, ±2 cm strokes; carrot = serrated-knife food (chopping failed, sawing succeeded).
zhang_2024	Zhang et al. (2024) Foods 13:397 (PMC10855756): rolls (80 g dough, 4 C) crumb hardness ~1.5 N (3 d) → ~4.0 N (7 d).
eu1333_2011	Commission Implementing Reg. (EU) 1333/2011 bananas: min length 14 cm along the convex face from the blossom end to where the peduncle joins the crown; min grade 27 mm (transverse thickness); skin defects Extra ≤ 1, I ≤ 2, II ≤ 4 cm2.
ciro05	Ciro Velasquez et al. (2005) Rev. Fac. Nal. Agr. Medellin 58:2975 (Cavendish Valery, 24 C, 12 d): outer-curve length 23.3 ± 1.13 cm; diameter 34/36/34 mm at 25/50/75 %; peel puncture 13.6-15.2 N (2 mm probe); pulp penetration 7.20 → 3.41 N; pulp E compression 1.75 → 1.09 MPa, bending 5.16 → 2.65 MPa; whole fruit fracture 129 N (bending, 14 cm span), 121 N (compression).
mal22	Malikongwa et al. (2022) Plant Science Today 9:364 (Grand Naine): length 19.0 cm; pedicel 17.3 x 10.15 mm; peel 2.56 mm; pulp 62.3 g / peel 23.2 g; IPGRI: slightly ridged, blunt tip, persistent style, slightly curved, cream pulp.
chiw23	Chiwate et al. (2023) Curr Agric Res J 11(1) (Cavendish, 22-27 C): day 1/4/6/8/10 mass 105/101/93/87/82 g, pulp:peel 1.69..1.78, Brix 4.5/6.8/8/16.7/23.7, firmness 44.8/36.8/34.2/28.1/22.1 N.
bant24	Bantayehu & Alemayehu (2024) Heliyon 10:e37143: Grand Naine 102 g, pulp:peel 2.18.
amn12	Amnuaysin et al. (2012) Trop. Nat. Hist. 12:127 (Hom Thong AAA): epidermis + cuticle, outer ground tissue (chloroplasts, starch), inner ground tissue with air spaces growing on ripening, bundles ringed by laticifers; peel 1.1-3.8 mm, thinning on ripening.
guapo22	Guapo & Ferreira (2022) Cadernos UniFOA 17(50): 3 carpels/locules; 'strings' = innermost mesocarp strands of parenchyma with two small vascular bundles and secretory spaces (not pure phloem).
promusa	ProMusa 'Banana flowers': tricarpellate inferior ovary, 200-300 ovules on a central placenta; the black dots in the centre are the ovule remnants.
khanal23	Khanal et al. (2023) Postharvest Biol Technol: outer peel layers are the mechanical backbone; peel splits axially in the fruit corners (ridges); peel fracture 53.0 → 2.1 N on ripening (abstract).
ringer21	Ringer & Blanke (2021) J Food Meas Charact / Tropentag 2021 (German supply chain): peel hue R2..R7 98.7/97.3/92.7/89.4/87.5/82.0 deg, b* 56-72; peel water 90.5 % (≤ R5) → 82.9 % (R7); Shore 82 → 42.
hou15	Hou et al. (2015) IJABE 8(6):99: ethylene-treated AAA at 20 C, stage 1 → 7 in 7 days; peel L*, b* peak at stage 5, a* rises.
ma22	Ma et al. (2022) Curr Res Food Sci 5:1808: stage day bands RS1 1-4, RS2 5-6, RS3 7-8, RS4 9-10, RS5 11-12, RS6 13-15, RS7 > 15 d.
markon	Markon banana colour chart: 1 green, 2 green trace yellow, 3 more green than yellow, 4 more yellow than green, 5 yellow green tips, 6 all yellow, 7 yellow flecked brown.
wflo	WFLO Commodity Storage Manual, Bananas (2018): chilling below 13 C (pulp); ripening rooms 14-18 C; specific heat 3.56 kJ/kg K; moisture 74.3 %.
mbeguie09	Mbeguie-A-Mbeguie et al. (2009) J Exp Bot 60:2021: pedicel rupture force falls > 2x between day 1 and 3 (Grande Naine, 20 C); no abscission zone (ketsa25).
imsabai	Thai finger-drop report (TSRI BGJ4380011): finger-drop pull resistance ~5 kg (~49 N) on day 1 (Hom Thong / Namwa).
bugaud14	Bugaud et al. (2014) PBT 87:113: Grande Naine no visible bruise up to 200 mJ; susceptibility rises with ripening (abstract).
col20	Colombian Musaceae ripening (PMC7984252), Gros Michel day 0/4/6 L* 39.9/45.0/56.4, a* -26/-12/+13.1, b* 40.9/62.1/85.4.
kim22	Kim, Ohmura, Kuniyoshi (2022) arXiv 2203.09749: dual-arm banana peeling, 40.9 % consecutive success (PeelNeck 95 %, PeelRight 59 %, PeelLeft 64 %); neck too stiff on fresh fruit, pre-cut.
banana_howto	Tasting Table (1552037), Chiquita blog: peel from the stem (snap the neck) or from the tip ('monkey method': pinch the tip); rounds ~6 mm (3-6 mm, MyPlate); lemon juice against browning; hang, keep apart from apples/avocados; refrigerate only when ripe.
sadka19	Sadka et al. (2019) Front. Plant Sci. 10:1167: hesperidium; flavedo (oil glands), albedo, endocarp segment membrane, juice vesicles on stalks from the endocarp; 3 bundles per segment (1 dorsal, 2 septal) ending in the albedo; central axis (columella).
smith18	Smith, Ebrahimi, Ghosh, Dickerson (2018) PNAS 115:E5887 (PMC6042112): oil jets on bending the peel 8.47 ± 4.03 m/s, 102 um; reservoir ~1 mm3 under an 84 um flavedo cap, burst at 30-140 kPa; flavedo E ~103 MPa (30-300), albedo 0.79 ± 0.58 MPa; albedo 3.9 mm in the FE model.
jentzsch22	Jentzsch et al. (2022) Front. Mater. 9:979151: 'Lane Late' peel 6.2 mm, density 746 kg/m3, compressive E ~0.02 MPa, Poisson -0.05, 0.51 MPa at 50 % strain.
ucr_citrus	UCR Givaudan Citrus Variety Collection data: Washington navel (ripe) 162-335 g, 61-89 x 61-88 mm, rind 4.1-6.4 mm, juice 45-54 %, TSS 9.3-14.1, acid 0.49-0.77 %, 0-6 seeds per 10 fruit; Olinda Valencia 113-299 g, L/W 1.0-1.1, rind 3.5-6.2 mm, juice 36.9-56.1 %, 1.2-6.6 seeds per fruit.
edis891	UF/IFAS HS891 Dooryard Citrus Guide: Valencia and navel 0-6 seeds; navel = rudimentary secondary fruit at the blossom end; navel juice turns bitter within several hours.
citrus_patents	USPTO PP21535 / PP27829 (sweet orange): 9.6-10.6 and 10-11 segments; PP27829 208 oil cells/cm2; PP21224 albedo 2-5 mm; PP6047 navel peel 5-5.1 mm.
knight01	Knight, Klieber, Sedgley (2001) Ann. Bot. 88:1039: 8,000-12,000 oil glands on a mature Washington navel (abstract) → ~40-60 /cm2 on ~200 cm2.
eu543_citrus	Reg. (EU) 543/2011 Annex I B 2 (oranges): min diameter 53 mm (size codes 0-13); min juice 33 % navel / 35 % others; light green on ≤ 1/5 of the surface (superseded by 2023/2429: verify).
orange_mech	Khoshnam et al. 2022 Iran. J. Biosyst. Eng. 53:109 (punch 15.5-79 N, 1.5-7.9 N/mm, abstract); Dadvar et al. 2015 (Valencia rupture 140-221 N, abstract); Montero et al. 2012 Sci. Hortic. (oil glands rupture under 10-250 N sphere compression; Valencia no oleocellosis).
ritenour13	Ritenour et al. ASHS 2013: peel-removal rig (16 mm strip, 75 mm): Valencia hard-peel (strips break), navel medium-easy; cold fruit (5 C) needs more force and tears more; no N values published.
orange_juice	BSDA QUID: juice SG 1.045 at 11.2 Brix; US4938985: single-strength viscosity ~4-6 mPa s at 8 C; alphafoodie: 60-80 mL per medium orange by hand, roll 10-15 s, halve at the equator, ream.
orange_storage	UC Davis Produce Facts (Arpaia & Kader): 3-8 C, 90-95 % RH, up to 3 months; chilling (pitting) after 3-6 weeks at ~4 C (UF PHNL194).
orange_howto	instructables 'Easy Orange Peeling' (score 6-8 mm deep round the navel, spiral to the stem); kitchprep; The Daily Meal / A Couple Cooks: supreme – peel and pith off, cut inside each membrane; zest the flavedo only (albedo and membranes bitter, limonin).
cci	Citrus colour index CCI = 1000 a*/(L* b*) (Jimenez-Cuesta 1981); market CI target ~ +10..+20 (snippet).
test25_toast	Stiftung Warentest 'Toastbrot im Test' (23.04.2025, test.de 4421655): baked ~30 min under a lid (thin soft crust), cooled in a clean room, sliced and bagged; most packs have no heels (organic packs: 2); best-before 12-20 d; no classic preservatives needed.
toast_pack	Open Food Facts / REWE / Friedrich Verlag (snippets): 500 g pack, 20 slices of ~25 g, pack ~0.25 m long.
nuss90	Nussinovitch, Roy, Peleg (1990) Cereal Chem 67:101: commercial white sliced bread, crumb 0.19 g/cm3; tensile failure 0.31 / 0.39 N (12 mm neck, 10 / 50 mm/min) at 9.2-9.7 mm; compressed bread rarely fails.
sehn20	Sehn & Steel (2020) J Food Sci Technol 57:557 (PMC7016051): white control pan loaf firmness 1.98 / 3.62 / 4.18 / 5.64 N at day 1 / 5 / 9 / 13 (25 C, PE bag); crumb moisture 39.35 → 36.01 %.
ishida14	Ishida & Steel (2014) Food Sci Technol Int: Brazilian commercial white pan bread: 4.72 ml/g, crumb 37.0 % moisture, firmness 276 gf day 1, crumb L*a*b* 74.73 / 0.37 / 15.51 (abstract).

Table 92: Sources held in food/params.py (SOURCES). (continued)

Key	Source
toaster26	Stiftung Warentest toaster test (2026): 80 s (fastest) to > 2.5 min to the best toast; higher settings = longer time, not more heat; browning scored as browned area.
toast_misc	EU 2017/2158 acrylamide benchmark 50 ug/kg (wheat soft bread); FSA 'Go for Gold'; Daily Meal: 'perfect toast' 216 s at 5/6, 14 mm; butter ~0.068 g/cm2 (Wedzicha); fridge stales ~6x faster (popular, weak); Verbraucherzentrale: no fridge, freeze slices airtight.
tins	Runex 500 g bread tin: top inside 280 x 100 mm, depth 90 mm; hobby lidded toast tin 36 x 11 x 8 cm (2 loaves).
mc20	Massana-Codina et al. (2020) Front Plant Sci 11:1287 (PMC7468465): periderm = phellem / phellogen / phelloderm; phellem 46-84 um, 6-10 cell layers (five European cultivars).
mag25	Magwaza et al. (2025) Front Plant Sci 16:1595828: lenticels ≤ 12 /cm2 (small tubers); apical buds ~7x the lateral density, 6-9 apical buds; respiration 1.9-2.75 mL CO2/kg h.
bsl21	Bundessortenamt Beschreibende Sortenliste Kartoffel 2021: Belana oval yellow-skinned, very flat-flat eyes; Gala round-oval, flat eyes; Linda long-oval, flat-medium eyes; flesh medium-yellow.
potato_mech	E 3.12 MPa (Cagnolati 2018; Celik 2019), 3.15 MPa / nu 0.45 / yield 1.18 MPa (Zhao 2023), 4.6-5.6 MPa, fracture 1.13-1.40 MPa (Polish clones, SV); fracture strain ~0.3; toughness ~212 J/m2 (1996, SV); blackspot above ~50 g impact (Hyde 1991, SV).
kk18	Krishna Kumar et al. (2018) J Food Meas Charact: raw potato 1053 kg/m3, k 0.56 W/m K, alpha 0.15e-6 m2/s, initial freezing -1.8 C (abstract); ETB: k 0.55, cp 3.43 kJ/kg K.
potato_cook	Lee (2017) Food Eng Prog 21:303 review: first-order softening, k(100 C) 0.28-0.40 /min, Ea 103-145 kJ/mol (Harada 1985); gelatinisation onset 61, peak 67.5, end 76-82 C (Witczak 2013, SV); PME firming at 60-70 C (Gonzalez-Martinez 2004).
gritzer	Serious Eats (Gritzer): cold start in 1-2 % salted water; diced ~15-20 min, large whole up to ~1 h; done when a skewer passes without resistance; German guides 20-25 min whole from cold.
potato_storage	<8 C cold-induced sweetening; home 7-13 C dark; ~3 % weight loss in month 1; greening under retail light 1-2.5 d at 20 C (Nofima), from day 4 / 2 at 10 / 15 C (JSHS 1990); BfR: remove green parts and eyes generously.
potato_peel	Peeler vs knife study (SV): hand peeler ~1.1 mm, ~9 % waste; peeled slices under light: glycoalkaloids 2 → 74 mg/kg in 48 h (OSU EM 9407).
ince16	Ince, Cevik, Vursavus (2016) IJABE 9(6):200 (Jadelo F1): 72 x 70 x 53 mm, 165 g; plate firmness initial G/P/R 15.1 / 6.0 / 4.5 N/mm, average 13.2 / 4.7 / 3.1; 2 mm puncture skin rupture 6.92 / 3.99 / 2.74 N; a*/b* bands green -0.59..-0.47, breaker ..-0.27, turning ..0.08, pink ..0.60, light red ..0.95, red ..1.21 (after Batu 2004).
fdc_tomato	USDA FoodData Central 170457: small whole 61 mm 91 g, medium 66 mm 123 g; water 94.5 g/100 g; slices 1/4 in 20 g, 1/2 in 27 g; wedge (1/4 medium) 31 g.
bn05	Bargel & Neinhuis (2005) J Exp Bot 56:1049: cuticle 9.9-15.7 um (red); skin E 252-293 MPa, failure ≤ 25-28 MPa at ~9-10 % strain (red).
meta19	Metabolites 9:93 (2019, Ailsa Craig red): pericarp 5.3 mm; pericarp ~50 %, exocarp 9.6 %, columella + placenta 10.0 %, locular tissue 23.3 % of FW.
ucd_rost	UC Davis Rost lab tomato anatomy: skin, pericarp, septa, columella, placenta, locular gel; most cultivated non-cherry types 4-5 locules; small types often 2 (Wikipedia).
cfr_tomato	7 CFR 51.1860 colour classes: green, breakers (≤10 %), turning (10-30 %), pink (30-60 %), light red (60-90 %), red (> 90 %).
wflo_tomato	WFLO Commodity Storage Manual, Tomatoes: ripen 18.3-20 C; chilling < 12.8 C; ripe 7-10 C 3-5 d; freezing -0.5 C; cp 4.08 kJ/kg K; mature-green seeds pushed aside by the knife (gel liquefied).
zhang16	Zhang et al. (2016) PNAS 113:12580: 8 d at 5 C cuts volatiles by 65 %, not recovered after 3 d at 20 C.
tomato_mech	Sirisomboon 2012 (whole stiffness 2000-10000 N/m, peel 70-90 % of firmness); Safitri 2022 (Hertz E 0.58-1.97 MPa, SV); Li 2017 (FEM: damage from ~3 % strain; 6.5 N at 10 %); Kultongkham 2021 (bio-yield 2.57 N, SV); Lana 2004 (slice firmness Ea 87.5 / 94.8 kJ/mol).
tomato_howto	Wikibooks Tomato Concasse: cone-core, X 2.5 cm, blanch 30-45 s, ice water, peel, seed, dice 0.75 cm; Jessica Gavin: serrated or very sharp knife, slices 6-13 mm, 8 wedges; room temperature.
hb66_onion	USDA AH-66 onion chapter (Adamicki): bulb = enlarged leaf bases + compact stem; 0 C, 65-75 % RH, 6-9 months; rest 4-6 weeks; curing 3-5 % loss; respiration 3 (0 C) .. 8 (20 C) mg CO2/kg h; freezing -0.8 C; medium 5.1-8.3 cm.
bah04	Bahnasawy et al. (2004) J Food Eng 62:255 (yellow Giza 20): 51.7 x 57.1 mm, 78.7 g, density 1.04 g/cm3, crushing 400-780 N, puncture 27.6-45.5 N, friction 0.67-1.34, rolling angle 20-31 deg (base).
kim_heb20	Hortic Environ Biotechnol (2020), 45 cultivars: 7.6 ± 0.2 fleshy scales, 2.3 growing centres, scale thickness 5.5 ± 0.2 mm (abstract).
wu25	Wu et al. (2025) arXiv 2505.06016 'Droplet outbursts from onion cutting': layer ~3 mm; tissue E 2.2 MPa (skin on) / 1.4 (off); blunt / fast knives up to ~40x / ~4x more droplets; chilling did not change droplet speed.
wei01	Wei, Lintilhac, Tanguay (2001) Plant Physiol 126:1129: adaxial epidermal peel 120 um, E 3.5-8.0 MPa (turgor 0.14-0.54 MPa), nu 0.18-0.30; strips break at ~40 / ~20 gf (long./trans.).
jafari16	Jafari Malekabadi et al. (2016) J Agric Machinery 6:114: whole yellow onion E 2.03-5.45 MPa, nu 0.26-0.45.
lokke12	Lokke et al. (2012) Sensors 12:16060: lachrymatory factor highest at ~1 min after chopping, gone by ~10 min; Imai 2002 Nature 419:685 (LF synthase).
onion_howto	America's Test Kitchen 'How to cut an onion' (2016): halve through the root, peel, keep the root on, horizontal then vertical cuts, claw grip; mince ≤ 3 mm, fine 3-6, medium 6-13, coarse 13-19 mm; rings: trim a flat base; HWQ: 2-3 horizontal cuts, remove tough layers; UNL: chill 30 min.
onion_misc	Gracie 2012 (6th true leaf forms the outer skin; skins tear easily); Karate 2025 (cured neck 7.4-10.9 mm); Alamar 2020 (lateral buds up to 1/3 of mass; dormancy break week 8 at 1 C); frying colour a* 5.7 → 17.9, b* 14.2 → 29.3 (SV); caramelising 30-45 min (typical).
agu20	Aguilar-Hernandez et al. (2020) Foods 10:62 (PMC7824522), Fino / Verna: 100-184 g, ED 55-67 mm, PD 72.7-91.9 mm, ED/PD 0.6-0.8, peel 4.9-6.8 mm, 8.8-10.2 segments, 1.3-11.3 seeds, juice 31.7-44.3 %, TA 56-75 g/L, TSS 7.8-10.5, peel L* 64-68, a* -7.5..3.9, b* 42-49, hue 88-99 deg.
jen22_lemon	Jentzsch et al. (2022) Plants 11:991 (PMC9002614), Primofiore: 91 mm, 166 g, peel 5.26 mm, flavedo ~1 mm (20 %), peel 36.4 % of mass, peel density 818 kg/m3, drop impact COR 0.31.
ben26	Bengisu et al. (2026) Biomimetics 11:451 (PMC13406879), retail lemon: puncture 41.2 N (5 mm), knife cutting 36.3 N, compression 165.8 N at 16 mm; peel 0.68 g/cm3, albedo 0.39 g/cm3; oil glands 1.25 /mm2.
eu2023_2429	Delegated Reg. (EU) 2023/2429 citrus: lemons min juice 20 %, min diameter 45 mm, size codes 0 (79-90) .. 7 (45-52 mm).
lemon_misc	MAR22 Verna (PMC9407548): 93.7-98.1 mm, peel 6.7-7.1, seeds 1.3-2.3; LIA22 Eureka chilling < 10 C, 3.5 % loss in 80 d; MIT20 degreening CCI -16.3 → -1.1; USDA FDC 167747 juice of 1 lemon 48 g, wedge 5.9 g; Wikipedia citric acid 5-6 %, pH 2.2; zest = flavedo only (albedo bitter).
dewiki_sg	de.wikipedia 'Salatgurke': normally three locules; trade classes 300-400 / 400-500 g.
unece_cuc	UNECE FFV-15 (2017): Extra / I max 10 mm arc height per 10 cm, II 20 mm; weight spread 150 g (≥ 400 g); EU 1677/88 (expired 2009): ≥ 250 g, ≥ 25 cm (250-500 g), ≥ 30 cm (≥ 500 g); seeds soft.
sakata11	Sakata et al. (2011) J Japan Soc Hort Sci 80:420: greenhouse type puncture (3 mm) flesh 8.7 N, placenta 2.8 N, skin 9.6-9.8 N; 25-30 cm, ~260 g.
jah16	Jahangiri et al. (2016) CIGR J 18:323: apparent modulus 0.85-1.39 MPa, failure stress 0.53-0.85 MPa, strain 0.45-0.53; firmness -49 % in 9 d; stem end 7.53 vs blossom end 5.01 N.
disect_cuc	Heiden et al. (2021) DiSECt (arXiv 2105.12244) Table A1: cucumber E 2.5 MPa, nu 0.37, 950 kg/m3; SliceIt! (2404.02569): 5 mm slices, stiffness jump cucumber → board.
ucd_cuc	UC Davis Produce Facts, Cucumber: 10-12.5 C, 95 % RH, < 14 d; chilling below 10 C after 1-3 d (7.2 C: 2-3 d); ethylene 1-5 ppm yellows; freezing -0.5 C; respiration 12-15 mL CO2/kg h at 10 C.
cuc_misc	rett24 cuticle 4.0 um (1.1-10.7); ando15 peel layer 1-2 mm; wang22 skin a* -20.6, b* 27.5 (yellowing to b* 35 in 20 d); dhall10 wrapped 0.66 vs unwrapped 11.1 % loss in 15 d at 12 C; shoji22 typical 350 g; wikibooks: fork scrape for wax, zebra peel, spoon out seeds, salt 30 min.
leitsaetze_fb	Leitsaetze fuer Feine Backwaren (2025) 2.2.3.4: Plunder ≥ 30 kg butter (or equivalent fat) per 100 kg flour, flaky structure; croissant = rolled Plunder, often crescent-shaped; 2.1.1: butter in the name → no other fats.

Table 92: Sources held in food/params.py (SOURCES). (continued)

Key	Source
ste25	Steinkellner, Kroll, Franke (2025) Curr Res Food Sci 10:101042 (PMC11995129): butter croissant ~50 % butter on flour, 3 single folds = 27 fat layers, specific volume 3.13 ± 0.66 cm3/g, pore surface 6.62 /cm; TPA firmness 5.16 N (1 h) → 19.2 N (24 h); bake 200/220 C 20 min; lamination 13-15 C.
croissant_misc	Wikipedia en/de/fr Croissant: rolled 3.5-4 turns; optimum 16-50 fat layers (Ooms 2016); butter melts 32-34 C; final sheet 3.5 mm; bake 210-220 C. USDA FDC 174987: 28 / 42 / 57 / 67 g; water 23.2, fat 21.0 g/100 g. Weekend Bakery: egg wash 2 coats; reheat from frozen 180 C 8 min.
wp_baguette	Wikipedia fr/en 'Baguette': ~65 cm, 4-6 cm wide, 3-5 cm high, ~250 g; tradition = flour, water, yeast/levain, salt (decret 93-1074); scores at 45 deg with a razor; stales faster than a boule; Grand Prix de Paris: 55-65 cm, 250-300 g, salt 18 g/kg.
wp_grigne	Wikipedia fr 'Grigne': coupe parisienne = 5 medium overlapping cuts along the axis, ~1 cm deep; the grignes open into raised lips ('oreilles').
dv2012	Della Valle et al. (2012) J Cereal Sci 56:440: French breads (64), mean crust 1.7 mm; crust ends ~160 C, ~5 % moisture.
zou2024	Zou et al. (2024) npj Sci Food (PMC11390739): baguette crust L*a*b* 57.5 / 10.5 / 34.9, crumb 66.4 / -1.7 / 15.7 (powders, SV); bake 230 C 30 min with steam.
baguette_misc	King Arthur baguettes (232 C, 24-28 min, steam; paper bag overnight, freeze); van Vliet 2006 crispness lost within 2 h at 80 % RH; Wikipedia Staling: fastest at ~4 C, partly reversed at 60 C.
he22	He et al. (2022) Plants 11:985 (PMC9003265): blocky red cv. Gina 237 ± 37 g, 77.6 x 81.3 mm, wall 7.5 ± 0.7 mm (orange 6.9); cuticle 10.4 / 8.7 um; bite firmness 11.6 / 6.9 N, -81 % after 15 d at 20 C.
alt20	Althaus & Blanke (2020) J Imaging 6:122: German retail peppers 230-245 g; no stomata; at 17 C / 55 % RH 2.9-4.8 g/d loss, shrivel from 6-8 % (4-5 d); roughness Sa 2 / 7 / 24 um; gloss 450-500 → 80-90; non-climacteric.
bar20	Barbosa et al. (2020) Molecules (PMC7287789): pericarp L*a*b* green 33.7 / -8.6 / 16.9, red 38.8 / 31.4 / 27.8, yellow 61.4 / 5.0 / 50.0; whole-fruit shelf life 12-28 d.
sha26	Sharma et al. (2026) Sci Rep (DUS): locules 2-4, mostly 3; 1000-seed weight 4 to > 6 g.
pepper_misc	Mensah 2026: 38-142 seeds per fruit; Afolabi 2023: whole-fruit 8 mm probe 43.5 N; Darre 2024 / Hanaei 2022: chilling below 7 C (7-10); EU 2023/2429 sweet peppers (classes, sizing); USDA ring (3 in x 1/4 in) 10 g; Harvey / Sweeper harvesting robots (36.9 s, 24 s cycles); Wikibooks: remove stem, core, seeds and white veins; char to loosen the skin.
eu730_1999	Commission Reg. (EC) 730/1999 carrot marketing standard (mirrors UNECE FFV-10): main crop ≥ 20 mm or 50 g, Extra ≤ 45 mm / 200 g; green/violet tops: Extra none, Class I ≤ 1 cm (≤ 10 cm long) / ≤ 2 cm (longer), Class II ≤ 2 / 3 cm; firm, not woody, not forked.
con22	Conversa et al. (2022) Foods (PMC8750513): Nantes 'Presto' 135.4 g, 210.8 mm, 31.4 mm; 'CPT' 110.4 g, 182.7 mm, 30.3 mm; hand peeler removes ~2 mm = 12.1 / 14.6 % of fresh mass.
jah18	Jahanbakhshi et al. (2018) Food Sci Nutr 6:1898 (PMC6189625): 154.6 mm, 72.7 g; density 1.04 g/cm3; moisture 87.2 %; 3-point bending Fmax 48.6 N at 21.9 mm, 0.50 J; blade shear 41.1 N; compression 71.9 N.
vob24	Vo-Bui, Rogers, Pegg (2024) R Soc Open Sci 11:230420 (PMC10805594): Nantes orthotropic constants after Stopa et al. 2017: xylem E 4.62-4.88 MPa, phloem 4.33-4.85 MPa, nu 0.47-0.48, G 1.47-1.66 MPa; E falls 1.32x in 7 d (fridge, zip bag); core/root diameter ratio falls crown → tip.
peng14	Peng, Tang, Barrett, Sablani, Powers (2014) J Food Eng 122:84: carrot softening, 2nd-order fractional conversion, k 0.035 (80 C) → 1.453 1/min (110 C), Ea 138.9 kJ/mol, Finf/F0 0.037-0.008; raw dice (12.7x12.7x6 mm, 70 % strain) 274 N; xylem core 4-7 mm in dice.
singh17	Singh et al. (2017) Indian J Hort 74:56: 40 genotypes, root 30.55 mm, core 11.74 mm (ratio ~0.38).
hb66	USDA Agric. Handbook 66 (2016), carrot chapter: 0-1 C / 98-100 % RH 7-9 months; bagged 2-3 weeks at 3-5 C; freezing point -1.2 C; respiration 46-95 mg CO2/kg h at 20 C; Nantes prone to cracking and tip breakage; parenchyma cells ~50 um; softening = turgor loss then pectin beta-elimination.
herp99_calbo00	Herppich et al. (1999) PBT; Calbo (2000) Bragantia: fresh turgor ~0.7 MPa; turgor firmness ~0 below relative water content 0.95 (~5 % water loss) – the limp carrot.
storage_loss	Snippets: unwrapped 20 C / 50 % RH unacceptable after ~3 d (US4919948); > 5 % loss in 4 d bulk unsaleable (Semina 2010); 5-8 % loss unsaleable.
sim09_cz95	Simoes et al. (2009) PBT; Cisneros-Zevallos et al. (1995) J Food Sci 60:320: white blush on cut carrot = surface dehydration (reversible within ~1-1.5 h) then lignification (irreversible after ~15 h).
pac16_dry	Paciulli et al. (2016) J Food Sci Technol; drying study IJFP 10.1080/10942912.2010.489209: raw carrot L*a*b* 50-55 / 25-29 / 30-40 (hue ~50-54 deg).
alv00	Alvarez et al. (2000) J Texture Stud: SENB on carrot/celery/apple, K_IC 10-50 kPa m^0.5, fracture energy 40-400 J/m2.
morpheus_2024	Ye et al. ICRA 2024 MORPHeus (arXiv 2404.06570): instrumented peeler, carrot 90.1 % area peeled; skin and flesh colours hard to tell apart → force/vibration sensing.
carrot_cuts	ATK 'How to cut carrots' (articles/3368), Leiths, Wikipedia culinary cuts: plank first so it does not roll; 5-7.5 cm lengths; julienne 3x3x40-50 mm, brunoise 3 mm, batonnet 6x6x50-60 mm, dice 6/12/19 mm; peel from the thick end, rotate after each stroke.
carrot_boil	Evolving Table / Food & Home snippets: 1/4 in rounds 4-5 min crisp-tender (+1-2 min soft); whole 10-15 min; not much more than 10 min for slices.
diarch_rows	Botany: lateral roots arise beside each xylem pole → a diarch root has 4 rows of lateral roots (general text; carrot-specific confirmation in Esau 1940 Hilgardia 13:175 to obtain).

C Files and rebuild

All paths are relative to ~/Documents/HouseholdTwin.

```
food/params.py               parameter register (status per value) and SOURCES
food/<item>.py               analytic anatomy, part tree, asset spec per item
food/<item>_models.py        hidden-state models and beliefs (wave 1-3)
food/dynamics.py, sdt.py     egg/apple/avocado truth dynamics and beliefs
food/egg_crack.py, egg_cook.py, egg_open.py   egg crack, boil and opening models
food/schema.py               discrete SDT schema (egg, apple, avocado, roll, banana, carrot)
food/partgraph.py, usd_partgraph.py           closed meshes, cuts, USD part graph
food/egg_tasks.py, wave1_tasks.py, wave2_tasks.py   task designators
household/garmi_rig.py       GARMI articulation rig (IK, force-limited grasp)
scripts/build_food_assets.py assets/food/<kind>_<seed>/ (whole + cut states, sdt.usda, truth.usda)
scripts/import_apartments.py assets/apartments/ (EASE, GARMI apartment, GARMI robot)
scripts/sim_*.py             Isaac Sim benches -> test_reports/*.json
scripts/render_*.py          path-traced renders -> renders/
tools/*.py                   CPU studies, plots and generated docs
food/avocado_vision.py, avocado_belief.py    photo -> node estimates; particle-filter belief of the avocado node
food/avocado_peel.py, peel_monitor.py        peeling outcomes as USD states; vision-based peeling monitor
food/avocado_process.py, avocado_emulator.py, avocado_tasks.py   halving / pitting mechanics, emulator, designators
scripts/avocado_from_image.py, avocado_observe.py, avocado_process_demo.py, *peel*.py   CLI and studies
cram/avocado_sdt.py          node, competence questions, mental simulation, value of information (CRAM side)
```

```
cram/avocado_world.py, avocado_motions.py   Avocado(Fruit) annotation, pieces, PR2 motions
cram/avocado_pr2_demo.py     scripted PR2 scenario; cram/record_run.py -> cram/results/pr2_run.json
cram/avocado_kitchen.py      interactive kitchen for the CRAMERA viewer
report/build_report.py       this document (CPU only; cache in report/cache, figures in report/figures)
report/avocado_reasoning.py  the section on the avocado as the robot's belief state
cram/avocado_agent.py        goal-directed agent: effects to enforce / prevent, missions, lifetime-based decisions
food/avocado_lifetime.py     ripening, ripe window, decline, spoilage, browning under storage treatments
theory/avocado_bws.py, avocado_policies.py, axioms.py   research SDT, task programs, relational axioms and audit
theory/run_microtheory.py    held-out evaluation, ablations, misspecified plants -> theory/results/
theory/package.py, learning.py, prl.py   theory packages and gate, theory construction, programmable RL
report/avocado_theory.py     the section on the agent, the microtheory, theory construction and RL
report/build_avocado_microtheory.py   the standalone microtheory (report/avocado_microtheory.pdf)

E=~/Documents/LabAutomationTwin/env_cad/bin/python
I=~/Documents/LabAutomationTwin/env_isaacsim/bin/python
$E scripts/build_food_assets.py
$E tools/egg_crack_study.py && $E tools/egg_cook_study.py && $E tools/anatomy_doc.py
OMNI_KIT_ACCEPT_EULA=YES $I scripts/sim_egg_bench.py        (GPU, one job at a time)
OMNI_KIT_ACCEPT_EULA=YES $I scripts/render_food.py
$E report/build_report.py [--refresh]
```